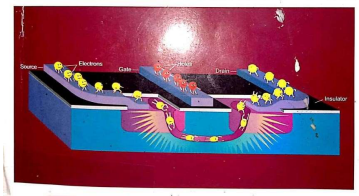
Eastern Economy Edition

Semiconductor Devices

Modelling and Technology



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1 Semiconductors

1.1 INTRODUCTION

In terms of electrical conductivity, that is, the ability to conduct current, all materials in the world can be classified under three categories, namely, conductors, insulators, and semiconductors. A *conductor* allows current to flow through it, that is, it has high electrical conductivity. On the other hand, an *insulator* tends to block the flow of current. In between these two extremes we have the *semiconductors*. Semiconductors are the materials whose conductivity can be modulated over a wide range. Numerically, the conductivity of a material is denoted by the symbol σ (the reciprocal of conductivity is resistivity which is denoted by the symbol ρ). Generally for conductors, $\rho < 10^{-3}$ ohm-cm while for insulators, $\rho > 10^{9}$ ohm-cm at room temperature. In its pure form, silicon, the most widely used semiconductor exhibits a resistivity of 2.31×10^{5} ohm-cm at 300 K. However, by introducing a very small amount of impurity (few parts per million) in silicon, it is possible to reduce this value by several orders of magnitude. This unique property of semiconductors has fuelled intense research in this field that has culminated in the stupendous success of semiconductor devices and integrated circuits. In fact, it is not an exaggeration to say that the "Electronic Revolution" of twentieth century is largely due to the ubiquitous semiconductor devices, which are used today in almost all spheres of life where electronic instruments make an entry. It is therefore, extremely important to understand the properties of semiconductor materials, as they hold the key to the performance of semiconductor devices.

Semiconductor elements generally belong to the group IV in the periodic table. Certain compounds can also be classified as semiconductors. It is important at this stage to answer a query: why is it that some materials behave as semiconductors while others are conductors and insulators? For a simple explanation, let us start with the definition of conductivity in a solid, and also look at the factors which affect conductivity.

Ohm's law states that the current I through a resistance R due to an applied voltage V is given by

$$I = \frac{V}{R} \tag{1.1}$$

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metal. A large number of free electrons causes higher conductivity. The covalent bond is formed by sharing of electrons between similar atoms. This bond is of particular interest to us since the group IV elements form covalent bond. These elements have four valence electrons which are shared with their four neighbours to form a stable configuration. A two-dimensional schematic representation of the silicon lattice with the covalent bonds is shown in Fig. 1.2. A covalent bond is very strong and hence, covalently bonded materials are generally reasonably good conductors at room temperature. The strength of the covalent bond also depends on the size of the atoms. For small atoms such as carbon, the valence electrons are tightly bound, while for the larger atoms such as germanium and tin, the bonds can be broken more easily. Hence at room temperature, the conductivity of the group IV elements increases with increase in atomic number due to the increased availability of the charge carriers. In the case of the semiconductors, silicon and germanium, the conductivities are more than diamond (an insulator) and less than tin (a conductor). An interesting property of semiconductors is the increase in conductivity with increase in temperature. As the temperature is raised, the electrons gain energy, resulting in more bonds being broken and free carriers being generated. In the case of metals, however, almost all the valence electrons are already free at room temperature. Hence, there is no further increase in the number of charge carriers with the increase in temperature. Their conductivity in fact reduces at high temperatures due to a fall in mobility of the carriers.

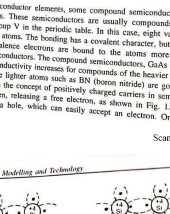

Figure 1.2 Schematic two-dimensional representation of silicon lattice The black dots represent valence electrons that participate in the formation of bonds.

In addition to the semiconductor elements, some compound semiconductors are now being used for various applications. These semiconductors are usually compounds formed from the elements in group III and group V in the periodic table. In this case, eight valence electrons are shared amongst neighbouring atoms. The bonding has a covalent character, but it is partially ionic as well. As a result, the valence electrons are bound to the atoms more strongly than the corresponding group IV semiconductors. The compound semiconductors, GaAs and InP have lower conductivities than Si. The conductivity increases for compounds of the heavier atoms, for example InSb, while compounds of the lighter atoms such as BN (boron nitride) are good insulators.

We shall now introduce the concept of positively charged carriers in semiconductors, called **holes**. When a bond is broken, releasing a free electron, as shown in Fig. 1.3(a), it also leaves behind a vacant site, called a hole, which can easily accept an electron. On application of an

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electric field, a neighbouring valence electron may move to occupy this hole, thus leaving behind a hole in its original position, as shown in Fig. 1.3(b). Hence, the hole has effectively moved in a direction opposite to the direction of movement of the electron. So far as the flow of current is concerned, the hole therefore behaves like a positively charged particle with a charge equal in magnitude to the electronic charge. We thus see that the breaking of a bond results in the creation of an *electron–hole pair* (*EHP*). Since they are oppositely charged, and they move in opposite directions under the influence of an electric field, the total current is the sum of the contributions due to the flow of electrons and holes.

It must however be emphasized that this simple picture of a hole as depicted in Fig. 1.3 has a number of serious limitations as neither a hole nor an electron is a particle in the classical sense. However, this elementary model helps one to visualize the thermal generation of electrons and holes. We can thus consider holes to be physical entities whose movement constitutes the flow of current.

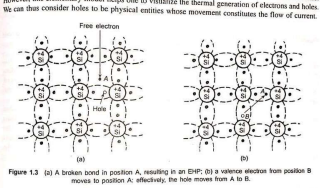

Figure 1.3 (a) A broken bond in position A, resulting in an EHP; (b) a valence electron from position B moves to position A; effectively, the hole moves from A to B.

In addition to the thermally generated intrinsic carriers, it is possible to create carriers in the semiconductors by deliberately introducing impurities in the lattice. This is called *doping*. For example, let us consider the introduction of a group V element such as phosphorus in silicon. In terms of the covalent bond formation, the situation can be explained in the following manner. A group V element has five valence electrons, four of which are used to form the covalent bonds. With very little energy, much less than what is required to break a covalent bond, the fifth electron can be made to move freely inside the crystal. This electron is thus "donated" to participate in electrical conduction. Such a group V impurity is called a *donor* as it donates electrons. In an analogous situation, a group III element has only three valence electrons. So there is a hole at the fourth covalent bond site. In order to complete this bond, the group III element can "accept" one more electron from some other covalent bond site. A group III impurity is therefore called an *acceptor* as it accepts electrons. In the process, a hole is created at the original electron position. Figures 1.4(a) and (b) represent these two situations schematically.

A semiconductor in its pure form is called an *intrinsic semiconductor*. In an intrinsic semiconductor, the number of free electrons is equal to the number of holes, as they are generated simultaneously by the breaking of bonds. A doped semiconductor is called an *extrinsic*

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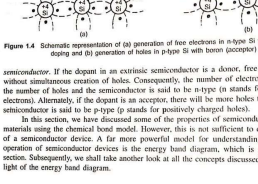

Figure 1.4 Schematic representation of (a) generation of free electrons in n-type Si with phosphorus (donor) doping and (b) generation of holes in p-type Si with boron (acceptor) doping.

semiconductor. If the dopant in an extrinsic semiconductor is a donor, free electrons are created without simultaneous creation of holes. Consequently, the number of electrons will be more than the number of holes and the semiconductor is said to be n-type (n stands for negatively charged electrons). Alternately, if the dopant is an acceptor, there will be more holes than electrons and the semiconductor is said to be p-type (p stands for positively charged holes).

In this section, we have discussed some of the properties of semiconductors and some other materials using the chemical bond model. However, this is not sufficient to describe the operation of a semiconductor device. A far more powerful model for understanding and describing the operation of semiconductor devices is the energy band diagram, which is presented in the next section. Subsequently, we shall take another look at all the concepts discussed in this section in the light of the energy band diagram.

1.2 ENERGY BANDS IN SOLIDS

1.2.1 Splitting of Discrete Energy Levels into Bands

The transport of charge in a solid depends not only on the properties of the charge carriers but also on the arrangement of atoms in a solid. Solids can be *single crystals*, where the atoms are arranged in a perfectly regular periodic structure in all three dimensions called the lattice; *polycrystalline*, where there are many small regions of single crystal material called the grains; and *amorphous*, where there is no periodic structure. The semiconductors that form the subject of our discussion are single crystals. As already mentioned, single crystals consist of a space array of atoms or molecules (or strictly speaking ions) constituted by regular repetition of a certain basic structural unit in three dimensions. This structural unit is called the *unit cell*. Silicon, GaAs, and most of the other semiconductors have a diamond or zinc blende structure. In this structure, each atom is surrounded by four of its nearest neighbours. Thus, the electrons in the outer orbit of one atom feel the influence of neighbouring atoms. Hence, the discrete energy levels for a single free atom (as in a gas where the atoms are sufficiently far apart so as not to exert any influence on one another) do

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As isolated atoms are brought closer, various interactions occur between neighbouring atoms. The forces of attraction and repulsion between atoms find a balance at the proper interatomic spacing for the crystal. In the process, important changes occur in the electron energy level configurations, which are responsible for various electrical properties of solids.

Qualitatively, we can say that as atoms are brought closer to each other, the application of *Pauli's Exclusion Principle* becomes important. When two atoms are completely isolated from each other (that is, they are far apart) so that there is no interaction of electron wavefunctions, they can have identical electronic structures. As the distance between atoms decreases, the wave functions begin to overlap. The exclusion principle dictates that *no two electrons in a given interacting system can occupy the same quantum state*. Thus, the discrete energy levels of the isolated atoms must split into new levels belonging to the pair rather than to the individual atoms. In a solid where many atoms are brought close together, the split energy levels form a large number of discrete but closely spaced energy levels called **Energy Bands**.

Let us consider the imaginary structure of a diamond crystal formed by isolated carbon atoms. Carbon has six electrons in $1s^2\ 2s^2\ 2p^2$ configuration. Consider a system of n equidistant carbon atoms. Each atom has two 1s states, two 2s states, and six 2p states. So for n carbon atoms, there will be $2n$ states of 1s type (all filled), $2n$ states of 2s type (all filled), and $6n$ states of 2p type (of which $2n$ are filled and $4n$ empty). When the interatomic spacing is large, the energy levels of these subshells are isolated. However, as the atoms are brought closer, these energy levels split into bands, beginning with the outermost subshell, as shown in Fig. 1.5. As the 2s and 2p bands grow, they first merge into a single band comprising of $8n$ states, of which $4n$ are occupied. As the distance between atoms approaches the equilibrium interatomic spacing of diamond, this single band splits into two bands separated by an energy gap E_g. The lower band contains $4n$ states all of which are occupied at 0 K. The upper band also contains $4n$ states but they are empty at 0 K. The energy gap contains no available states for the electrons to occupy and is hence, also referred to as the *forbidden gap*. The lower band is called the *valence band* since it contains all the valence electrons. The upper band is called the *conduction band* since excitation of electrons into this band is primarily responsible for electrical conduction.

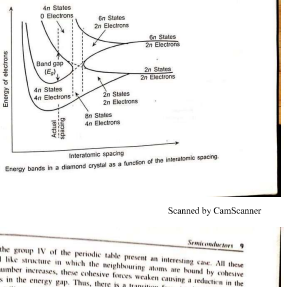

Figure 1.5 Energy bands in a diamond crystal as a function of the interatomic spacing.

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1.2.2 Metals, Semiconductors, and Insulators

For electrical conduction, that is, for electrons to experience acceleration in an applied electric field, they must be able to move into new energy states. That is, there must be empty (not already occupied) energy states available to electrons. For diamond, however, the valence band is almost completely filled and therefore hardly any empty states are available for electrons. On the other hand, there are very few electrons present in the conduction band and hence, there is little possibility of charge transport. So the electrical conductivity is very poor and diamond is an insulator.

Semiconductors have similar energy band structure as insulators, that is, a filled valence band and an empty conduction band at 0 K. The only difference lies in the size of the energy gap, which is smaller in semiconductors. For example, the band gap in diamond is 5 eV while that in the case of silicon is only 1.1 eV. The energy gaps of some common semiconductors are given in Table 1.1. The relatively small separation between valence and conduction bands in semiconductors allows electrons to be excited from the filled valence band into the empty conduction band by thermal excitation at room temperature. This introduces electrons in an almost empty conduction band as well as vacant states in an almost full valence band, thereby increasing the conductivity of the material.

Table 1.1 Band gaps of some common semiconductors at 300 K

Semiconductor	Band gap (eV)
Germanium (Ge)	0.66
Silicon (Si)	1.12
Indium Phosphide (InP)	1.35
Gallium Arsenide (GaAs)	1.42
Silicon Carbide (SiC)	2.69
Gallium Nitride (GaN)	3.36

On the other hand, in the case of metals, bands either overlap or they are partially filled. Thus, electrons have empty states available to them into which they can move under the influence of an electric field. This results in good electrical conductivity of most metals. The differences in the energy band structure of metals, semiconductors, and insulators are shown in Fig. 1.6.

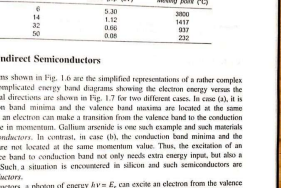

Figure 1.6 Typical band structures for insulators, semiconductors, and metals.

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The elements in the group IV of the periodic table present an interesting case. All these elements have diamond like structure in which the neighbouring atoms are bound by cohesive forces. As the atomic number increases, these cohesive forces weaken causing a reduction in the melting point as well as in the energy gap. Thus, there is a transition from insulating (diamond) through semiconducting (silicon, germanium) to metallic (tin) behaviour as we consider successive elements along with their atomic numbers are presented in Table 1.2.

Table 1.2 Atomic number, band gap, and melting points of some group IV elements

Element	Atomic number	Band gap (eV)	Melting point (°C)
Carbon	6	5.30	3800
Silicon	14	1.12	1417
Germanium	32	0.66	937
Tin (α–Sn)	50	0.08	232

1.2.3 Direct and Indirect Semiconductors

The energy band diagrams shown in Fig. 1.6 are the simplified representations of a rather complex band structure. More complicated energy band diagrams showing the electron energy versus the momentum in two crystal directions are shown in Fig. 1.7 for two different cases. In case (a), it is seen that the conduction band minima and the valence band maxima are located at the same momentum value. Thus, an electron can make a transition from the valence band to the conduction band without any change in momentum. Gallium arsenide is one such example and such materials are called *direct semiconductors*. In contrast, in case (b), the conduction band minima and the valence band maxima are not located at the same momentum value. Thus, the excitation of an electron from the valence band to conduction band not only needs extra energy input, but also a change in momentum. Such a situation is encountered in silicon and such semiconductors are called *indirect semiconductors*.

In direct semiconductors, a photon of energy $h\nu \approx E_g$ can excite an electron from the valence band to the conduction band (direct transition). However, in indirect semiconductors, this type of direct transition is not possible. This is because the photons have very small momentum while the electron has to undergo a large change in momentum. In such cases, electron transition from valence band to conduction band can occur by involving a lattice phonon (thermal energy), which can support the required momentum change (indirect transition). Direct transitions are also possible but the minimum photon energy required to excite electrons will be larger than E_g as shown in Fig. 1.7(b).

In a direct semiconductor, when an electron in the conduction band falls to occupy an empty state in the valence band, the energy is usually given off as a photon of light. However, in the case of an indirect semiconductor, such as silicon, this type of transition involves a change in momentum in addition to a change in energy, and the energy difference is generally given up as heat to the lattice rather than a photon of light. Therefore, light emitting devices are generally made of direct semiconductors.

Subsequently, in this book, we shall use a simplified energy band diagram where E_c is the minimum energy in the conduction band and E_v is the maximum energy in the valence band. The band gap $E_g = E_c - E_v$ is therefore the minimum energy separating the conduction band and the valence band as shown in Fig. 1.7.

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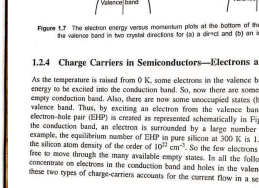

Figure 1.7 The electron energy versus momentum plots at the bottom of the conduction band and top of the valence band in two crystal directions for (a) a direct and (b) an indirect semiconductor.

1.2.4 Charge Carriers in Semiconductors—Electrons and Holes

As the temperature is raised from 0 K, some electrons in the valence band receive enough thermal energy to be excited into the conduction band. So, now there are some electrons in the previously empty conduction band. Also, there are now some unoccupied states (holes) in the normally filled valence band. Thus, by exciting an electron from the valence band to conduction band, an electron-hole pair (EHP) is created as represented schematically in Fig. 1.8. After excitation into the conduction band, an electron is surrounded by a large number of unoccupied states. For example, the equilibrium number of EHP in pure silicon at 300 K is 1.5×10^{10} cm^{-3} compared to the silicon atom density of the order of 10^{22} cm^{-3}. So the few electrons in the conduction band are free to move through the many available empty states. In all the following discussions, we shall concentrate on electrons in the conduction band and holes in the valence band. The movement of these two types of charge-carriers accounts for the current flow in a semiconductor.

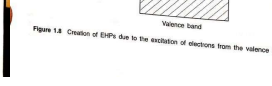

Figure 1.8 Creation of EHPs due to the excitation of electrons from the valence to the conduction band.

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We can now relate to the discussion in section 1.1. At 0 K, all valence electrons are tightly bound through the covalent bonds between neighbouring atoms. As the temperature increases, thermal vibration of the lattice can impart sufficient energy to some of the electrons allowing them to break the bonds and move freely inside the crystal. These free electrons can now participate in electrical conduction. The energy needed to break the bond is actually the band gap energy in the energy band diagram.

The energy band diagram actually reflects the electronic energy, that is, if an electron gains energy, it moves up in the energy band diagram. On the other hand, when a hole gains energy, it moves down due to its opposite charge. Thus, holes seeking the lowest energy state are available at the top of the valence band (E_v) while electrons are available at the bottom of the conduction band (E_c). The unit of energy generally used is electron volt (eV), which is equal to 1.6×10^{-19} J. The unit electron volt is so named due to the fact that if an electron falls through a potential of 1 V, the kinetic energy gained, which is equal to the decrease in its potential energy, is

$$E = qV = 1.6 \times 10^{-19}\ \text{C} \times 1\ \text{V} = 1.6 \times 10^{-19}\ \text{J} = 1\ \text{eV} \quad (\because \text{Coulomb} \times \text{Volt} = \text{Joule})$$

So in the energy band diagram if electron energy E goes up by 1 eV, it means equivalently that it has fallen through a potential V of 1 volt. Note that although V and E are dimensionally different, they have the same magnitude. So, we may say that the energy band diagram also reflects (what is known as) the *electronic potential*, which is the negative of the electrostatic potential (the electrostatic potential is defined with respect to a positive charge). For an electron in the conduction band having an energy E, the energy corresponding to the bottom of the conduction band (E_c) is its potential energy, while ($E - E_c$) is its kinetic energy. This is shown in Fig. 1.9(a).

If we apply a voltage V across a semiconductor, the energy band diagram looks as shown in Fig. 1.9(b). As the electron moves under the influence of the applied potential V, its loss in potential energy (qV) is equal to the gain in kinetic energy. However, as the electron moves, it suffers a number of collisions, giving its kinetic energy to the lattice (generating heat), and falls to the bottom of the conduction band. Consider the analogy of a stone of mass m at a height h, which has a potential energy mgh, where g is the acceleration due to gravity. When it falls and reaches the ground level under the influence of gravitational field, its entire potential energy is converted into kinetic energy. However, as it 'collides' with earth, it ultimately loses all its kinetic energy and comes to rest.

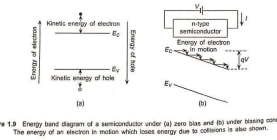

Figure 1.9 Energy band diagram of a semiconductor under (a) zero bias and (b) under biasing conditions. The energy of an electron in motion which loses energy due to collisions is also shown.

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1.2.5 Intrinsic and Extrinsic Semiconductors

A perfect semiconductor crystal with no impurities or defects is called an intrinsic semiconductor. In such a material, the valence band is filled and the conduction band is empty at 0 K as already mentioned in section 1.2.2. At higher temperature, electron–hole pairs (EHPs) are generated as the covalent bonds break and valence band electrons are excited to E_C as explained in section 1.2.4. These EHPs are the only carriers in an intrinsic semiconductor. Since electrons and holes are created in pairs, the electron concentration, n in the conduction band is equal to the hole concentration p in the valence band, that is,

$$n = p = n_i \tag{1.9}$$

where n_i is called the intrinsic carrier concentration. As the temperature increases, lattice vibrations also increase and more EHPs are created. Thus, n_i increases with temperature. Also, if the energy band gap is smaller, less energy is required for EHPs to be generated. So at a given temperature, n_i will be higher for a material with smaller band gap. For example, for silicon $n_i = 1.5 \times 10^{10}$ per cm^3 at 300 K while at 500 K it is about 3×10^{13} per cm^3. On the other hand, GaAs which has a larger band gap than silicon has $n_i = 1.79 \times 10^6$ per cm^3 at 300 K. The exact dependence of n_i on temperature and band gap energy are derived later in the chapter.

We have already discussed in section 1.1 that it is possible to create additional free electrons or holes by introducing a suitable dopant in the semiconductor. This concept is now explained with the help of energy bands. When impurities are introduced in a perfect crystal, additional energy levels are created in the energy band structure. For example, a group V element (P, As, and Sb) generates an energy level E_D which is very close to E_C in silicon or germanium. This level is filled with electrons at 0 K and with very little thermal energy these electrons can be excited into the conduction band. Such an impurity is therefore called a *donor* as it donates electrons. Thus, even when the temperature is low so that the number of thermally generated carriers is small, there is a large concentration of electrons in the conduction band of this type of doped semiconductor, that is, $n >> n_i$, p. Such a semiconductor is called n-type semiconductor. Again, if an element from group III is used to dope silicon or germanium, an energy level E_A is created very close to E_V. This level is empty at 0 K but with little thermal energy, electrons from the valence band can be excited into this level, creating a large number of holes in the valence band. As this level accepts electrons, this type of dopants are termed *acceptors* and the material is called p-type, where $p >> n$, n. Energy band diagrams of n-type and p-type semiconductors with the donor and acceptor levels are shown in Fig. 1.10. It may be mentioned here that the process of donating or accepting electrons ionizes the impurity atoms, the donors becoming positively charged, while the acceptors become negatively charged (refer Fig. 1.10). This is detailed further in section 1.3.7.

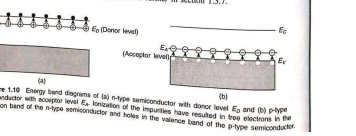

Figure 1.10 Energy band diagrams of (a) n-type semiconductor with donor level E_D and (b) p-type semiconductor with acceptor level E_A. Ionization of the impurities have resulted in free electrons in the conduction band of the n-type semiconductor and holes in the valence band of the p-type semiconductor.

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$f(E) = 0.5$. It may also be noted that since $f(E)$ is the probability that a particular state is occupied, the probability that it is vacant is $[1 - f(E)]$.

Some properties of Fermi–Dirac function are listed here as follows:

(a) At $T = 0$ K, the distribution function takes a simple rectangular form as shown in Fig. 1.11. This implies that all states above E_F are empty and all states below E_F are filled with electrons, that is, for $E > E_F$, $f(E) = 0$ and for $E < E_F$, $f(E) = 1$.

(b) As shown in Fig. 1.11, for $T > 0$ K, there exists some probability for states above E_F to be filled, that is $f(E)$ is nonzero for $E > E_F$. There is also a corresponding probability that states below E_F are empty since $f(E) < 1$ for $E < E_F$. For $E > E_F$, the state at a higher energy has a lesser probability of being occupied by an electron. On the other hand, for $E < E_F$, the state at a lower energy has a lesser probability of being vacant. Also the probability that a state at $E > E_F$ is occupied or a state at $E < E_F$ is vacant increases with temperature.

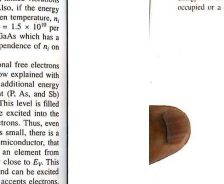

Figure 1.11 The Fermi distribution function f(E) versus $(E - E_F)$ at various temperatures.

(c) It can be shown that the probability of a state at an energy ΔE above E_F being occupied is exactly the same as the probability of a state at an energy ΔE below E_F being vacant. In other words, $f(E)$ is symmetrical about E_F at all temperatures, that is, $f(E_F + \Delta E) = 1 - f(E_F - \Delta E)$.

(d) A filled state in the conduction band indicates the presence of a free electron and a vacant state in the valence band indicates the presence of a hole. The free electron density around an energy level E in the conduction band is the product of the density of states $g(E)$ and the probability $f(E)$ that the states are occupied. On the other hand, the density of holes around an energy level E in the valence band would be the product of the density of states $g(E)$ and the probability $[1 - f(E)]$ that the states are vacant. Since the number of free electrons is the same as the number of holes in an intrinsic semiconductor, the symmetry of $f(E)$ about E_F implies that for an intrinsic semiconductor, if the density of available states in the conduction and valence band are the same, then the Fermi level should lie exactly at the middle of the band gap. This is shown clearly in Fig. 1.12(a).

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(e) In an n-type material, $n > p$. This implies that there are more filled states in the conduction band than there are empty states in the valence band. The symmetry of $f(E)$ implies that E_F should be closer to E_C for n-type material. As shown in Fig. 1.12(b), in this case the area under the $g(E)$ $f(E)$ curve in the conduction band (which represents the concentration of electrons in the conduction band) is more than that under the $g(E)[1 - f(E)]$ curve in the valence band (which represents the concentration of holes in the valence band). Similarly for p-type material, E_F should be closer to E_V. This is discussed further in Fig. 1.12(c). (For mathematical proof and further discussion of the last two properties, the reader should refer to sections 1.3.5 and 1.3.6).

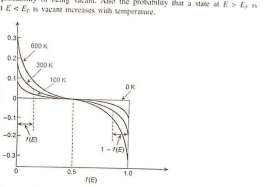

Figure 1.12 The density of states (t)E), the Fermi distribution function f(E), and the electron and hole concentrations in (a) intrinsic semiconductor, (b) n-type semiconductor and (c) p-type semiconductor

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When a semiconductor is n- or p-doped, only one type of carriers dominate. These are called *the majority carriers*. Carriers of the other type, whose concentration is much less, are called the *minority carriers*. Thus, electrons in an n-type semiconductor and holes in a p-type semiconductor are majority carriers while electrons in p-type semiconductor and holes in n-type semiconductor are the minority carriers.

HELP DESK 1.1

Why is the intrinsic carrier concentration in a semiconductor a constant and not an increasing function of time, inspite of EHPs being continuously generated at room temperature?

Till now we have only discussed the process of generation of EHPs, where an electron in the valence band gains sufficient energy to reach the conduction band. A simultaneous process called *recombination* also takes place in the semiconductor, where an electron in the conduction band releases energy to fall back to the valence band and recombines with a hole, thereby annihilating an electron–hole pair. The generation rate G, defined as the number of EHPs generated per unit volume per second, at a given temperature is more for lower band gap materials. Also, for a particular semiconductor, G is higher for a higher temperature. On the other hand, the recombination rate R, defined as the number of EHPs which recombine per unit volume per second, depends on the number of available carriers, and is actually proportional to the product of the number of free electrons and holes. If $G > R$, the number of carriers will increase as the number of carriers generated is more than the number of carriers recombining. On the other hand, if $G < R$, the number of carriers reduces. An equilibrium is reached and the number of carriers becomes constant only when $G = R$. Now let us conduct a thought experiment. Let the temperature of a semiconductor sample be raised instantaneously at $t = 0$ from 0–300 K and maintained at 300 K for $t > 0$. Let the generation rate at 300 K be given by G_{300}. Initially, there are no carriers present and $R = 0$. As $G_{300} > R$, the number of carriers increases and R keeps-increasing with time. This process continues till $G_{300} = R$ and the carrier concentration reaches an equilibrium value, say $(n_i)_{300}$. Now at $t = t_1$, the temperature of the semiconductor is suddenly raised to 400 K. Let the generation rate at 400 K be G_{400}. Since $G_{400} > G_{300}$, initially the generation rate is more than the recombination rate ($G > R$), and the carrier concentration keeps increasing with a consequent increase in R. Again an equilibrium is reached when $G = R$, and let the equilibrium concentration at 400 K be $(n_i)_{400}$. It is quite obvious that $(n_i)_{400} > (n_i)_{300}$. Thus, the intrinsic carrier concentration increases with temperature. Similarly, we can also argue that the intrinsic carrier concentration will be lower for higher band gap materials. We also argue that the intrinsic carrier concentration will be lower for higher band gap materials. We shall develop quantitative relations for the dependence of n_i on band gap and temperature, as well as discuss recombination in more detail later in this chapter.

1.3 ELECTRON AND HOLE DENSITIES IN EQUILIBRIUM

1.3.1 Distribution of Quantum States in the Energy Band

To develop quantitative expressions for electron and hole densities (n and p), we shall introduce two mathematical results without proof. The first of these results is derived from quantum

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mechanics and it deals with the distribution of quantum states in energy in the conduction band, the energy gap, and the valence band of the semiconductor. According to this, the number of states per unit volume (a state is an energy which can be occupied by an electron) in the conduction band with energies between E and $E + dE$ is given by

$$g(E)dE = 4\pi\left(\frac{2m_n^*}{h^2}\right)^{3/2}(E - E_C)^{1/2}\,dE \quad \text{for } E \geq E_C \tag{1.10a}$$

Similarly, the number of states per unit volume in the valence band with energies between E and $E + dE$ is given by

$$g(E)dE = 4\pi\left(\frac{2m_p^*}{h^2}\right)^{3/2}(E_V - E)^{1/2}\,dE \quad \text{for } E \leq E_V \tag{1.10b}$$

where

h = Planck's constant
m_n^* = effective mass of electrons
m_n^* = effective mass of holes

As there are no allowable states in the forbidden gap, $g(E) = 0$ for $E_V < E < E_C$. It is clear from the above equations that $g(E)$ is the density of states, that is, the number of states per unit volume per unit energy around an energy level E.

Since the electrons in a crystal are not completely free and instead interact with the periodic potential of the lattice, their behaviour cannot be expected to be the same as that of electrons in free space. Thus, in applying usual equations of electrodynamics to charge carriers in a solid, we must use altered values of particle mass called "effective mass". In doing so, the influence of lattice is taken into account so that electrons and holes can be considered as 'almost free charge carriers' in most computations. The effective masses of electrons and holes are different in different materials. For example, $m_n^* = 0.067m_0$ in GaAs, where m_0 is the free electron mass. From Eq. (1.10), we see that at the band edges, that is, at $E = E_C$ and $E = E_V$, the density of states $g(E) = 0$. The density of states increases continuously as we move deeper into the bands, that is, for increasing E in the conduction band and decreasing E in the valence band.

1.3.2 Fermi–Dirac Statistics

The second result that we follow is the Fermi–Dirac distribution function. Electrons in solids follow Fermi–Dirac statistics, which can be expressed as follows—the probability $f(E)$ that an electron occupies an available state at an energy level (E) at thermal equilibrium at a temperature $T(K)$ is given by

$$f(E) = \frac{1}{1 + \exp\left(\dfrac{E - E_F}{kT}\right)} \tag{1.11}$$

where

k = Boltzmann's constant and
E_F = Fermi energy level.

Equation (1.11) shows that the maximum value of $f(E)$ is 1 when $E \to -\infty$ and the minimum value is 0 when $E \to \infty$. For an energy $E = E_F$, the occupation probability at any temperature is

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1.3.3 Electron Concentration in the Conduction Band

Based on the discussion so far, we can now express the electron concentration in the conduction band as

$$n = \int_{E_C}^{E_{top}} f(E)\, g(E)\, dE \tag{1.12}$$

where E_{top} = energy at the top of the conduction band.
Substituting Eqs. (1.10a) and (1.11) into Eq. (1.12), we have

$$n = \int_{E_C}^{E_{top}} \frac{4\pi(2m_n^*/h^2)^{3/2}(E - E_C)^{1/2}}{1 + \exp\left(\dfrac{E - E_F}{kT}\right)}\, dE \tag{1.13}$$

However, it is very difficult to solve this equation analytically. Hence, the following approximations are made:

(i) Let us assume that $(E_C - E_F) > 3kT$. Since kT is 0.026 eV at room temperature, it is usually a good approximation for normal doping levels. Using this, since $(E - E_F) > 3kT$ for any energy level E in the conduction band, $f(E)$ given by Eq. (1.11) can be approximated as $f(E) = \exp[-(E - E_F)/kT]$. In other words, the Fermi–Dirac distribution can now be approximated by a Boltzmann distribution. However, it must be remembered that when the semiconductor is very heavily doped (degenerate), E_F is very close to E_C and this approximation may not be valid.

(ii) Typically, the width of the conduction band is several electron volts. However, most of the electrons in the conduction band are confined close to E_C since for energies above E_C, the probability for electron occupancy quickly approaches zero. So the upper limit of the integration in Eq. (1.13) can be changed to ∞ without any loss of accuracy.
Therefore, Eq. (1.13) can now be modified as

$$n = \int_{E_C}^{\infty} 4\pi\left(\frac{2m_n^*}{h^2}\right)^{3/2}(E - E_C)^{1/2}\exp\left[-\left(\frac{E - E_F}{kT}\right)\right]dE$$

$$= \frac{4\pi}{h^3}(2m_n^*)^{3/2}\exp\left[-\left(\frac{E_F - E_C}{kT}\right)\right]\int_{E_C}^{\infty}(E - E_C)^{1/2}\exp\left[-\left(\frac{E - E_C}{kT}\right)\right]dE \tag{1.14}$$

The solution of Eq. (1.14) is given by

$$n = N_C \exp\left[-\left(\frac{E_C - E_F}{kT}\right)\right] \tag{1.15}$$

where $N_C = 2(2\pi kTm_n^*/h^2)^{3/2}$ and is called the effective density of states in the conduction band. The concept of N_C can be explained in the following manner. From Eq. (1.15), we see that the expression for n is of the form $n = N_C f(E_C)$. Therefore, if all the states in the conduction band were to be confined at the particular energy level $E = E_C$, then N_C represents the number of states required to be present at E_C. As a result, after multiplying N_C with the probability of electron occupancy at $E = E_C$, the number of holes in the valence band can be obtained. From the expression of N_C and N_V, it can be seen that they both increase with temperature. Also as m_n^* and m_p^* do not have the same values, N_C and N_V are not equal (although they are of the same order). For a particular semiconductor, at a given temperature N_C and N_V are constants. In case of silicon at room temperature (300 K), these values are $N_C = 2.8 \times 10^{19}$ per cm^3 and $N_V = 1.02 \times 10^{19}$ per cm^3.

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occupancy at $E = E_C$, we get the number of electrons in the conduction band. However, it must be noted that the actual density of states at $E = E_C$ is zero and the concept of effective density of states only helps us to obtain a simple expression for the free electron concentration in terms of the separation between the Fermi level and the conduction band edge. Since N_C depends on m_n^*, its value is different for different materials.

1.3.4 Hole Concentration in the Valence Band

To compute the hole concentration in the valence band, we note that the probability of a hole occupying a level in the valence band (f_h) is actually the probability that a state is not occupied, or $f_h(E) = 1 - f(E)$. Hence, from Eq. (1.11), we can write that

$$f_h(E) = 1 - f(E) = \frac{\exp\left(\dfrac{E - E_F}{kT}\right)}{1 + \exp\left(\dfrac{E - E_F}{kT}\right)} = \frac{1}{1 + \exp\left(\dfrac{E_F - E}{kT}\right)} \tag{1.16}$$

The hole concentration in the valence band can now be written as

$$p = \int_{E_{bot}}^{E_V} f_h(E)\, g(E)\, dE \tag{1.17}$$

where E_{bot} is the energy corresponding to the bottom of the valence band. After making analogous assumptions as in the previous case, that is, $(E_F - E_V) > 3kT$, from which we have $f_h(E) \approx \exp[-(E_F - E)/kT]$ in the valence band and replacing E_{bot} with $-\infty$, we have

$$p = \int_{-\infty}^{E_V} 4\pi\left(\frac{2m_p^*}{h^2}\right)^{3/2}(E_V - E)^{1/2}\exp\left[-\left(\frac{E_V - E}{kT}\right)\right]dE$$

$$= \frac{4\pi}{h^3}(2m_p^*)^{3/2}\exp\left[-\left(\frac{E_F - E_V}{kT}\right)\right]\int_{-\infty}^{E_V}(E_V - E)^{1/2}\exp\left[-\left(\frac{E_V - E}{kT}\right)\right]dE \tag{1.18}$$

From Eq. (1.18), we obtain the concentration of holes in the valence band at thermal equilibrium as

$$p = N_V \exp\left[-\left(\frac{E_F - E_V}{kT}\right)\right] \tag{1.19}$$

where $N_V = 2(2\pi kTm_p^*/h^2)^{3/2}$ is referred to as the effective density of states in the valence band. From Eq. (1.19), we see that the expression for p is of the form $p = N_V f_h(E_V)$. Therefore, N_V represents the number of states required to be present at E_V, so that after multiplying with the probability of vacancy at $E = E_V$, the number of holes in the valence band can be obtained. From the expressions of N_C and N_V, it can be seen that they both increase with temperature. Also as m_n^* and m_p^* do not have the same values, N_C and N_V are not equal (although they are of the same order). For a particular semiconductor, at a given temperature N_C and N_V are constants. In case of silicon at room temperature (300 K), these values are $N_C = 2.8 \times 10^{19}$ per cm^3 and $N_V = 1.02 \times 10^{19}$ per cm^3.

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1.3.5 Carrier Concentration in Intrinsic Semiconductor

If E_i denotes the position of the Fermi energy level in an intrinsic semiconductor, then from Eqs. (1.15) and (1.19), we have

$$n_i = N_C \exp\left[-\left(\frac{E_C - E_i}{kT}\right)\right] \tag{1.20}$$

$$p_i = N_V \exp\left[-\left(\frac{E_i - E_V}{kT}\right)\right] \tag{1.21}$$

where

n_i = concentration of electrons in intrinsic semiconductor and
p_i = concentration of holes in the intrinsic semiconductor.

Since the electron concentration is equal to the hole concentration in an intrinsic semiconductor, that is, $n_i = p_i$, from Eqs. (1.20) and (1.21), we obtain

$$E_i = \frac{E_C + E_V}{2} - \frac{kT}{2}\ln\left(\frac{N_C}{N_V}\right) \tag{1.22}$$

Equation (1.22) mathematically proves our observation in section 1.3.2 that if the effective densities of states in the conduction and valence bands are equal, the Fermi level for an intrinsic material would be located exactly at the centre of the band gap. However, since the quantity ($kT/2$) in (N_C/N_V) is rather small (a few meV) in practical situations, E_i **is assumed to be located approximately at the middle of the band gap for all intrinsic materials.**

Again, since n_i is equal to p_i, from Eqs. (1.20) and (1.21), we can write

$$n_i p_i = n_i^2 = N_C N_V \exp\left[-\left(\frac{E_C - E_V}{kT}\right)\right] = N_C N_V \exp\left[-\left(\frac{E_g}{kT}\right)\right] \tag{1.23}$$

Therefore,

$$n_i = \sqrt{N_C N_V}\exp\left[-\left(\frac{E_g}{2kT}\right)\right] \tag{1.24}$$

So, we see that the intrinsic carrier concentration for a particular semiconductor is constant at a given temperature. **The value of the intrinsic carrier concentration is a strong function of the band gap and is higher for lower band gap materials.** This statement agrees well with our discussion in section 1.1. Since the band gap is actually the energy needed to break a covalent bond, a smaller band gap would imply a larger number of broken bonds at room temperature and consequently a larger number of carriers. The band gaps and the intrinsic carrier concentrations for some semiconductors at room temperature are listed in Table 1.3.

Table 1.3 Band gap and intrinsic carrier concentrations in some common semiconductors at 300 K

Semiconductor	Band gap (eV)	Intrinsic carrier concentration (cm^{-3})
Indium Arsenide	0.35	1.3×10^{15}
Germanium	0.66	2.3×10^{13}
Silicon	1.12	1.5×10^{10}
Indium Phosphide	1.34	1.2×10^{8}
Gallium Arsenide	1.42	1.8×10^{6}

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HELP DESK 1.2

How does the intrinsic carrier concentration vary with temperature?

In the expression for n_i given by Eq. (1.24), in addition to the presence of temperature T in a variable, N_C, N_V and E_g are all functions of temperature. Ignoring the small variation in energy gap with temperature, and considering the expressions for N_C and N_V together with Eq. (1.24), we can write

$$n_i \propto T^{3/2}\exp\left[-\left(\frac{E_g}{2kT}\right)\right]$$

Thus, we see that n_i increases very rapidly with temperature, the increase being almost exponential in nature. This again agrees with our discussion in section 1.1. With increase in temperature, more bonds are broken giving rise to more free carriers. Figure 1.13 shows the variation of n_i with temperature for Ge, Si, and GaAs.

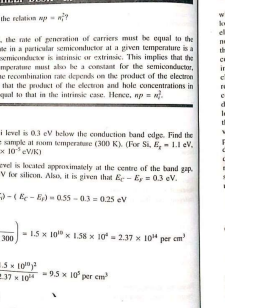

Figure 1.13 Intrinsic carrier concentrations in Ge, Si, and GaAs plotted as a function of temperature

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obtain the carrier concentrations or the position of the Fermi level in a semiconductor with a known dopant concentration. A simple approximation which is often used is $n = N_D$, where N_D is the donor concentration in an n-type semiconductor, or $p = N_A$, where N_A is the acceptor concentration in a p-type semiconductor. Although this approximation holds in most situations, in some cases it may lead to erroneous results. Another situation where we may have difficulty with this approximation is when both donor and acceptor impurities are present. In a semiconductor, the position of Fermi level (or the values of n and p) is determined from the **charge neutrality condition** (the total charge in the semiconductor is zero), which requires a knowledge of the degree of ionization of donor and acceptor impurities. This section elucidates the ionization of impurities, while the following section 1.3.8, shows how the values of n and p in an extrinsic semiconductor can be calculated using the charge neutrality condition once the degree of ionization is known.

Let us at first consider the ionization of donor impurities. We have already mentioned that when a group V impurity is introduced in a group IV semiconductor, the fifth valence electron is loosely bound and can easily be set free. This process of creation of a negatively charged free electron results in the simultaneous formation of a positively charged donor ion, so that overall neutrality of charge is maintained. If N_D is the concentration of donor atoms, then we may write that $N_D = N_D^+ + N_D^0$, where N_D^+ represents the concentration of ionized donors and N_D^0 is the concentration of neutral donors. As shown in the energy band diagram in Fig. 1.10, the introduction of the donors results in the creation of N_D states at an energy level E_D, which is very close to the energy level E_C at the conduction band edge. The ionization of a donor atom can be represented by an electron at the donor energy level E_D which absorbs energy to move to the conduction band, thereby creating a free electron in the conduction band and a vacancy at the donor energy level. It is clear that N_D^+ represents the number of occupied states at the energy level E_D. The probability of occupancy of the donor level E_D by an electron can be obtained from the Fermi–Dirac statistics. However, in doing so, we must also consider a 'degeneracy factor'. The value of the degeneracy factor is calculated by noting that an electron can be placed in a particular energy level either with spin up or spin down. Thus the leaving of an electron from the donor level causes the availability of two empty states (one for spin up and the other for spin down). Only one of these two states may be filled by a recaptured electron; nevertheless, this receptore can take place in two distinctly different ways. Thus the effective number of available states for electrons is twice the actual value ($= 2N_D^+$) resulting in a spin degeneracy factor of 2 and using the Fermi–Dirac function given by Eq. (1.11), we can write

$$f(E_D) = \frac{N_D^+}{N_D^+ + 2N_D^+} = \frac{1}{1 + \exp\left(\dfrac{E_D - E_F}{kT}\right)} \tag{1.31}$$

On simplification, Eq. (1.31) gives

$$\frac{N_D^+}{N_D^-} = 2\exp\left(\frac{E_F - E_D}{kT}\right) \tag{1.32}$$

We can now obtain a relation for the ionization coefficient of donors (η_D). The *ionization coefficient* is defined as the ratio of the concentration of ionized donors to the total concentration of donors and is given by

$$\eta_D = \frac{N_D^+}{N_D} = \frac{1}{1 + \dfrac{N_D^0}{N_D^+}} = \frac{1}{1 + 2\exp\left(\dfrac{E_F - E_D}{kT}\right)} \tag{1.33}$$

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We see that η_D depends on the position of the Fermi level with respect to the donor level. However, the position of the Fermi level can be determined if we know the free electron concentration, which again depends on η_D. This problem can be analytically solved by assuming that $\eta_D N_D = N_D^+ = n$, or the electron concentration is equal to the number of ionized donors. We have noted that $\eta_D N_D = N_D^+ = n$, or the electron concentration is equal to the number of ionized donors. We have neglected the thermally generated carriers. This is a good approximation since the probability of an electron moving from the valence band to the conduction band is much less than the transition from the donor level to the conduction band. Therefore from Eqs. (1.15) and (1.33), we have

$$\eta_D = \frac{1}{1 + 2\exp\left(\dfrac{E_F - E_D}{kT}\right)} = \frac{1}{1 + 2\exp\left(\dfrac{E_C - E_D}{kT}\right)\exp\left[-\left(\dfrac{E_C - E_F}{kT}\right)\right]}$$

$$= \frac{1}{1 + \dfrac{2n}{N_C}\exp\left(\dfrac{E_{DB}}{kT}\right)} = \frac{1}{1 + \dfrac{2\eta_D N_D}{N_C}\exp\left(\dfrac{E_{DB}}{kT}\right)} \tag{1.34}$$

where $E_{DB} = E_C - E_D$ is the separation between the conduction band edge and the donor level, or the energy required to ionize a donor atom. From Eq. (1.34), we get a quadratic equation in η_D as

$$\frac{2\eta_D^2 N_D}{N_C}\exp\left(\frac{E_{DB}}{kT}\right) + \eta_D - 1 = 0 \tag{1.35}$$

The solution of the above equation is given by

$$\eta_D = \frac{\sqrt{\dfrac{8N_D}{N_C}\exp\left(\dfrac{E_{DB}}{kT}\right) + 1} - 1}{\dfrac{4N_D}{N_C}\exp\left(\dfrac{E_{DB}}{kT}\right)} = \frac{2}{1 + \sqrt{\dfrac{8N_D}{N_C}\exp\left(\dfrac{E_{DB}}{kT}\right)} + 1} \tag{1.36}$$

From this expression of the ionization coefficient, it is evident that η_D is a strong function of temperature. In practice, for most dopants, η_D is very close to unity at room temperature signifying that at this temperature almost all the donors are ionized.

Following a similar line of reasoning, it can be shown that for an acceptor level E_A in a p-type semiconductor with a concentration of acceptor impurities given by N_A, the ionization coefficient of the acceptor impurities (η_A) is given by

$$\eta_A = \frac{N_A^-}{N_A} = \frac{1}{1 + 2\exp\left(\dfrac{E_A - E_F}{kT}\right)} \tag{1.37}$$

We now assume that $\eta_D N_A = N_A^- = p$, and use Eq. (1.19) to eliminate E_F in Eq. (1.37) and get an expression for η_A as

$$\eta_A = \frac{\sqrt{\dfrac{8N_A}{N_V}\exp\left(\dfrac{E_{AB}}{kT}\right) + 1} - 1}{\dfrac{4N_A}{N_V}\exp\left(\dfrac{E_{AB}}{kT}\right)} = \frac{2}{1 + \sqrt{\dfrac{8N_A}{N_V}\exp\left(\dfrac{E_{AB}}{kT}\right)} + 1} \tag{1.38}$$

where E_{AB} is the energy required to ionize an acceptor impurity and is given by $E_{AB} = E_F - E_V$.

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Example 1.1

The intrinsic carrier concentration of silicon ($E_g = 1.12$ eV) at 300 K is 1.5×10^{10} per cm^3. Calculate n_i for silicon at 400 K and 500 K. ($k = 8.62 \times 10^{-5}$ eV/K)

Solution: Since $n_i \propto T^{3/2}\exp\left[-\left(\dfrac{E_g}{2kT}\right)\right]$, we can write

$$\frac{(n_i)_{400\text{K}}}{(n_i)_{300\text{K}}} = \left(\frac{400}{300}\right)^{3/2}\exp\left[-\frac{1.12}{2 \times 8.62 \times 10^{-5}}\left(\frac{1}{400} - \frac{1}{300}\right)\right] = 1.54 \times 224.475 = 345.7$$

Therefore,

$$(n_i)_{400\text{ K}} = 1.5 \times 10^{10} \times 345.7 = 5.185 \times 10^{12} \text{ per cm}^3$$

Similarly,

$$(n_i)_{500\text{ K}} = 1.5 \times 10^{10} \times \left(\frac{500}{300}\right)^{3/2}\exp\left[-\frac{1.12}{2 \times 8.62 \times 10^{-5}}\left(\frac{1}{500} - \frac{1}{300}\right)\right]$$

$$= 1.5 \times 10^{10} \times 2.15 \times 5779.23 = 1.86 \times 10^{14} \text{ per cm}^3$$

1.3.6 Position of Fermi Level in Extrinsic Semiconductors

From Eqs. (1.15) and (1.19), we obtain

$$np = N_C N_V \exp\left[-\left(\frac{E_C - E_F}{kT}\right)\right] = N_C N_V \exp\left[-\left(\frac{E_g}{kT}\right)\right] \tag{1.25}$$

Now, from Eqs. (1.24) and (1.25), we have

$$np = n_i^2 \tag{1.26}$$

This is an important fundamental relationship which implies that **for any semiconductor at thermal equilibrium, the product of electron and hole concentrations is a constant and is independent of doping.** From Eqs. (1.15) and (1.20), it can also be written that

$$n = n_i \exp\left(\frac{E_F - E_i}{kT}\right) \tag{1.27}$$

and similarly from Eqs. (1.19) and (1.21), we have

$$p = n_i \exp\left(\frac{E_i - E_F}{kT}\right) \tag{1.28}$$

From the above expressions, the position of the Fermi level in an extrinsic semiconductor can be easily obtained. Rearranging Eqs. (1.27) and (1.28), we can also write

$$E_F = E_i + kT\ln\left(\frac{n}{n_i}\right) \tag{1.29}$$

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and

$$E_F = E_i - kT\ln\left(\frac{p}{n_i}\right) \tag{1.30}$$

As already discussed, for an n-type (p-type) semiconductor, the electron (hole) concentration is much larger than the intrinsic carrier concentration. Therefore, Eqs. (1.29) and (1.30) mathematically prove that the Fermi energy level for an n-type semiconductor is above the intrinsic Fermi level (closer to the conduction band edge) while for a p-type semiconductor, the Fermi level is below the intrinsic level (closer to the valence band edge), as already discussed in section 1.3.2.

HELP DESK 1.3

What is the physical significance of the relation $np = n_i^2$?

For any equilibrium condition, the rate of generation of carriers must be equal to the recombination rate. The generation rate in a particular semiconductor at a given temperature is a constant, irrespective of whether the semiconductor is intrinsic or extrinsic. This implies that the recombination rate at a particular temperature must also be a constant for semiconductors, whether intrinsic or extrinsic. Since the recombination rate depends on the product of the electron and hole concentration, we can infer that the product of the electron and hole concentrations in an extrinsic semiconductor must be equal to that in the intrinsic case. Hence, $np = n_i^

Example 1.5

Consider a silicon sample with electron concentration $n = 10^{11}$ per cm^3 at 300 K. Show that this material is strongly extrinsic at 300 K but becomes nearly intrinsic at 473 K.

Solution: At 300 K, n_i for silicon is 1.5×10^{10} per cm^3. Therefore, the hole concentration in this sample is $p = n_i^2/n = 2.25 \times 10^9$ per cm^3. Assuming complete ionization at this temperature and using Eq. (1.40), we get $N_D - N_A = n - p = 10^{11} - 2.25 \times 10^9 \approx 10^{11}$ per cm^3. This is the net dopant concentration and its value is independent of temperature. At 300 K, evidently $(N_D - N_A) \gg 2n_i$ and therefore, the material is deemed to be strongly extrinsic.

At 473 K, however, the value of n_i in silicon becomes 1.5×10^{14} per cm^3. Obviously at this temperature, $N_D - N_A$ is much less than n_i and the material can no longer be deemed strongly extrinsic. From Eqs. (1.42) and (1.43), the values of n and p at 473 K are calculated to be $n = 1.55 \times 10^{14}$ per cm^3 and $p = 1.45 \times 10^{14}$ per cm^3. Since the values of n and p are nearly equal to each other, the material displays nearly intrinsic behaviour at this temperature.

A plot of electron concentration versus temperature for an n-type semiconductor sample is shown in Fig. 1.14. As already discussed, at low temperatures the electron concentration is low since complete ionization has not taken place. As the temperature increases, the degree of ionization of the donors increases and the electron concentration increases consequently. The electron concentration now remains nearly constant over a range of temperature signifying that the ionization is complete. This is the range of temperature at which the material is strongly extrinsic. Finally at higher temperatures, n_i increases thereby increasing the electron as well as the hole concentrations as thermal generation of electron and hole pairs dominates. When the thermally generated electron and hole concentrations become comparable to the dopant concentration, the material exhibits nearly intrinsic behaviour.

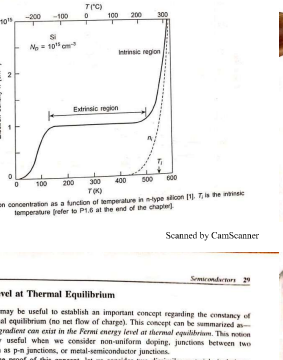

Figure 1.14 Electron concentration as a function of temperature in n-type silicon [1]. T_i is the intrinsic temperature [refer to P1.6 at the end of the chapter].

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As a corollary, it will be interesting to note the variation in the position of the Fermi level with respect to the conduction band edge (that is, $E_c - E_F$) with temperature. At very low temperatures, we know that the value of the ionization coefficient is small, that is, most of the donors are not ionized. In other words, almost all donor states are filled. From our discussion of the Fermi energy in section 1.3.2, we can therefore reason that at such low temperatures, Fermi level of this semiconductor must lie above the donor level, since levels below E_F are more likely to be occupied by electrons. Indeed it can be shown (given as Problem) that at very low temperature, for an n-type semiconductor, the position of the Fermi level is given by

$$E_F = \frac{E_c + E_D}{2} - \frac{kT}{2} \ln\left(\frac{2N_c}{N_D}\right)$$

As the temperature increases, the value of the ionization coefficient approaches unity. This implies that now the donor levels are empty and consequently the Fermi level lies below the donor level. Then, at very high temperatures when the electron and hole concentrations become nearly equal, the Fermi level comes close to the intrinsic Fermi level. Thus, with increase in temperature, the position of the Fermi level shifts from a level close to the conduction band edge to approximately the middle of the band gap. A similar explanation can be given for a p-type material and it can be seen that for this case, the position of the Fermi level varies from a level close to the valence band edge to the middle of the bandgap E_i with increase in temperature. Figure 1.15 illustrates the nature of variation of Fermi level with temperature for various doping levels.

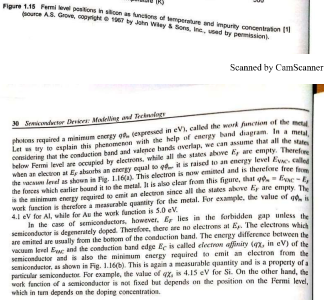

Figure 1.15 Fermi level positions in silicon as functions of temperature and impurity concentration [1] (source A.S. Grove, copyright © 1967 by John Wiley & Sons, Inc., used by permission).

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1.3.9 Fermi Level at Thermal Equilibrium

At this juncture, it may be useful to establish an important concept regarding the constancy of Fermi level at thermal equilibrium (no net flow of charge). This concept can be summarized as—*No discontinuity or gradient can exist in the Fermi energy level at thermal equilibrium*. This notion will be particularly useful when we consider non-uniform doping, junctions between two semiconductors such as p-n junctions, or metal-semiconductor junctions.

To establish the proof of this concept, let us consider two dissimilar material, in intimate contact, so that electrons can flow between the two. Let $f_1(E)$ and $f_2(E)$ be the Fermi–Dirac distribution function and distribution function of available states respectively for material 1, while $f_2(E)$ and $g_2(E)$ are the Fermi–Dirac distribution function and distribution function of available states respectively for material 2. At thermal equilibrium, there is no current and therefore no net flow of charge. Also, there is no net transfer of energy. Therefore, for each energy value E, transfer of electrons from material 1 to material 2 must be exactly balanced by a corresponding transfer of electrons from material 2 to material 1. The rate of transfer of electrons from material 1 to material 2 at energy E will be proportional to the product of the number of filled states in material 1 and the number of empty states in material 2 at energy E. Thus, the electron flux from material 1 to material 2 is

$$F_{12} \propto g_1(E)\, f_1(E)\, g_2(E)\{1 - f_2(E)\} \tag{1.44}$$

Similarly, the electron flux from material 2 to material 1 is

$$F_{21} \propto g_2(E) f_2(E)\, g_1(E)\{1 - f_1(E)\} \tag{1.45}$$

At equilibrium, $F_{12} = F_{21}$. This means

$$g_1(E) f_1(E)\, g_2(E)\{1 - f_2(E)\} = g_2(E) f_2(E)\, g_1(E)\{1 - f_1(E)\} \tag{1.46}$$

which gives $f_1(E) = f_2(E)$. That is,

$$\frac{1}{1 + \exp\left(\dfrac{E - E_{F1}}{kT}\right)} = \frac{1}{1 + \exp\left(\dfrac{E - E_{F2}}{kT}\right)} \tag{1.47}$$

where E_{F1} and E_{F2} are Fermi energy levels in material 1 and material 2 respectively.

From Eq. (1.47), we can conclude that $E_{F1} = E_{F2}$. In other words, there is no discontinuity in the Fermi level at thermal equilibrium or more generally, *the equilibrium Fermi level is constant throughout the materials in intimate contact*. Another way of expressing this is to say that there is no gradient in the Fermi level at equilibrium. This can be expressed as an important relation as

$$\frac{dE_F}{dx} = 0 \tag{1.48}$$

1.3.10 Vacuum Level, Work Function, and Electron Affinity

Let us consider Einstein's experiments on the photoelectric effect. He observed that electrons can be emitted from a metal by shining light on it. However, for electron emission, the incident

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photons required a minimum energy $q\phi_0$ (expressed in eV), called the *work function* of the metal. Let us try to explain this phenomenon with the help of energy band diagram. In a metal, considering that the conduction band and valence bands overlap, we can assume that all the states below Fermi level are occupied by electrons, while all the states above E_F are empty. Therefore, when an electron at E_F absorbs an energy equal to $q\phi_0$, it is raised to an energy level E_{VAC}, called the *vacuum level* as shown in Fig. 1.16(a). This electron is now emitted and is therefore free from the forces which earlier bound it to the metal. It is also clear from this figure, that $q\phi_0 = E_{VAC} - E_F$ is the minimum energy required to emit an electron since all the states above E_F are empty. Thus, work function is therefore a measurable quantity for the metal. For example, the value of $q\phi_0$ is 4.1 eV for Al, while for Au the work function is 5.0 eV.

In the case of semiconductors, however, E_F lies in the forbidden gap unless the semiconductor is degenerately doped. Therefore, there are no electrons at E_F. The electrons which are emitted are usually from the bottom of the conduction band. The energy difference between the vacuum level E_{VAC} and the conduction band edge E_C is called *electron affinity* ($q\chi_e$, in eV) of the semiconductor and is also the minimum energy required to emit an electron from the semiconductor, as shown in Fig. 1.16(b). This is again a measurable quantity and is a property of a particular semiconductor. For example, the value of $q\chi_e$ is 4.15 eV for Si. On the other hand, the work function of a semiconductor is not fixed but depends on the position on the Fermi level, which in turn depends on the doping concentration.

Figure 1.16 Band diagram of (a) metal and (b) semiconductor showing work function and electron affinity.

The concept of vacuum level is very useful for drawing the energy band diagram of multi-individual materials. This will be clear when heterojunctions and MOSFETs are discussed in later chapters.

1.4 EXCESS CARRIERS—NON-EQUILIBRIUM SITUATION

In the case of the semiconductor in thermal equilibrium, the product of electron and hole concentrations is a constant for a particular material at a given temperature. However, if excess

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carriers are introduced in a semiconductor, so that $np > n_i^2$, a non-equilibrium situation arises. Excess carriers can be introduced in a semiconductor by shining light (optical excitation) or forward-biasing a p-n junction (discussed in detail in Chapter 4). This process of introducing excess carriers is called *injection*. In case of optical excitation, if the energy of photons is greater than the energy gap, that is, $h\nu > E_g$, they are absorbed in the semiconductor, resulting in excitation of electrons from the valence band to the conduction band. Thus, the optical excitation results in additional EHPs being generated, and a new steady state is reached where the recombination rate is equal to the total generation rate (thermal + optical). The electron and hole concentrations in this steady state are more than their equilibrium values. These additional carriers are called *excess carriers*. The magnitude of the excess carriers relative to the equilibrium majority carrier concentration determines the level of injection. For example, let us consider an n-type silicon sample with $N_D = 10^{15}$ per cm^3 at 300 K. The majority carrier concentration at thermal equilibrium is $n_{n0} \approx 10^{15}$ per cm^3 assuming complete ionization and the minority carrier concentration is $p_{n0} = 2.25 \times 10^5$ per cm^3 (The subscript n indicates that we are referring to an n-type semiconductor and 0 refers to thermal equilibrium condition). Now let us suppose that by optical excitation 10^{12} excess minority carriers per cm^3 are injected in this sample. The excess electron concentration n'_n must be equal to the excess hole concentration p'_n since excess holes and electrons are generated in pairs. Thus, while the minority carrier concentration in this sample has increased by nearly seven orders of magnitude (from 2.25×10^5 per cm^3 to 10^{12} per cm^3), the increase in the electron concentration is negligible. This condition where the excess carrier concentration is small compared to the majority carrier concentration, that is, $n'_n = p'_n \ll n_{n0}$ is referred to as *low level injection*. The other case where $n'_n = p'_n$ is approximately equal to or greater than n_{n0} is called *high level injection*, and is more difficult to analyze. We shall restrict our discussion mostly to low level injection.

1.4.1 Quasi-Fermi Level or IMREF

When excess carriers are generated, since $np \neq n_i^2$, Eqs. (1.27) and (1.28) are no longer valid and the concept of a Fermi energy level becomes meaningless. However, it is desirable to retain the same form of expressions for the electron and hole concentrations. Therefore, we can define separate *quasi-Fermi levels* (E_{Fn} and E_{Fp}) for electrons and holes for the non-equilibrium cases. These energy levels are also referred to as IMREFs (Fermi spelt-backwards). Writing the expressions for the electron and hole concentrations for the non-equilibrium case in steady state, in the same manner as Eqs. (1.27) and (1.28), we have

$$n = n_i \exp\left(\frac{E_{Fn} - E_i}{kT}\right) \tag{1.49}$$

$$p = n_i \exp\left(\frac{E_i - E_{Fp}}{kT}\right) \tag{1.50}$$

The product of Eqs. (1.49) and (1.50), gives

$$np = n_i^2 \exp\left(\frac{E_{Fn} - E_{Fp}}{kT}\right) \tag{1.51}$$

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Therefore,

(a) If $E_{Fn} > E_{Fp}$, then $np > n_i^2$
$\Rightarrow$ Excess carrier injection.

(b) If $E_{Fn} = E_{Fp}$, then $np = n_i^2$
$\Rightarrow$ Equilibrium with single Fermi level.

(c) If $E_{Fn} < E_{Fp}$, then $np < n_i^2$
$\Rightarrow$ carrier extraction occurs.

Quasi-Fermi levels are discussed in detail in Chapter 4.

1.4.2 Generation and Recombination of Carriers and the Concept of Lifetime

Whenever the thermal equilibrium condition is disturbed, processes exist to restore the equilibrium condition. In the case of excess carrier injection, equilibrium is restored by the mechanism of recombination of the injected minority carriers and majority carriers. The recombination can be *direct* (also called band-to-band) which involves a transition of carriers between E_C and E_V or *indirect* involving an intermediate trap level (called the recombination centre) in the band gap. Generally, direct recombination is dominant in a direct band gap semiconductor (for example GaAs, InP) while indirect recombination is prevalent where the material has an indirect band gap (for example Si, Ge). The energy released from the recombination process can be emitted as a photon (radiative recombination) or as a phonon (that is as heat to the lattice, a non-radiative recombination). Let us consider direct recombination first. In a direct band gap semiconductor, the continuous thermal vibration of lattice atoms causes some bonds between neighbouring atoms to break. EHPs are thus generated, that is, thermal energy enables an electron to make an upward transition from E_V to E_C leaving a hole in E_V. This is called generation of carriers and is represented by a generation rate G_{th} (that is, number of EHPs generated per unit volume per second). On the other hand, when an electron makes a transition from E_C to E_V, an EHP is annihilated and this is called recombination of carriers. This recombination is represented by recombination rate R_{th}. At thermal equilibrium, $G_{th} = R_{th}$, so as to keep the product np constant. As mentioned earlier, R_{th} is proportional to the electron concentration n and the hole concentration p. Hence, we can write

$$R_{th} = \beta np \tag{1.52}$$

where β is a proportionality constant.

Since R_{th} is equal to G_{th} at thermal equilibrium, for an n-type semiconductor we can write

$$G_{th} = R_{th} = \beta n_{n0} p_{n0} \tag{1.53}$$

where n_{n0} and p_{n0} denote the electron and hole concentrations respectively in an n-type semiconductor under thermal equilibrium.

When this semiconductor is optically excited to generate EHP at a rate of G_L per cm^3 per second, the carrier concentrations rise above their thermal equilibrium value to n_n and p_n. Then, the new recombination rate is given by

$$R = \beta n_n p_n = \beta(n_{n0} + n'_n)(p_{n0} + p'_n) \tag{1.54}$$

where n'_n and p'_n are the excess electron and hole concentrations respectively.

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The new generation rate is given by

$$G = G_{th} + G_L \tag{1.55}$$

The net rate of change in the minority carrier concentration (holes for n-type material) is the difference of the generation and recombination rates, and is given by

$$\frac{dp_n}{dt} = G - R = G_{th} + G_L - R \tag{1.56}$$

Now at steady state, $dp_n/dt = 0$. Therefore from Eq. (1.56), we can write

$$G_L = R - G_{th} = U \tag{1.57}$$

where U = *excess recombination rate*.

Substituting the values of G_{th} and R from Eqs. (1.53) and (1.54) respectively into Eq. (1.57) and noting that $n'_n = p'_n$ as holes and electrons are generated in pairs, we have

$$U = \beta(n_{n0} + n'_n)(p_{n0} + p'_n) - \beta(n_{n0}p_{n0}) = \beta(n_{n0} + p'_n + p_{n0})p'_n \tag{1.58}$$

Now, for the low injection case in an n-type semiconductor, $p'_n \ll n_{n0}$ and $p_{n0} \ll n_{n0}$. Therefore, the net recombination rate can be approximated as

$$U = \beta n_{n0} p'_n = \beta n_{n0}(p_n - p_{n0}) = \frac{p_n - p_{n0}}{\tau_p} = \frac{\text{excess carrier concentration}}{\text{lifetime}} \tag{1.59}$$

where τ_p = lifetime of the minority carriers. It is a constant for the particular semiconductor sample.

Since U represents the excess recombination rate over and above the thermal equilibrium value, we see from Eq. (1.59) that $U = 0$ when $p_n = p_{n0}$, that is, the minority concentration is equal to the thermal equilibrium value. In Eq. (1.59), we have considered excess carriers. However, if carriers are extracted, so that $p_n < p_{n0}$, U is negative. This implies that the thermal generation rate is more than the recombination rate, and $-U$ represents the excess generation rate.

Physically, the concept of lifetime can be explained in the following manner. Let us consider an n-type semiconductor illuminated by light, which results in additional EHPs being generated throughout the sample at a constant rate of G_L per unit volume per unit time. In the steady state, from Eqs. (1.57) and (1.59), $G_L = U = (p_n - p_{n0})/\tau_p$. Therefore,

$$p_n = p_{n0} + \tau_p G_L \tag{1.60}$$

We see that the excess carrier concentration is proportional to the lifetime. If we consider the majority carriers, we can write $n_n = n_{n0} + n'_n$. In the low injection case, since $n'_n = p'_n < n_{n0}$, we have $n_n \approx n_{n0}$. In other words, the minority carrier concentration changes significantly while majority carrier concentration is almost unaffected.

At an arbitrary time instant ($t = 0$), let the light be switched off. Now, since $G_L = 0$, from Eqs. (1.56), (1.57), and (1.59) we can write that for $t > 0$,

$$\frac{dp_n}{dt} = G_{th} - R = -U = -\frac{p_n - p_{n0}}{\tau_p} \tag{1.61}$$

The solution of the above equation is of the form

$$p_n - p_{n0} = C \exp\left[-\left(\frac{t}{\tau_p}\right)\right] \tag{1.62}$$

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where C = integration constant.

In order to evaluate C, we note that at $t = 0$, $(p_n - p_{n0}) = \tau_p G_L$. Therefore, we can write

$$p_n = p_{n0} + \tau_p G_L \exp\left[-\left(\frac{t}{\tau_p}\right)\right] \tag{1.63}$$

So we see that once the light is switched off, with the passage of time, the optically-generated minority carriers recombine with the majority carriers to decrease the carrier concentrations back to their thermal equilibrium values. This decay of excess minority carriers occurs exponentially with a time constant τ_p, referred to as the **lifetime of the minority carriers**. If the excess carriers have a longer lifetime, it takes more time for the carrier concentration to revert to the thermal equilibrium value. On the other hand, if the excess carriers have a shorter lifetime, they 'die' quickly and thermal equilibrium values are attained faster. It is to be noted that although in a recombination process, the majority and minority excess carriers decay at the same rate, that is, $\tau_n = \tau_p$, the pace of recombination is governed by the minority carriers as they are in short supply. Thus, the process of recombination is ultimately characterized by the lifetime of the minority carriers. Figure 1.17 shows the variation of the majority and minority carrier concentrations for this case as a function of time.

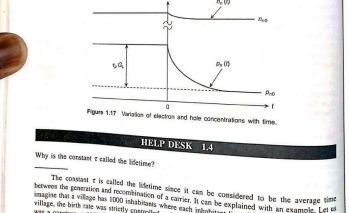

Figure 1.17 Variation of electron and hole concentrations with time.

HELP DESK 1.4

Why is the constant τ called the lifetime?

The constant τ is called the lifetime since it can be considered to be the average time between the generation and recombination of a carrier. It can be explained with an example. Let us imagine that a village has 1000 inhabitants, where each inhabitant lives for exactly 50 years. In this village, the birth rate was strictly controlled so as to be equal to the death rate, so that the population was a constant at 1000. In a particular year, say 1900, the villagers decided to allow 5 additional babies to be born every year. So for the next 50 years, 250 additional babies were born, raising the

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total population of the village to $1000 + 50 \times 5 = 1250$. But after 50 years elapsed, the death rate suddenly went up by 5 as the 5 additional babies born 1900 onwards started dying from 1950. So again the birth rate became equal to the death rate and a steady state was reached at a population of 1250. In this example, the analogy to the optical generation case in semiconductor is clearly visible. The eventual steady state population in the village can be obtained using Eq. (1.60), by taking $p_{n0} = 1000$, $G_L = 5$, and $\tau = 50$. This also makes it clear why τ is referred to as the lifetime.

The concept of lifetime can also be explained with the help of Eq. (1.59), where the rate of recombination is given as the excess carrier concentration divided by lifetime. This clearly implies that the lifetime is the average time before a carrier recombines.

Example 1.6

Consider an n-type silicon sample with a doping concentration $N_D = 10^{16}$ per cm^3. If it is illuminated such that EHPs at a rate of 10^{18} per cm^3 per second are generated, find out the majority and minority carrier concentrations at steady state. Assuming $\tau_p = 10^{-6}$ s, plot the decay of the excess carriers with time once the light source is switched off.

Solution: Assuming complete ionization, the electron concentration at thermal equilibrium is given as

$$n_{n0} = N_D = 10^{16} \text{ per cm}^3$$

Therefore, the hole concentration,

$$p_{n0} = n_i^2/n_{n0} = 2.25 \times 10^4 \text{ per cm}^3$$

Now, when the sample is illuminated such that $G_L = 10^{18}$ per cm^3 per second in steady state, from Eq. (1.60), we have

$$p'_n = n'_n = \tau_p G_L = 10^{12} \text{ per cm}^3$$

Therefore,

$$p_n = p_{n0} + p'_n \approx p'_n = 10^{12} \text{ per cm}^3$$

and

$$n_n = n_{n0} + n'_n \approx n_{n0} = 10^{16} \text{ per cm}^3$$

Note that there is a large change in the minority carrier concentration while the majority carrier concentration remains virtually unaffected.

Now when the light is switched off, excess carriers decay with a time constant τ_p as shown in the Fig. 1.17. The minority carrier concentration decays exponentially from 10^{12} per cm^3 to the thermal equilibrium value of 2.25×10^4 per cm^3. However, the majority carrier concentration remains almost constant throughout at about 10^{16} per cm^3.

Example 1.7

For the n-type silicon sample in Example 1.6, give a pictorial representation of the Fermi level positions before and after the light source is switched on at room temperature. (Given: $kT = 0.026$ eV and $n_i = 1.5 \times 10^{10}$ per cm^3 at room temperature.)

Solution: Before the light is switched on, the sample is in thermal equilibrium and the Fermi level position is given by Eq. (1.29). Assuming complete ionization, we have $n = N_D = 10^{16}$ per cm^3. Therefore, from Eq. (1.29), at room temperature

$$E_F = E_i + 0.026 \ln\left(\frac{10^{16}}{1.5 \times 10^{10}}\right) = (E_i + 0.35) \text{ eV}$$

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When light is switched on, from Example 1.6, we have $n_n = 10^{16}$ per cm^3 and $p_n = 10^{12}$ per cm^3. Since the sample is no longer in thermal equilibrium (that is, $np \neq n_i^2$), the Fermi level splits into the electron and hole quasi-fermi levels, whose positions are given by Eqs. (1.49) and (1.50). From these equations, we have

$$E_{Fn} = E_i + kT \ln\left(\frac{n_n}{n_i}\right) = E_i + 0.026 \ln\left(\frac{10^{16}}{1.5 \times 10^{10}}\right) = (E_i + 0.35) \text{ eV}$$

$$E_{Fp} = E_i - kT \ln\left(\frac{p_n}{n_i}\right) = E_i - 0.026 \ln\left(\frac{10^{12}}{1.5 \times 10^{10}}\right) = (E_i - 0.11) \text{ eV}$$

It is observed that the position of E_{Fn} coincides with E_F, since the majority carrier concentration is almost unchanged. Also, the quantity $(E_{Fn} - E_{Fp})$ is positive showing the presence of excess carriers. The positions of the Fermi level and the quasi-Fermi levels are shown as follows.

1.4.3 Indirect Recombination

In an indirect band gap semiconductor, the process of direct recombination with an associated photon emission is unlikely to occur. This is because not only the energy but also the momentum needs to be conserved when the electron makes a transition from E_C to E_V. Photons, being light particles (literally!), cannot assist in momentum conservation. In such cases, the dominant recombination process is indirect transition that occurs through localized energy states between E_C and E_V. As the lifetimes in indirect semiconductors are generally high, these states, which act as intermediate steps, are sometimes deliberately created by introducing special impurities (Au and Pt in silicon; N in GaAsP). As the transition probability depends on the energy difference between the intermediate steps and E_C (or E_V), they can substantially increase the recombination rate and consequently reduce the lifetime.

1.4.4 Surface Recombination

We have essentially discussed in sections 1.4.2 and 1.4.3, about the recombination process in the bulk of a semiconductor. A similar process also occurs at the semiconductor surface. The discontinuity of the lattice structure at the semiconductor surface introduces a large number of energy states in the forbidden gap, called *surface states*. The presence of these states greatly enhances the recombination rate at the surface. In addition, there may be adsorbed ions, molecules,

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or damages in the surface layer, which increase the recombination rate. The recombination rate at the surface per unit area (U_s) can be expressed in a similar form as Eq. (1.59) in the following manner:

$$U_s = S(p_s - p_{s0}) \tag{1.64}$$

Since U_s and p_s have the dimensions of cm^{-2}s^{-1} and cm^{-3}, the dimensions of the constant S are given by cm/s. Since S has the same dimension as velocity, it is called the *surface recombination velocity* and it plays a role similar to lifetime in bulk recombination. A higher value of S indicates a higher recombination rate.

1.5 MOBILITY OF CARRIERS

1.5.1 Effect of Electric Field on Carrier Movement

In a semiconductor, the charge carriers (that is, electrons and holes) are in constant motion even at thermal equilibrium. The thermal motion of an electron at room temperature may be visualized as a random scattering from lattice atoms, impurities, other electrons and defects. This random movement of an electron leads to a net zero displacement over a sufficiently long period of time. Schematically, this can be shown as in Fig. 1.18(a), which depicts the trajectory of an electron consisting of a series of straight lines between collisions. The average time between collisions is called the *mean free time* and the average distance travelled between collisions is the *mean free path*. The root mean square (rms) value of the random *thermal velocity* v_{th} is of the order of 10^7 cm/s for most semiconductors.

When an electric field $\mathscr{E}$ is applied to the semiconductor, the situation changes. Each electron now experiences a force $-q\mathscr{E}$ due to the field and is consequently accelerated between each collision in a direction opposite to the field, showing a net displacement of the electron. This is schematically represented in Fig. 1.18(b). Thus, we see that an additional velocity component is superposed on the random thermal velocity. This additional component is called the *drift velocity* v_d which is defined as the net displacement per unit time.

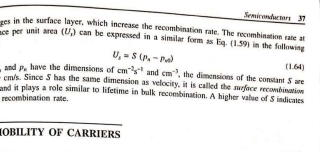

Figure 1.18 Possible trajectory of an electron (a) in the absence of electric field and (b) in the presence of electric field.

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Thus, in presence of an electric field, an electron acquires a drift velocity and its resulting velocity is therefore the vector sum of v_{th} and v_d. Since the mean free path does not alter due to the application of field, this increase in electron velocity reduces the mean free time. Consequently, the electron suffers more frequent collisions and therefore loses more energy to the lattice. Hence, relative force develops which does not allow a continuous increase of the carrier velocity. Rather, at equilibrium, an electron acquires an average drift velocity. Assuming that in steady state, a momentum gained between collisions is lost to the lattice, we can write

$$m^*_n v_d = -q\mathscr{E}\tau_c \tag{1.65}$$

where
m^*_n = effective mass of the electron and
τ_c = mean free time between two successive collisions.

Therefore,

$$v_d = \frac{q\tau_c}{m^*_n}\mathscr{E} = -\mu_n \mathscr{E} \tag{1.66}$$

Thus, it is seen that the average drift velocity of the electrons is proportional to the applied electric field and this constant of proportionality μ_n is called the mobility of electrons. The negative sign in the above equation indicates that the direction of v_d is opposite to that of the applied field. In a similar manner, for holes we can write

$$v_d = \frac{q\tau_c}{m^*_h}\mathscr{E} = \mu_p \mathscr{E} \tag{1.67}$$

where
m^*_h = effective mass of holes.

Since the holes carry positive charge, their movement is in the same direction as the electric field and hence there is no negative sign. Usually $m^*_n < m^*_h$, and therefore the mobility of electrons is larger than that of the holes for any semiconductor. Typical values of electron and hole mobilities at 300 K for some common semiconductors are listed in Table 1.4.

Table 1.4 Electron and hole mobilities in some common semiconductors

Semiconductor	Electron mobility (cm^2/Vs)	Hole mobility (cm^2/Vs)
Silicon	1350	480
Germanium	3900	1900
Gallium Arsenide	8500	400
Indium Phosphide	4600	150
Indium Arseride	28000	400

HELP DESK 1.5

While calculating effective density of states of silicon in section 1.3, m^*_h has been considered greater than m^*_n so that N_C is greater than N_V. On the other hand, while calculating mobilities of electrons and holes, m^*_n has been taken to be less than m^*_h, so that μ_n is greater than μ_p. Why are the effective masses different in the two cases?

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An electron in a semiconductor experiences various forces within the crystal lattice. The effects of these forces are taken into account by assigning an effective mass m^*_n to the electron in place of its true mass (m_0). In this way, the effects of the internal forces (while evaluating the effect of the external force on the electron) can be ignored. Since the internal forces acting on the electron are different in the various crystal directions, the effective mass is also dependent on direction. As a result, to obtain an overall effective mass, a suitable average of the direction dependent effective masses must be taken.

From Eq. (1.10), we see that $g(E) \propto (m^*_n)^{3/2}$. Also N_C is proportional to $(m^*_n)^{3/2}$. Therefore, the effective density of states is calculated by taking the effective mass as

$$(m^*_n)^{3/2} = (m_{n_1}{}^{3/2} m_{n_2}{}^{3/2} m_{n_3}{}^{3/2})^{1/3}$$

Here m_{n_1}, m_{n_2} and m_{n_3} are the effective masses in different directions. On the other hand, from Eq. (1.60), we see that $\mu_n \propto (1/m^*_n)$. Therefore, to calculate mobility, effective mass is taken as

$$\frac{1}{m^*_n} = \frac{1}{3}\left(\frac{1}{m_{n_1}} + \frac{1}{m_{n_2}} + \frac{1}{m_{n_3}}\right)$$

It is very clear that the effective mass for calculating $g(E)$ can be widely different from that for calculating μ_n. By a similar argument, it can be shown that the effective mass for holes m^*_h used for calculating $g(E)$ is different from that for calculating μ_p. It so happens that in silicon, $m^*_n > m^*_h$ for calculating N_C and N_V, while $m^*_n < m^*_h$ while calculating μ_n and μ_p.

1.5.2 Effect of Temperature and Doping on Carrier Mobility

So far we have learnt that the drift velocity of the carriers stabilizes to an average value (instead of increasing continuously) proportional to the applied electric field, as the carriers lose their energy gained from the field by collision (scattering). The two basic scattering mechanisms that affect the carrier mobility are *lattice scattering* and *impurity scattering*. In lattice scattering, the carriers moving through the crystal are scattered by the thermal vibration of the lattice atoms. With an increase in temperature, the thermal vibration becomes greater, and hence the frequency of such scattering events goes up. Thus, lattice scattering effect decreases with temperature and the carrier mobility reduces as the sample is heated. Theoretically, the mobility due to lattice scattering μ_l expressed as a function of temperature is given by this simple relationship

$$\mu_l \propto T^{-3/2} \tag{1.68}$$

On the other hand, impurity scattering is the dominant factor at low temperatures. When a charge carrier travels past an ionized impurity, its path gets deflected due to Coulomb force interaction. The probability of impurity scattering depends on the total concentration of ionized impurity and therefore, the carrier mobility is reduced for a highly doped semiconductor. However, unlike lattice scattering, at higher temperatures, the effect of impurity scattering becomes less and therefore, spend less time in the vicinity of an ionized impurity. Thus, they are less effectively scattered by the ionized dopants. The dependence of carrier mobility due to impurity scattering μ_i on the temperature as well as doping concentration N is expressed as

$$\mu_i \propto \frac{T^{3/2}}{N} \tag{1.69}$$

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In general, the carrier mobility in a semiconductor can be expressed as

$$\frac{1}{\mu} = \frac{1}{\mu_l} + \frac{1}{\mu_i} \tag{1.70}$$

The electron mobility in silicon for various doping concentrations of the sample is plotted as a function of temperature in Fig. 1.19. It can be seen that for lightly doped samples, mobility decreases with an increase in temperature, clearly depicting the dominance of lattice scattering. However for heavily doped samples, mobility is low at low temperatures, where the impurity scattering dominates. It increases with temperature, reaches a peak, till finally at higher temperatures lattice scattering takes over and mobility decreases again. Also, at any temperature, mobility is lower for samples with higher doping concentration. The variation of electron and hole mobility in silicon as a function of the doping concentration is shown in Fig. 1.20.

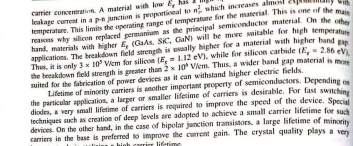

Figure 1.19 Variation of electron mobility in silicon as functions of doping concentration and temperature [2].

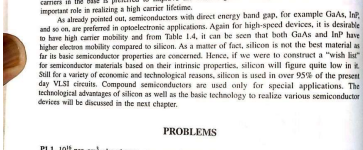

Figure 1.20 Electron and hole mobilities in silicon at room temperature plotted as functions of impurity concentration [2].

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1.5.3 Effect of High Electric Field on Mobility

In section 1.5.1, we have shown that the average drift velocity is proportional to the applied electric field; the proportionality constant is termed mobility, and it is independent of the applied field. However, this is true only when the applied field is small, so that $v_d \ll v_{th}$. At higher electric fields, when $v_d \approx v_{th}$, the drift velocity shows a sublinear dependence of the electric field. This represents the situation when the additional energy imparted by the field is transferred to the lattice rather than increasing the carrier velocity. So, the velocity approaches a saturation velocity $v_

P1.4 A silicon wafer is doped with 10^{16} atoms per cm^3 of Indium, which introduces an acceptor level E_A which is 0.16 eV above E_V. Determine the temperature at which 60% of Indium is ionized. If the wafer was doped with Boron instead of Indium, what would have been the percentage of ionization at this temperature? For Boron, $E_A - E_V = 0.045$ eV.

P1.5 Assuming $n = N_D^+$, show that at very low temperatures the Fermi level in an n-type semiconductor is given by

$$E_F = \frac{E_C + E_D}{2} - \frac{kT}{2} \ln\left(\frac{2N_C}{N_D}\right)$$

Hence, show that in this temperature range

$$n = \sqrt{\frac{N_C N_D}{2}} \exp\left(-\frac{E_{ion}}{2kT}\right)$$

where $E_{ion} = E_C - E_D$.
Also find E_F at 0 K.

P1.6 (a) At moderately high temperatures, assuming $N_D^+ \approx N_D$, show that the electron concentration in an n-type semiconductor is given by

$$n = \frac{N_D}{2}\left[1 + \sqrt{1 + \left(\frac{2n_i}{N_D}\right)^2}\right]$$

(b) The intrinsic temperature T_i is defined as the temperature at which the intrinsic concentration n_i equals the doping concentration N. Show that the ratio T_i/T_0 where T_0 is 300 K, is given by

$$\frac{T_i}{T_0} = \frac{E_g}{2kT_0 \ln\left(\sqrt{N_C N_V}/N\right)}$$

Determine T_i for Si and Ge, assuming N_C and N_V to be independent of temperature because of their logarithmic interdependence.

P1.7 Draw the energy band diagram of silicon doped with 10^{15} arsenic atoms per cm^3 at 77 K, 300 K and 600 K. Show the Fermi level diagrammatically and use the intrinsic Fermi level as reference. Assume complete ionization.

P1.8 A silicon wafer is doped with 2×10^{18} boron and 10^{16} phosphorus atoms per cm^3. Calculate n, p, and E_F at room temperature assuming complete ionization. Repeat the same for 8×10^{15} boron atoms per cm^3 and 10^{16} phosphorus atoms per cm^3.

P1.9 An n-type silicon wafer, doped with $N_D = 10^{16}$ per cm^3 is illuminated such that 10^{18} per cm^3 per second excess EHPs are generated. If the lifetime of the excess minority carriers to drop to 10% of its value at time required for the excess minority carriers to drop to 10% of its value at $t = 0$, assuming $\tau_p = 10^{-6}$ s.

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2.2 DOPING AND IMPURITIES

In order to fabricate semiconductor devices, a controlled amount of impurities has to be introduced (doped) selectively into single crystal wafers. There are three basic methods used for controlled doping of a semiconductor, namely epitaxy, diffusion, and ion-implantation. These methods are dealt with in the subsequent sections.

2.2.1 Epitaxy

The term epitaxy literally means "arranged upon". In this process, a thin layer of single crystal semiconductor (typically a few nanometers to a few microns) is grown on an already existing crystalline substrate such that the film has the same lattice structure as the substrate. Epitaxy can be further classified into *Vapour Phase Epitaxy* (VPE), *Liquid Phase Epitaxy* (LPE), and *Molecular Beam Epitaxy* (MBE).

Epitaxial growth of silicon is almost exclusively carried out by VPE. In this method, silicon is deposited by chemical vapour deposition (CVD) from source materials such as $SiCl_4$, $SiHCl_3$, and SiH_2Cl_2. The silicon wafer (on which epitaxial growth occurs) is placed on a graphite susceptor kept in a quartz chamber. Hydrogen gas is passed through liquid $SiCl_4$ and the mixture of $SiCl_4$ and H_2 is passed through this quartz tube. The system is rf-heated to a temperature above 1100°C. The schematic diagram of a VPE reactor is shown in Fig. 2.2. The basic reaction that occurs on the silicon surface in the process is

$$SiCl_4 + 2H_2 \Leftrightarrow Si + 4HCl$$

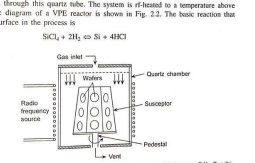

Figure 2.2 Schematic diagram of a VPE reactor with barrel-shaped susceptor [1] (source: S.K. Gandhi, copyright © 1983 by John Wiley & sons, Inc., used by permission).

The reaction is surface-catalyzed and silicon is deposited on the wafer surface. However, the deposition temperature is very high. Also, as the reaction is reversible and can proceed in both directions, etching, instead of deposition, may sometimes occur. Alternatively, SiH_4 may be used as a source material. Pyrolytic decomposition of SiH_4 at 1000–1100°C results in deposition of epitaxial silicon by the following reaction:

$$SiH_4 \rightarrow Si + 2H_2\uparrow$$

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In order to grow epitaxial layer of silicon with desired doping, gases containing dopants such as PH_3, AsH_3, or B_2H_6 are introduced into the system.

VPE of compound semiconductors, for example, GaAs can also be performed using Ga and $AsCl_3$ (or Ga and AsH_3) as source materials. However, ternary compounds such as AlGaAs cannot be grown by this technique since the growth temperatures of GaAs and AlAs are widely different. A modified VPE technique, in which metallorganic compounds and hydrides are used as source materials, is extensively used for such cases. This is called *metallorganic VPE* (MOVPE). MOVPE allows controlled growth of very thin layers.

Another method extensively used for the growth of multiple ternary and quaternary layers is *liquid phase epitaxy* (LPE), also referred to as *solute growth*. The basic principle is that the solubility of material A in material B (for example, As in Ga) decreases as the temperature is reduced and the excess material is precipitated. For example, let us consider a saturated solution of arsenic in gallium at temperature T_1 and the substrate (also at T_1) is brought in contact with the solution. Now if the system is slowly cooled down to temperature T_2, GaAs is precipitated on the substrate as an epitaxial layer till the solution again becomes saturated at this lower temperature. Growth takes place at a comparatively lower temperature (typically 600–900°C). In a typical LPE system, a multiple-well graphite block containing different solutions is used and the sample is brought in contact with the solutions successively in order to achieve multiple layer growth. A calculated amount of dopant is added to the solvent in order to grow doped layers.

Molecular beam epitaxy (MBE) uses vacuum evaporation technique to grow epitaxial layers with very high degree of control. The substrate is held in ultra-high vacuum and epitaxial layers are deposited on the heated substrate from molecular beams impinging upon its surface. These beams are thermally generated in effusion cells (Knudsen cells) containing the constituent elements (such as Ga, As, and so on) of the desired layer. Each cell has a shutter to control the composition and/or doping of the film. The temperature of the cell is also accurately controlled to maintain the required intensity of the beams. Because of the ultra-high vacuum and precise control requirements, MBE systems are very expensive. However, the quality of the films is very good with precise control of doping. A schematic diagram of an MBE system is shown in Fig. 2.3.

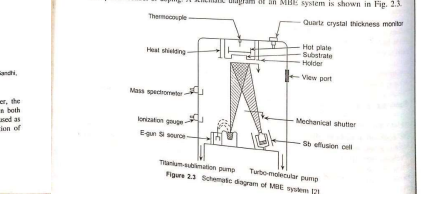

Figure 2.3 Schematic diagram of MBE system [2].

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2 Integrated Circuits Fabrication Technology

In an integrated circuit, all the circuit components, that is diodes, transistors, resistors, capacitors as well as the interconnections among them are realized on a single semiconductor chip. A large number of process steps are involved in the fabrication of semiconductor devices for integrated circuits. To begin with, the semiconductor material must be in the form of single crystals with defect-free surfaces. Controlled amount of impurities must then be introduced in the substrate to achieve proper doping. This may involve protecting particular regions of the substrate (masking), so that the doping occurs only in selected regions. This is followed by metallization to realize electrical contacts, scribing the devices into individual dies, attaching leads and finally encapsulation (packaging). This chapter discusses the various unit processes used in IC fabrication. Though most of the discussions are centred around silicon technology, the technology of compound semiconductors is also mentioned at appropriate places. The technological advantages of silicon over compound semiconductors are also highlighted. The process of fabrication for realizing the three basic devices in silicon technology, that is diodes, bipolar junction transistors, and MOSFETs are discussed later in Chapters 4, 6, and 10 respectively.

2.1 CRYSTAL GROWTH

As already pointed out, for the fabrication of semiconductor devices, the substrate must be in single crystal form. Silicon, the most commonly used semiconductor, is abundantly available on the earth's surface in the form of sand, which is almost pure silica (SiO_2). After chemical purification and reduction of silicon, very pure (99.9999%) polycrystalline silicon is obtained, which is used as the starting material for single crystal growth. Single crystal silicon for integrated circuits is mostly grown by two methods, namely the *Czochralski* (CZ) and the *Float Zone* (FZ) techniques. In the CZ technique, the polycrystalline material is kept in a quartz crucible held in a graphite susceptor and is heated by rf or resistive heating. Once the polycrystalline charge melts, a seed of single crystal suspended above the melt, is slowly lowered and brought into contact with the melt. The system is then evacuated, sealed, and placed in a diffusion furnace. After diffusion, the tube has to be heated to take out the samples. Hence, this is not a suitable and cost effective technique for mass production of doped semiconductors. Silicon, selenium, and tellurium are commonly used as n-type dopants while zinc is used as p-type dopant in GaAs.

HELP DESK 2.1

Although gallium is a group III element, why is it not commonly used as a p-type dopant in silicon?

Theoretically, group III elements such as Al or Ga, which have sufficiently low ionization energies can be used as p-type dopants in silicon. However, the diffusivity of these materials in SiO_2 which is the most commonly used masking material, is very high. During diffusion, Ga therefore has a tendency to diffuse in the regions protected by SiO_2. As the diffusivity of boron in SiO_2 is very small, such problems are not encountered in boron diffusion. Boron is thus exclusively used as the p-type dopant in silicon technology.

2.2.3 Ion Implantation

Ion implantation is an alternative technique used for selective doping of semiconductors. The doping profile is more precisely controlled by this technique. Also, it is a low temperature process and hence suitable for compound semiconductors as well. Another advantage of ion implantation is that it allows the doping profile to be tailored with a greater degree of flexibility. For example, from Eqs. (2.1) and (2.3) as well as Figs. 2.4(a) and (b), we see that in a diffusion process, the peak impurity concentration always occurs at the surface. However, by ion implantation, it is possible to attain the peak concentration below the surface. These advantages make ion implantation a very important process step in the present day MOS as well as Bipolar Transistor technologies.

In ion implantation, a beam of high-energy dopant ions is made to impinge on the semiconductor surface. When these ions enter the semiconductor, they lose their kinetic energy through collisions with the electrons as well as the nuclei of the lattice atoms. Finally, an impurity atom comes to rest when its kinetic energy falls to zero. The distance perpendicular to the surface covered by the impurity atom before coming to rest is called its *projected range* (R_p). Obviously, the projected range for a particular impurity species depends upon the energy (E) of the ion beam. In an amorphous material the doping after implantation is given by

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2.2.2 Diffusion

Although, it is possible to grow a layer with controlled doping by epitaxy, it is not possible to control the doping of particular regions of the semiconductor surface. In other words, epitaxial growth takes place on the entire surface, that is, it is nonselective. There have been some reports of selective epitaxy, but the layer quality is not as good and the process is therefore not very popular. In order to achieve selective doping, the technique most commonly used in silicon processing is called *diffusion*. The basic principle underlying this process is that the dopant atoms migrate from a region of high concentration to a region of low concentration. Some portions of the semiconductor are covered by a masking material, while the rest is left unprotected. Now if the semiconductor is held in an ambience of high dopant concentration and the temperature is raised, dopant atoms migrate into the unprotected regions of the semiconductor while many semiconductor atoms move out of their regular lattice sites. The dopant atoms may either move into these vacant sites (**substitutional impurities**) or occupy the empty space in between the lattice atoms (**interstitial impurities**). On cooling the sample, interstitial atoms may occupy substitutional positions and thus, become electronically active. Most common dopants in silicon, for example, phosphorus and boron, occupy substitutional sites, that is, they replace silicon atoms in the lattice. In practice, usually a two-step process is adopted to dope silicon.

Predeposition: In the first step (also called *predeposition*), the sample is heated in the presence of a very high concentration of the dopant. Under this condition, the diffusion profile is given by

$$N(x,t) = N_B \operatorname{erfc}\left(\frac{x}{2\sqrt{Dt}}\right) + N_A \tag{2.1}$$

where
N_B = original doping concentration of the sample,
N_A = solid solubility of the dopant in the semiconductor at the process temperature,
D = diffusion coefficient of the dopant in the semiconductor at the process temperature, and
t = diffusion time
x = distance from the sample surface.

From Eq. (2.1), we can see that after predeposition, the surface concentration $N(0, t)$ will always be N_A (since usually $N_A >> N_B$), which is a constant for a given temperature. The doping profiles for different durations of predeposition are shown in Fig. 2.4(a). The total number of impurity atoms per unit area of the semiconductor surface introduced in this step is

$$Q(t) = \int_0^\infty [N(x,t) - N_B]\,dx = 2N_A\sqrt{\frac{Dt}{\pi}} \tag{2.2}$$

Drive-in: In the next step (called *drive-in*), the dopant source is shut off and the sample is heated further. The doped layer already present on the sample surface now acts as the dopant source and the doping profile is given by

$$N(x,t) = \frac{Q}{\sqrt{\pi Dt}} \exp\left(-\frac{x^2}{4Dt}\right) + N_B \tag{2.3}$$

Equation (2.3) represents a Gaussian plot. An examination of Eq. (2.3) reveals that for longer durations of drive-in, the surface impurity concentration decreases while the impurities move deeper into the substrate increasing the junction depth. The doping profiles for different drive-in durations are shown in Fig. 2.4(b).

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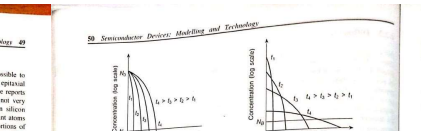

Figure 2.4 Doping profiles with different durations of (a) predeposition and (b) drive-in [1].

Example 2.1

Phosphorus is diffused into a uniformly doped p-type silicon with original doping concentration of the sample being 10^{16} per cm^3 at 1150°C. Given that the solid-solubility of phosphorus in silicon at 1150°C is 10^{20} per cm^3 and the diffusion coefficient at this temperature is 1×10^{-13} cm^2/s, (a) calculate the total number of phosphorus atoms per unit area of the silicon surface after a predeposition time of 1 hour. (b) If after this, drive-in is carried out for 2 hours at the same temperature, what will be the final junction depth and the surface concentration?

Solution: (a) From Eq. (2.2), we obtain the total amount of phosphorus introduced in silicon per unit area after predeposition as

$$Q = 2 \times 10^{20}\sqrt{\frac{10^{-13} \times 3600}{\pi}} = 6.77 \times 10^{13} \text{ per cm}^2$$

(b) Now after drive-in, the surface concentration is given by setting $x = 0$ in Eq. (2.3). So, if the drive-in is carried out for 2 hours at the same temperature, we have

$$N_s = \frac{6.77 \times 10^{13}}{\sqrt{\pi \times 10^{-13} \times 7200}} - 10^{16} = 4.5 \times 10^{19} \text{ per cm}^3$$

The negative sign for N_B implies that the original substrate dopant and the diffused impurity are of opposite types.

In order to obtain the junction depth, we note that at the junction the phosphorus concentration becomes equal to the original background doping concentration of the sample, that is, $N(x_j, t) = 0$. Substituting this in Eq. (2.3), we get

$$\frac{\exp\left(-\dfrac{x_j^2}{4 \times 10^{-13} \times 7200}\right)}{\sqrt{\pi \times 10^{-13} \times 7200}} = \frac{4.5 \times 10^{19}}{10^{16}}$$

This gives the junction depth as $x_j = 4.92$ μm.

The common n-type dopants in silicon are the group V elements such as phosphorus, arsenic, and antimony while the group III elements such as boron is almost exclusively used for p-type doping. Usually, diffusion in silicon is carried out in an open-tube furnace. A gas mixture carrying the dopant is made to flow over the sample kept in a carefully controlled atmosphere. The dopant

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source may be a solid, liquid, or gas. $POCl_3$ is the preferred liquid source used for doping phosphorus, though gaseous dopants such as PH_3 may also be used. BN (solid source) or BBr_3 (liquid) are the commonly used dopant sources for boron diffusion. Whatever be the actual dopant source in all the cases, it is made to react with oxygen at high temperature to form an oxide (D_2O_5, where D is the dopant element, say, P or B). This oxide then reacts with the silicon surface forming SiO_2 and releases the dopant into the semiconductor.

For compound semiconductors such as GaAs or InP, open-tube diffusion is not possible due to the high volatility of arsenic and phosphorus. For such materials, sealed-tube diffusion is used. In this method, the samples, along with the dopant source, are loaded into a quartz tube. The tube is then evacuated, sealed, and placed in a diffusion furnace. After diffusion, the tube has to be broken to take out the samples. Hence, this is not a suitable and cost effective technique for mass production of doped semiconductors. Silicon, selenium, and tellurium are commonly used as n-type dopants while zinc is used as p-type dopant in GaAs.

$$N(x) = \frac{Q_0}{\Delta R_p\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x - R_p}{\Delta R_p}\right)^2\right] \tag{2.4}$$

where
ΔR_p = standard deviation of the projected range and
Q_0 = total implantation dose (number of impurity ions introduced per unit area).

The implantation dose depends on the ion beam current density J and the implantation time t as

$$Q_0 = \frac{Jt}{q} \tag{2.5}$$

Since the implantation dose depends only on the current and time, it can be very precisely controlled from Eq. (2.4), it is evident that the peak impurity concentration occurs at $x = R_p$, that is, where the maximum number of dopant ions come to rest. By varying the energy of the ion beam, it is possible to obtain implantation very close to the surface or deep inside as the requirements may be. Figure 2.5 shows the actual implantation profiles for boron in silicon for different ion-beam energies.

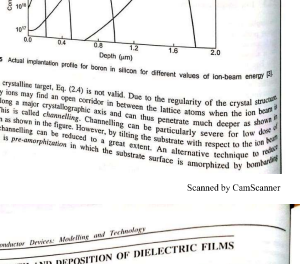

Figure 2.5 Actual implantation profile for boron in silicon for different values of ion-beam energy [3].

For a crystalline target, Eq. (2.4) is not valid. Due to the regularity of the crystal structure, the impurity ions may find an open corridor in between the lattice atoms when the ion beam is projected along a major crystallographic axis and can thus penetrate much deeper as shown in Fig. 2.6. This is called *channelling*. Channelling can be prevented to a great extent by tilting the substrate with respect to the ion beam direction, channelling can be reduced to a great extent. An alternative technique to reduce channelling is *pre-amorphization* in which the substrate surface is amorphized by bombarding

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In this regime, the oxidation rate is primarily governed by the diffusion of the oxidizing agent through the already existing oxide layer and is characterized by the *parabolic rate constant* (B). The values of the linear rate constant and the parabolic rate constant increase with temperature as can be seen from Table 2.1. In addition, the oxidation rate is much faster for wet oxidation than for dry oxidation. The crystal orientation also plays a role in the oxide growth process with {111} samples having thicker oxide films than {100} samples when oxidation is carried out under identical conditions.

Table 2.1 Rate constants for dry oxidation of silicon [4]

Oxidation temperature (°C)	A (μm)	B (μm^2/h)	B/A (μm/h)
920	0.235	0.0049	0.0208
1000	0.165	0.0117	0.071
1100	0.09	0.027	0.3
1200	0.04	0.045	1.12

Example 2.3

For a particular oxidation process, $B = 0.287$ μm^2/h, $A = 0.226$ μm, and τ is zero. Calculate, till what time the oxidation rate can be approximated to be linear, such that the oxide thickness calculated by using the linear rate approximation corresponds to within 10% of the value obtained by the more rigorous analysis.

Solution: Using the linear rate of oxidation, the oxide thickness is given by Eq. (2.7). If the oxidation is carried out for t hours, the oxide thickness (d_1) for this particular oxidation process, assuming linear growth rate, is given as

$$d_1 = \frac{0.287\,t}{0.226} \text{ μm}$$

On the other hand, considering the more rigorous analysis, the oxide thickness for the same time is obtained using Eq. (2.6) as

$$d_2 = 0.5\{(0.226^2 + 4 \times 0.287t)^{1/2} - 0.226\} \text{ μm}$$

Clearly d_1 is greater than d_2. According to the problem, the value of t is such that d_2 should be within 10% of d_1, that is, $d_1 = 1.1\,d_2$. Further solving the equation, the value of t is obtained as 0.02197 h, or 1.318 min.

It must be noted that oxide grown in dry O_2 ambient is denser, free of defects and therefore has paramount importance in MOS technology, where the quality of gate oxide has to be carefully controlled. Sometimes trace amounts of chlorine are introduced in the gas flow during oxidation to improve the quality of gate oxide. However, for growing thick oxide within a reasonable time, wet oxidation has to be carried out. The common practice for growing an oxide film thick enough to be used as a mask against diffusion (> 500 nm) is to follow a dry–wet–dry sequence.

2.3.2 Deposition of Dielectric Films

In compound semiconductors such as GaAs and InP, thermal oxidation is not a feasible option. This is because, the high vapour pressure of arsenic and phosphorus causes severe degradation of

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the material when it is exposed to high temperature. Also, the oxide is non-stoichiometric (that is, the different constituents of the material oxidize at different rates). Even in silicon devices, where a dielectric film is needed as a final passivating layer, high temperature oxidation process may not be feasible. In such cases, silicon dioxide or silicon nitride are usually deposited by *chemical vapour deposition* (CVD) technique. For example, a layer of SiO_2 can be deposited by the reaction of SiH_4 and O_2 at temperatures between 250–500°C. The reaction is

$$SiH_4 + 2O_2 \rightarrow SiO_2 + 2H_2O$$

Usually a gas mixture containing N_2 or Ar with about 1% SiH_4 and O_2 is used in a CVD process. For Si_3N_4 deposition, SiH_4 and NH_3 with N_2 or Ar as the carrier gas are used. The chemical reaction that occurs in the temperature range of 700–1100°C is expressed as

$$3SiH_4 + 4NH_3 \rightarrow Si_3N_4 + 12H_2\uparrow$$

Plasma-enhanced CVD (PECVD) process can also be used to deposit SiO_2 and Si_3N_4. In this technique, the presence of a gas plasma allows reactions (which would otherwise occur only at high temperatures) to take place at a comparatively lower temperature. However, silicon nitride deposited by PECVD contains a lot of hydrogen in the film and the film is usually non-stoichiometric.

2.4 MASKING AND PHOTOLITHOGRAPHY

As already pointed out, for the fabrication of semiconductor devices, selective doping is often necessary. This means that certain regions of the wafer have to be protected against doping during diffusion or ion-implantation. In general, this is done by covering the entire sample by a protective (masking) layer and then removing this mask layer at some selected regions by a process called *photolithography*. Afterwards, diffusion or ion-implantation is carried out and doping takes place only in the regions not protected by the mask. The most commonly used mask material in silicon technology is silicon dioxide (SiO_2) and it can be easily grown on silicon by thermal oxidation as discussed in the previous section. For compound semiconductors, SiO_2 and/or Si_3N_4 deposited by chemical vapour deposition can be used as masks. Since ion-implantation is a relatively low temperature process, photoresist itself can be used as a mask against implantation.

Once the mask layer is grown or deposited on the semiconductor surface, it must be patterned. That is, the mask should be retained only over certain selected regions and removed from the rest of the surface. Patterning is a two-step process. In the first step, a photosensitive material (photoresist) is spin-coated on the entire sample surface. There are two types of photoresist, namely positive and negative. In *optical photolithography*, the photoresist-coated wafer is exposed to UV light through an appropriate mask plate (or photomask). Certain regions on the mask plate are transparent, and the rest is opaque. In case of positive photoresist, the photoresist exposed to UV light is softened and is therefore selectively removed in a developer solution. In case of negative photoresist, only the exposed resist remains and the unexposed resist is removed by the developer solution. Thus, the mask pattern (or its negative) is transferred onto the resist-coated sample after the exposed (or unexposed) resist is removed in the developer solution. Figures 2.7(a), (b), and (c) illustrate the process schematically.

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with self-ions (that is, silicon surface bombarded with Si$^+$) before the actual implantation of the dopants. Sometimes, the implantation is also carried out through a thin (amorphous) oxide layer grown or deposited on the semiconductor surface to reduce channelling.

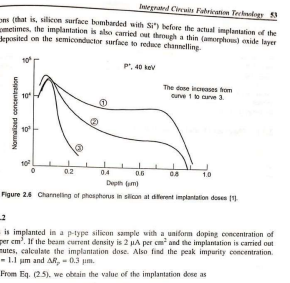

Figure 2.6 Channelling of phosphorus in silicon at different implantation doses [1].

Example 2.2

Phosphorus is implanted in a p-type silicon sample with a uniform doping concentration of 10^{16} atoms per cm^3. If the beam current density is 2 μA per cm^2 and the implantation is carried out for ten minutes, calculate the implantation dose. Also find the peak impurity concentration. Assume $R_p = 1.1$ μm and $\Delta R_p = 0.3$ μm.

Solution: From Eq. (2.5), we obtain the value of the implantation dose as

$$Q_0 = \frac{2 \times 10^{-6} \times 10 \times 60}{1.6 \times 10^{-19}} = 7.5 \times 10^{13} \text{ cm}^{-2}$$

The peak impurity concentration occurs at $x = R_p$. Substituting this value of x in Eq. (2.4), we get the peak concentration as

$$N_p = \frac{7.5 \times 10^{13}}{0.3 \times 10^{-4} \times \sqrt{2\pi}} = 6.267 \times 10^{20} \text{ cm}^{-3}$$

Even though ion-implantation offers a lot of advantages over diffusion, the process of implantation creates a lot of damages in the implanted region. These defects have to be thermally annealed in order to make the doped region electronically active. Annealing can be carried out in a conventional furnace at a temperature range of 800–1000°C for 20–30 minutes. However, this may alter the doping profile considerably by driving the dopants in. *Rapid thermal annealing* (RTA) is an alternative technique in which the sample temperature is raised to a high value quickly, held constant for a brief period, and then cooled down fast. The entire annealing cycle may take a few seconds to a few minutes and the doping profile remains essentially unaltered. RTA is therefore preferred as an annealing technique in present day VLSI technology over conventional furnace annealing.

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2.3 GROWTH AND DEPOSITION OF DIELECTRIC FILMS

In semiconductor processing, dielectric films essentially serve three purposes. They can be (i) a part of the active device (for example, gate oxide of a MOS transistor), (ii) used as masks to protect against diffusion (or ion-implantation), and (iii) used as a protecting layer (also known as passivation) at the end of device fabrication so as to protect the device from harsh ambience and ensure reliable operation. The most commonly used dielectric materials in semiconductor technology are SiO_2 and Si_3N_4. These films can either be grown on the semiconductor or deposited by using various techniques.

2.3.1 Thermal Oxidation of Silicon

Thermal oxidation is the most widely used technique to grow SiO_2 on silicon. This is usually carried out in an open-tube quartz furnace at a temperature range of 900–1200°C in an atmosphere of dry oxygen (*dry oxidation*) or water vapour (*wet oxidation*). For dry oxidation, oxygen is passed through the tube where it reacts with silicon to form SiO_2 by means of the following reaction

$$Si + O_2 \rightarrow SiO_2$$

In case of wet oxidation, high purity de-ionized water, kept in a quartz bubbler at the inlet of the furnace, is heated to a temperature close to its boiling point. High purity oxygen or nitrogen is passed through it so that the gas flowing into the furnace is saturated with water vapour. Water then acts as the oxidizing agent and the oxidation reaction is given by

$$Si + 2H_2O \rightarrow SiO_2 + 2H_2\uparrow$$

The oxide thickness grown on silicon by the process of dry or wet oxidation is dependent on the oxidation time and temperature. This thickness can be expressed by a linear-parabolic relation represented as

$$d^2 + Ad = B\,(t + \tau) \tag{2.6}$$

where
d = oxide thickness,
A and B = coefficients that depend on oxidation temperature and ambient conditions,
t = oxidation time, and
τ = a parameter for fitting the initial-value of oxide thickness.

For short duration of oxidation, $d \ll A$ and Eq. (2.6) can be approximated as

$$d = \frac{B}{A}(t + \tau) \tag{2.7}$$

which signifies that when the oxidation is carried out for a short time, the oxide thickness grows linearly with time. The oxidation rate in this regime is limited by the surface reaction and is characterized by the *linear rate constant* (B/A).

The solution of Eq. (2.6) shows that for longer oxidation times, such that $(t + \tau) >> (A^2/4B)$, the oxide thickness can be expressed as

$$d = \sqrt{Bt}$$

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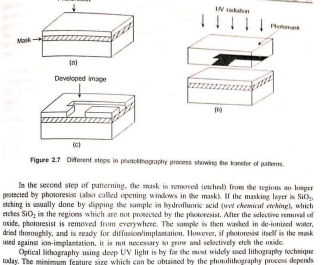

Figure 2.7 Different steps in photolithography process showing the transfer of patterns.

In the second step of patterning, the mask is removed (etched) from the regions no longer protected by photoresist (also called opening windows in the mask). If the masking layer is SiO_2, etching is usually done by dipping the sample in hydrofluoric acid (*wet chemical etching*), which etches SiO_2 in the regions which are not protected by the photoresist. After the selective removal of oxide, photoresist is removed from everywhere. The sample is then washed in de-ionized water, dried thoroughly, and is ready for diffusion/implantation. However, if photoresist itself is the mask used against ion-implantation, it is not necessary to grow and selectively etch the oxide.

Optical lithography using deep UV light is by far the most widely used lithography technique today. The minimum feature size which can be obtained by the photolithography process depends on the wavelength of the UV radiation, and lower wavelengths are used for better resolution. *Electron beam lithography* and *X-ray lithography* have also been used to reduce the minimum feature size. Electron beam lithography allows direct writing on the sample (no patterned mask plate is needed as the electron beam is directly raster-scanned on the photoresist). Due to the very small electron beam size, high resolution can be achieved. However, scanning the entire wafer is a very slow process and hence not suitable for large scale production. On the other hand, it is difficult to prepare a suitable mask plate for X-ray lithography. Thus, these techniques so far have found only limited application.

2.5 METALLIZATION

After the fabrication of the device, ohmic contacts to the different regions of the device have to be provided by metal deposition. Also, in case of integrated circuits, different devices on the chip must be suitably connected (interconnection) to perform the desired circuit operation. Aluminium is most widely used in silicon technology for providing ohmic contacts. Metal deposition is usually carried out in a vacuum deposition chamber where the source and the substrate are placed. The system is then evacuated with the help of a combination of pumps to a pressure of 10^{-6}–10^{-7} torr. The source

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metal is placed either in a tungsten filament or a boat heated by resistance heating and evaporation takes place. For better uniformity of deposition, a provision for rotating the substrate is available in many commercial deposition systems. Instead of resistance heating, electron beam (e-beam) evaporation technique is also used. In this process, a high-intensity electron beam is generated and focussed on the source material. This e-beam melts the metal and causes evaporation. Contamination from the heat is significantly reduced with e-beam deposition compared to the resistance heating process.

Aluminium, however, reacts with silicon, and in some instances may cause problems during deposition. Electromigration is another problem encountered in aluminium. When an electric field is applied during circuit operation, aluminium ions move in the direction of the field causing breaks in the metal lines. To prevent this, copper–aluminium alloy is used in many present day ICs. Quite recently, only copper interconnections have also been used successfully and the superior performance of these ICs shows the immense potential of copper to be used as an interconnect.

One more disadvantage associated with aluminium is that it has a low melting point. Therefore, after deposition of aluminium, the substrate cannot be subjected to any processing at high temperature. In MOSFET technology, this poses a great problem. Therefore, instead of aluminium, most modern MOSFETs use doped polycrystalline silicon, commonly referred to as polysilicon, as the gate metal. Polysilicon is deposited in a CVD reactor (similar to a VPE reactor) usually at a low pressure by decomposition of SiH_4. The typical deposition temperature is 600–800°C. In-situ doping of the polysilicon layer can be carried out by introducing PH_3 or AsH_3 in the gas stream during deposition. Alternatively after deposition, polysilicon can be doped by standard diffusion or implantation techniques.

In compound semiconductors such as GaAs and InP, metal contact formation is more difficult and a multi-layer structure has to be used. The most common ohmic contact in n-GaAs is Ni/Au–Ge with an overlayer of gold and a barrier material like titanium in between the ohmic contact and the gold overlayer. After metal deposition, the wafer is heated to allow formation of an alloy of metal and the semiconductor to ensure proper ohmic contact. Rapid thermal annealing is preferred for compound semiconductors to counter the high vapour pressure problem at elevated temperatures.

2.6 TECHNOLOGICAL ADVANTAGES OF SILICON

As already pointed out in Chapter 1, silicon may not be the most desirable semiconductor as far as its characteristic properties are concerned. Still more than 95% of VLSI circuits are based on silicon. As a matter of fact, the electronics revolution of the twentieth century has become possible mainly because of silicon devices. In this section, a few technological advantages of silicon over other compound semiconductor devices are enumerated.

(1) Cost factor: Silicon is easily available on the surface of the earth in the form of nearly pure silica (sand). The reduction of silica into semiconductor grade silicon (99.9999%) is also a comparatively simple process. In addition, the single crystal growth of silicon poses no vapour pressure problem unlike that of GaAs or InP (discussed in section 2.1). As a result, large crystals of silicon can be grown. In contrast, GaAs and InP crystal growth is more difficult and crystals are usually smaller. Larger wafer size means that many more IC chips can be realized per wafer. Since

ge cost of processing depends only marginally on the wafer size, the cost per chip is much less for silicon than for any compound semiconductor substrate. Also, silicon is a material with excellent mechanical properties and can withstand high temperature processing. Consequently, fabrication of silicon devices is relatively simple. For example, in silicon, a p-n junction can be easily realized by subjecting the sample to an open tube diffusion process. However, compound semiconductors are severely degraded when subjected to high temperature without proper protection and sealed-tube diffusion (low throughput) or ion-implantation (more sophisticated technique) is required in order to realize a junction. Thus, the processing cost of compound semiconductors is also higher. Unless specific performance requirements are to be met, compound semiconductor ICs cannot be an economically viable alternative to silicon ICs because of this high substrate and processing cost.

(2) Native oxide: The success of silicon ICs (particularly the MOSFETs) is largely due to the excellent dielectric properties of SiO_2, the almost ideal interface between silicon and SiO_2, and the ease of thermal oxide growth on silicon surface. SiO_2 also acts as a protection layer (mask) against most common dopants of silicon and hence used almost exclusively as mask in silicon technology. On the other hand, native oxide growth in GaAs and InP poses various problems and the quality of the native oxide as well as the interface between the oxide and the semiconductor is not good. Consequently, MOSFETs using compound semiconductors is still not commercially viable. Even for masking and passivation, chemical vapor deposition of SiO_2 or Si_3N_4 has to be carried out. Despite extensive research for the last thirty years, the problems of oxidation in compound semiconductors have not been solved and this has adversely affected the growth of compound semiconductor technology.

So far, we have discussed the basic fabrication processes used to realize semiconductor devices. The present day semiconductor devices are fabricated using *planar technology* in which the fabrication is carried out from one surface plane (usually the mirror-polished top surface). The actual sequence of steps required to fabricate particular devices such as diodes, bipolar transistors and MOSFETs in silicon IC will be discussed in Chapters 4, 6, and 10 respectively, along with the operating principles of these devices. It will then become clear as to how the process technology has to be modified in order to enhance the performance of the devices.

PROBLEMS

P2.1 An n-type silicon substrate has 0.5 μm oxide covering the entire surface area. Now 100 μm × 100 μm windows are opened in the oxide and boron is diffused through the windows. The drive-in after diffusion is carried out in an oxidizing ambient for which $B = 0.3$ μm²/h and $A = 0.2$ μm. This drive-in is continued for two hours. After this step,

(a) what is the thickness of oxide on the windows?

(b) what is the thickness of oxide elsewhere on the substrate?

P2.2 Phosphorus is diffused into silicon from an infinite source at 1000°C for 30 min.

(a) Find out the total amount of phosphorus per unit area that has gone into silicon.

(b) After step (a), the source is shut off and the sample is subjected to drive-in at 1200°C. If the final surface concentration is to be maintained at 5×10^{19} per cm³, what should be the duration of drive-in?

(c) If the original substrate doping was 10^{15} per cm³, what is the junction depth after step (b)?

[Given: for phosphorus diffusion, $D_{1000} = 3 \times 10^{-14}$ cm²/s, $D_{1200} = 2.5 \times 10^{-12}$ cm²/s, solubility of phosphorus at 1000°C = 10^{20} per cm³]

P2.3 In a particular oxidation process carried out on unoxidized silicon wafers, the following data was noted:

Oxidation time (t)	Oxide thickness (d)
30 minutes	0.12 μm
60 minutes	0.20 μm

How much time will it take to grow an oxide of thickness 0.3 μm in this process?

P2.4 Boron is implanted at 100 KeV into n-type silicon with a background concentration of 10^{15} per cm³.

(a) If the beam current is 1 mA and the target area is 100 cm², how long should the implantation be carried out to realize a dose of 4×10^{15} per cm²?

(b) Calculate the location of the p-n junction thus formed and the peak doping concentration. Assume $R_p = 3$ μm/MeV and $\Delta R_p = 0.3R_p$.

(c) Sketch the doping profile after the above process.

P2.5 Windows of dimensions 5 μm × 5 μm are to be etched in an oxide layer grown on silicon substrate. Draw the mask patterns neatly if (a) positive photoresist (b) negative photoresist are to be used.

P2.6 (a) What is the principal source from which oxygen is unintentionally incorporated during Czochralski growth of silicon?

(b) By which technique is it possible to grow single crystal silicon with lower oxygen incorporation?

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3

Charge Transport in Semiconductors

In this chapter, we shall discuss the basic mechanisms by which current can flow in a semiconductor. Current flow in a semiconductor can be either due to an applied electric field (drift current) or due to a difference (gradient) in the carrier concentration (diffusion current). These two currents form our subject of discussion in this chapter. Subsequently, the fundamental semiconductor equations governing the flow of charge carriers in a semiconductor will be derived considering both these current components.

3.1 DRIFT CURRENT

We have already seen in section 1.5.1 of Chapter 1 that when a small electric field is applied across a bar of semiconductor, electrons and holes acquire a drift velocity (v_d) proportional to the magnitude of the electric field. While electrons move in a direction opposite to that of the applied field, holes move in the direction of the field. This directional movement of the charge carriers constitutes a current, which is usually referred to as the drift current. It has already been derived in Chapter 1 that the current density J in a semiconductor sample having n electrons per unit volume travelling at a velocity v is given by (refer Eq. (1.5))

$$J = qnv \tag{3.1}$$

Considering only free electrons, so that n is now the free electron concentration, and replacing v with the drift velocity v_{dn}, the electron drift current density can be expressed as

$$J_{ndr} = -qnv_d \tag{3.2}$$

The minus sign in Eq. (3.2) indicates that the direction of the electron drift current J_{ndr} is opposite to that of the drift velocity. From Eq. (1.66), we know that the drift velocity is proportional to the electric field $\mathscr{E}$ for moderate electric fields and for electrons it can be expressed as $v_d = -\mu_n \mathscr{E}$. Substituting Eq. (1.66) in Eq. (3.2), we get

$$J_{ndr} = qn\mu_n \mathscr{E} \tag{3.3a}$$

Following a similar analysis for holes, the hole drift current density can be expressed as

$$J_{pdr} = qnv_d = qnv_d \mathscr{E} \tag{3.3b}$$

However, the above equations are valid only for moderate electric fields. For high fields, as already pointed out in section 1.5.3, the mobility is no longer a constant and $\mu\mathscr{E}$ has to be replaced with v_{sat}.

As the semiconductor contains both types of carriers, the total drift current density (J_{dr}) is the sum of the electron and hole current densities. Since electrons and holes move in directions opposite to each other, the total current actually adds up, and so

$$J_{dr} = qn\mu_n \mathscr{E} + qp\mu_p \mathscr{E} = \sigma \mathscr{E} \tag{3.4}$$

where σ is the conductivity of the semiconductor sample given by

$$\sigma = qn\mu_n + qp\mu_p \tag{3.5}$$

The resistivity ρ, which is the reciprocal of conductivity, can therefore be expressed as

$$\rho = \frac{1}{qn\mu_n + qp\mu_p} \tag{3.6}$$

We know that the majority carrier concentration in a strongly extrinsic semiconductor is many orders of magnitude higher than the minority carrier concentration. Therefore, from Eqs. (3.4), (3.5), and (3.6), we can write

$$\left.\begin{aligned} J_{dr} &\approx J_{ndr} = qn\mu_n \mathscr{E} \\ \sigma &\approx \sigma_n = qn\mu_n \\ \rho &\approx \rho_n = \frac{1}{qn\mu_n} \end{aligned}\right\} \text{ for n-type} \tag{3.7}$$

$$\left.\begin{aligned} J_{dr} &\approx J_{pdr} = qp\mu_p \mathscr{E} \\ \sigma &\approx \sigma_p = qp\mu_p \\ \rho &\approx \rho_p = \frac{1}{qp\mu_p} \end{aligned}\right\} \text{ for p-type} \tag{3.8}$$

HELP DESK 3.1

Show that the maximum resistivity for silicon is obtained when the sample is slightly p-type.

From Eq. (3.5), we see that at thermal equilibrium, the conductivity of a semiconductor sample where the electron and hole concentrations are comparable to each other is given by

$$\sigma = qn\mu_n + qp\mu_p = q\left(n_i^2\left(\frac{1}{p}\mu_n + p\mu_p\right)\right)$$

For maximum resistivity (that is, minimum conductivity)

$$\frac{d\sigma}{dp} = 0$$

Differentiating the above expression with respect to the hole concentration and equating it to zero, we obtain

$$\frac{n_i^2}{p^2}\mu_n = \mu_p \Rightarrow p = n_i\sqrt{\frac{\mu_n}{\mu_p}}$$

Since $\mu_n > \mu_p$, $p > n_i$, which means that the sample is slightly p-type.

Example 3.1

(a) Calculate the resistivity of intrinsic silicon at room temperature.

(b) Calculate the resistivity of an n-type semiconductor with a doping concentration $N_D = 10^{16}$ per cm³ at room temperature.

Solution: (a) As given in Table 1.3, we know that the intrinsic carrier concentration for silicon at room temperature is 1.5×10^{10} cm^{-3}. Also, from Table 1.4, the electron mobility for intrinsic silicon is 1350 cm²/Vs and the hole mobility is 480 cm²/Vs. The resistivity of intrinsic silicon is therefore calculated from Eq. (3.6) as

$$\rho = \frac{1}{qn_i(\mu_n + \mu_p)} = \frac{1}{1.6 \times 10^{-19} \times 1.5 \times 10^{10}(1350 + 480)} = 2.3 \times 10^5 \ \Omega \text{ cm}$$

(b) Assuming complete ionization, in this case the electron concentration, $n \approx N_D = 10^{16}$ cm^{-3}.

The hole concentration $p = n_i^2/N_D = 2.25 \times 10^4$ cm^{-3}. Since $n >> p$, the resistivity of the sample is effectively given by Eq. (3.7). Assuming the electron mobility in this sample to be equal to that in an intrinsic silicon sample, the resistivity is calculated to be

$$\rho = \frac{1}{qn\mu_n} = \frac{1}{1.6 \times 10^{-19} \times 10^{16} \times 1350} = 0.462 \ \Omega \text{ cm}$$

Example 3.1 highlights the fact that by introducing a small amount of impurity (< 1 ppm), it is possible to change the resistivity of silicon by many orders of magnitude. As already mentioned in Chapter 1, this is an important property of semiconductors. Evidently then, an accurate measurement of doping concentration in a semiconductor is essential to determine its basic properties. It is to be noted that the electron (and hole) mobility is also dependent on doping and is reduced considerably for high concentrations of doping, although this has not been taken into account in Example 3.1. Therefore, in practice, a straightforward current–voltage measurement does not yield an accurate value of carrier concentration unless the value of electron (and hole) mobility is also known. In the next section, we shall discuss a simple way of measuring carrier concentration and mobility simultaneously in an extrinsic semiconductor.

3.2 HALL EFFECT

The Hall effect is widely used for directly measuring the carrier concentration. The underlying principle in this measurement is that when a magnetic field is applied in a direction perpendicular to the flow of charge carriers, the path of the carriers gets deflected. Figure 3.1 shows a p-type semiconductor bar of length L, width W, and thickness d. Electric and magnetic fields are applied as shown in the figure. Due to the magnetic field $\mathbf{B}_z$, an upward Lorentz force is exerted on the carriers moving along x-direction with a drift velocity v_x. This upward force results in accumulation of holes at the top of the sample which gives rise to a corresponding downward electric field $\mathscr{E}_y$. Since there is no net current flow in the y-direction in the steady state, the force due to electric field must exactly balance the Lorentz force. That is,

$$q\mathscr{E}_y = qv_x\mathbf{B}_z$$

$$\mathscr{E}_y = v_x\mathbf{B}_z \tag{3.9}$$

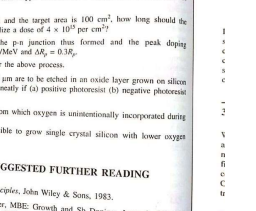

Figure 3.1 Basic set-up to measure carrier concentration using Hall effect. The polarity of V_H and the direction of v_x correspond to that for a p-type semiconductor.

The physical significance of this effect is that the magnetic field results in an increase in the hole concentration at the top surface of the sample shown in Fig. 3.1, till the electric field $\mathscr{E}_y$ becomes as large as $v_x\mathbf{B}_z$. Once this electric field is established, no net lateral force is experienced by the holes as they drift along the bar in the x-direction. The establishment of this electric field is known as the *Hall effect* and the field itself is called the 'Hall field'. The voltage across the terminals A and B is called the Hall voltage V_H. As seen from Fig. 3.1, $V_H = \mathscr{E}_y W$. With the help of Eqs. (3.9) and (3.3b), we can now express this Hall field as

$$\mathscr{E}_y = v_x\mathbf{B}_z = \frac{J_x\mathbf{B}_z}{qp} = J_x\mathbf{B}_z R_H \tag{3.10}$$

The above equation signifies that the Hall field is proportional to the product of current density and applied magnetic field with a proportionality constant $R_H = 1/qp$, called the 'Hall coefficient'. A measurement of Hall voltage for a known current (I) and magnetic field ($\mathbf{B}_z$) is carried out to obtain the hole concentration p in the sample. From Eq. (3.10), we have

$$p = \frac{1}{qR_H} = \frac{J_x\mathbf{B}_z}{q\mathscr{E}_y} = \frac{\left(\frac{I}{Wd}\right)\mathbf{B}_z}{q\left(\frac{V_H}{W}\right)} = \frac{I\mathbf{B}_z}{qV_H d} \tag{3.11}$$

Since all the quantities on the right hand side of Eq. (3.11) are measurable, accurate values of the hole concentration can be obtained directly from Hall measurement. Now substituting $v_x = \mu_p\mathscr{E}_x$ in Eq. (3.10), we can write

$$\mathscr{E}_y = \frac{V_H}{W} = \mu_p\mathscr{E}_x\mathbf{B}_z = \frac{\mu_p V_{DS}\mathbf{B}_z}{L} \tag{3.12}$$

Therefore, the mobility can also be obtained by direct measurement as

$$\mu_p = \frac{LV_H}{WV_{DS}\mathbf{B}_z} \tag{3.13}$$

Hall measurement carried out over a wide range of temperatures yields plots of majority carrier concentration and mobility versus temperature, which comprise very useful data in analysis of semiconductor materials. It is evident that Hall measurement can also be carried out for n-type semiconductors and similar results with a change of sign are obtained. Indeed, the sign of Hall voltage is used to determine whether the sample is n-type or p-type.

3.3 DIFFUSION CURRENT

So far in our discussions, we have considered a homogeneous semiconductor material, that is, a material with uniform doping concentration. However, if there is a spatial variation of carrier concentration in the semiconductor, in addition to the drift current, there exists another important current component due to this concentration gradient. When there is a concentration gradient, the carriers tend to diffuse from the region of higher concentration to the region of lower concentration, thereby constituting a current flow called the *diffusion current*.

In order to derive a relation for the diffusion current, let us consider an n-type semiconductor sample where the electron concentration varies from left to right as shown in Fig. 3.2. To carry out a one-dimensional analysis, let us assume that the doping concentration varies only in the x-direction, and is constant along the cross-section. The diffusion of carriers is actually a result of their constant random motion, which we discussed in section 1.5.1. The electrons are also constantly scattered due to collisions with lattice atoms, impurities, defects, or other electrons. Let the mean free path of the electrons between collisions be l and the mean free time be τ_c so that

$$l = v_{th}\tau_c$$

where v_{th} = average thermal velocity (Since the sample is at uniform temperature, v_{th} is same at every point).

Let $n(x)$ denote the concentration of electrons in the sample at any point x. Then the average concentration of electrons in the sample from $x = -l$ to $x = 0$ is

$$\frac{n(-l) + n(0)}{2}$$

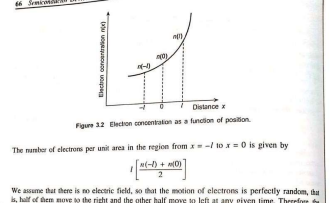

Figure 3.2 Electron concentration as a function of position.

The number of electrons per unit area in the region from $x = -l$ to $x = 0$ is given by

$$l\left[\frac{n(-l) + n(0)}{2}\right]$$

We assume that there is no electric field, so that the motion of electrons is perfectly random, that is, half of them move to the right and the other half move to left at any given time. Therefore, the number of electrons per unit area crossing the plane at $x = 0$ from left in time τ_c is

$$\frac{l}{4}[n(-l) + n(0)]$$

The current density at $x = 0$ is given by the net number of electrons crossing the plane at $x = 0$ per unit time per unit area. The flux F_1, defined as the average rate of electron flow per unit area per unit time moving from left to right, is given by

$$F_1 = \frac{l}{4\tau_c}[n(-l) + n(0)] \tag{3.14}$$

Similarly, the flux F_2 defined as the average rate of electron flow per unit area per unit time moving from right to left is given by

$$F_2 = \frac{l}{4\tau_c}[n(l) + n(0)] \tag{3.15}$$

Therefore, the net flux F can be written as

$$F = F_1 - F_2 = \frac{l}{4\tau_c}[n(-l) - n(l)] \tag{3.16}$$

The concentration gradient of electrons around $x = 0$ can be approximated as

$$\frac{dn}{dx} = \frac{n(l) - n(-l)}{2l} \tag{3.17}$$

From Eqs. (3.16) and (3.17), we now have

$$F = -\frac{dn}{dx}\frac{l^2}{2\tau_c} = -D_n\frac{dn}{dx} \tag{3.18}$$

where $D_n = l^2/(2\tau_c)$ is the **diffusion coefficient** or **diffusivity** of electrons.

The electron flux F constitutes an electron diffusion current density given by

$$J_{ndiff} = -qF = qD_n\frac{dn}{dx} \tag{3.19}$$

A similar analysis as above can be carried out considering holes. This analysis results in a relation for the hole diffusion current density given by

$$J_{pdiff} = qF = -qD_p\frac{dp}{dx} \tag{3.20}$$

The negative sign in Eq. (3.20) signifies that the direction of the hole current is opposite to the direction of the increasing concentration gradient, since the movement of holes is from a level of higher concentration to a level of lower concentration. Electrons also move from higher concentration to lower concentration. However, since the direction of the conventional current is opposite to the direction of electron movement, J_{ndiff} is in the same direction as the increasing concentration gradient as shown in Eq. (3.19).

3.4 CURRENT DENSITY EQUATIONS

When an electric field as well as a concentration gradient is present across a semiconductor sample, both drift and diffusion currents flow and the total current density is given as the sum of the drift and diffusion components. Thus, the total electron and hole current densities are given by

$$J_n = J_{ndr} + J_{ndiff} = qn\mu_n\mathscr{E} + qD_n\frac{dn}{dx} \tag{3.21a}$$

$$J_p = J_{pdr} + J_{pdiff} = qp\mu_p\mathscr{E} - qD_p\frac{dp}{dx} \tag{3.21b}$$

An important aspect of Eq. (3.21) is that the minority carriers can contribute significantly to the total current through diffusion. As the drift current is proportional to the carrier concentration, the contribution of minority carriers is usually negligible. However, even when the actual concentration of minority carriers is very small, the gradient may be significant. Thus, the minority carrier diffusion current may become as large as majority carrier currents in certain situations as we shall see later.

3.5 EINSTEIN'S RELATION CONNECTING μ AND D

Although drift and diffusion are two seemingly different processes, a close relationship exists between mobility and diffusion coefficient. This is because both these parameters are determined by the thermal motion and scattering of the free carriers. In order to establish the relationship between μ and D, the use of the important concept of constant Fermi Energy level at thermal equilibrium (already demonstrated in section 1.3.9) is made.

From Eqs. (3.16) and (3.17), we now have

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Let us consider a bar of non-uniformly doped n-type semiconductor under thermal equilibrium with a doping concentration as shown in Fig. 3.3(a). The bar is intrinsic for $x \le 0$, while from $x = 0$, the doping concentration gradually increases upto $x = l$. Beyond $x = l$, the doping becomes constant again. Let us assume complete ionization. Now, because of the concentration gradient, electrons diffuse from $x = l$ towards $x = 0$, leaving behind positively charged ionized donors. This charge separation gives rise to an electric field, which tends to pull the electrons back towards $x = l$. Thus, in thermal equilibrium, electrons tend to diffuse down the concentration gradient causing a diffusion flow of electrons from right to left. Again, the resulting electric field causes a drift flow of electrons in the opposite direction. Since no net current flows, these two current components must add up to zero. An equilibrium is then reached and it is easy to see why it is established. Suppose, for some reason, the equilibrium is disturbed so that the diffusion current is more than the drift current. This accounts for a net flow of electrons from right to left, resulting in an increase in charge separation. Consequently, the built-in electric field and the drift of electrons from left to right increase, till the equilibrium is re-established.

Therefore, we can write

$$J_n = qn(x)\mu_n\mathscr{E} + qD_n\frac{dn}{dx} = 0 \tag{3.22}$$

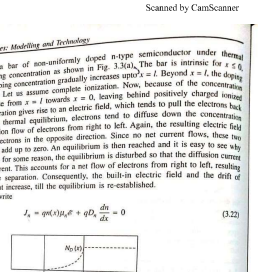

Figure 3.3 (a) A bar of n-type semiconductor showing graded impurity distribution; (b) the energy band diagram for this bar at thermal equilibrium.

Now, at $x = 0$, the Fermi level is at the intrinsic level, that is, $E_i(0) = E_F$. This is because the semiconductor is intrinsic up to that point. Since at thermal equilibrium, Fermi level must be constant, we can write from Fig. 3.3(b),

$$E_F - E_i(x) = E_i(0) - E_i(x) = q\psi(x) \tag{3.23}$$

where $\psi(x)$ is the potential difference between the points x and 0. From Eq. (1.27), we can express the electron concentration variation $n(x)$ as

$$n(x) = n_i \exp\left[\frac{E_F - E_i(x)}{kT}\right] \tag{3.24}$$

Now, substituting Eq. (3.23) in Eq. (3.24), we can write

$$n(x) = n_i \exp\left[\frac{q\psi(x)}{kT}\right] \tag{3.25}$$

Differentiating Eq. (3.25) with respect to x, we get

$$\frac{dn}{dx} = \frac{q}{kT}n(x)\frac{d\psi}{dx} = -\frac{n(x)\mathscr{E}}{V_T} \tag{3.26}$$

where $V_T = kT/q$ is the thermal voltage and $\mathscr{E}$ is the built-in field created due to the diffusion of electrons. Substituting Eq. (3.26) in Eq. (3.22), we have

$$D_n = \mu_n V_T \tag{3.27a}$$

Similarly, for holes, we can show that

$$D_p = \mu_p V_T \tag{3.27b}$$

Equations 3.27(a) and 3.27(b) are known as *Einstein's relations*. This important set of equations shows that there is a definite relationship between drift and diffusion.

3.6 CONTINUITY EQUATION

We shall now consider the overall effect when drift, diffusion, generation as well as recombination of carriers occur in a semiconductor. The combined effect is expressed by the continuity equation. Let us consider a bar of semiconductor with cross-sectional area A as shown in Fig. 3.4. Consider an element of infinitesimal thickness (dx) located at x as shown. The volume of the infinitesimal element is therefore (Adx). The essence of the *continuity equation* is that the 'overall rate of increase in the number of electrons in the given volume is given by the algebraic sum of four components: the rate of flow of electrons into the infinitesimal element at x, the negative of the rate of flow of electrons out of it at $x + dx$, the rate at which electrons are generated in the element, and negative of the rate at which they recombine in it.' Thus, the number of electrons in the slice may increase due to the net inflow of electrons into the element considered and/or net generation. Expressed mathematically, this can be written as

$$Adx\frac{\partial n}{\partial t} = J_n(x + dx)\frac{A}{q} - J_n(x)\frac{A}{q} + (G_n - R_n)Adx \tag{3.28}$$

where G_n and R_n are the generation and recombination rates of electrons respectively.

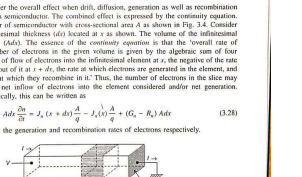

Figure 3.4 A semiconductor bar with cross-sectional area A showing an infinitesimal slice of thickness dx.

Also, since dx is extremely small, we can express

$$J_n(x + dx) = J_n(x) + \frac{\partial J_n}{\partial x}dx \tag{3.29}$$

Substituting Eq. (3.29) into (3.28), we get

$$\frac{\partial n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} + (G_n - R_n) \tag{3.30a}$$

Similarly, for the flow of holes

$$\frac{\partial p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} + (G_p - R_p) \tag{3.30b}$$

The difference in sign of the current density terms in Eqs. (3.30a) and (3.30b) is due to electrons and holes being oppositely charged. Equations (3.30a) and (3.30b) are referred to as the *continuity equations* for electrons and holes respectively. Now, substituting the expressions for hole and electron current densities from Eqs. (3.21a) and (3.21b) and the expressions for the net recombination rate as given by Eq. (1.59), the continuity equations for minority carriers (n_p for electrons in p-type semiconductor and p_n for holes in n-type semiconductor) can be expressed as

$$\frac{\partial n_p}{\partial t} = n_p\mu_n\frac{\partial \mathscr{E}}{\partial x} + \mu_n\mathscr{E}\frac{\partial n_p}{\partial x} + D_n\frac{\partial^2 n_p}{\partial x^2} + G_n - \frac{n_p - n_{p0}}{\tau_n} \tag{3.31a}$$

$$\frac{\partial p_n}{\partial t} = -p_n\mu_p\frac{\partial \mathscr{E}}{\partial x} - \mu_p\mathscr{E}\frac{\partial p_n}{\partial x} + D_p\frac{\partial^2 p_n}{\partial x^2} + G_p - \frac{p_n - p_{n0}}{\tau_p} \tag{3.31b}$$

There are three unknown quantities to be evaluated, namely n_p, p_n, and $\mathscr{E}$ and to solve for these three unknowns, three equations are needed. In principle, the two continuity equations (for electrons and holes) along with the Poisson's equation (discussed in Chapter 4) with appropriate boundary conditions should have a unique solution. However, because of the nonlinear nature of the equations, analytical solutions are not possible unless simplifying assumptions suitable for the particular case are made. The following section discusses a typical example that solves the carrier transport equation using suitable assumptions. In the case when there is a flow of carriers in the absence of electric field, (that is, $\mathscr{E} = 0$) and there is no other generation in addition to thermal generation (that is, $G_n = G_p = 0$), the continuity equation given by Eq. (3.31) can be simplified as

$$\frac{\partial n_p}{\partial t} = D_n\frac{\partial^2 n_p}{\partial x^2} - \frac{n_p - n_{p0}}{\tau_n} \tag{3.32a}$$

$$\frac{\partial p_n}{\partial t} = D_p\frac{\partial^2 p_n}{\partial x^2} - \frac{p_n - p_{n0}}{\tau_p} \tag{3.32b}$$

3.7 A TYPICAL EXAMPLE LEADING TO AN EXPRESSION FOR DIFFUSION LENGTH

Let us consider an n-type semiconductor (Fig. 3.5) where excess carriers are injected from one side as a result of illumination. This generation of excess carriers increases the carrier concentration at one surface ($x = 0$) and causes diffusion of the carriers inwards (that is, into the bulk of the semiconductor). In the steady state condition, the surface generation rate must be balanced by the flux of electrons and holes to the right of $x = 0$. As the electrons and holes diffuse into the bulk of the semiconductor, they recombine, resulting in a continuous decrease in their concentrations. Finally, deep inside the bulk, the carrier concentrations reduce to their thermal equilibrium values.

Since electrons and holes are generated in pairs, equal number of electrons and holes are

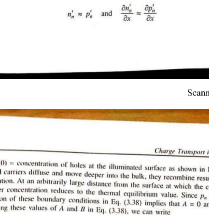

Figure 3.5 Injection of excess carriers due to illumination from one side of an n-type semiconductor bar and variation of minority carriers as they recombine inside the semiconductor.

injected into the bulk of the semiconductor at $x = 0$. This implies that the electron current is exactly equal and opposite to the hole current, or in other words, the total current is zero. Also, since these carriers recombine in pairs as they diffuse, even in the bulk $J_p(x) + J_n(x) = 0$.

Therefore, from Eqs. (3.21a) and (3.21b), we have

$$n_n\mu_n\mathscr{E} + D_n\frac{\partial n'_n}{\partial x} + p_n\mu_p\mathscr{E} - D_p\frac{\partial p'_n}{\partial x} = 0 \tag{3.33}$$

where n'_n and p'_n are the excess electron and hole concentrations respectively over the thermal equilibrium values.

Let us now make a simplifying assumption (called the *quasi-neutrality condition*), in order to obtain a solution of Eq. (3.33). This assumption states that at any time, the excess electron and hole concentrations, as well as their concentration gradients are almost equal at all points in the semiconductor, that is,

$$n'_n \approx p'_n \quad \text{and} \quad \frac{\partial n'_n}{\partial x} \approx \frac{\partial p'_n}{\partial x}$$

Clearly, the excess electron and hole concentrations would have been exactly equal if D_n and D_p were equal. However, in practical situations, there is only a small imbalance in excess electron and hole concentrations, and the quasi-neutrality condition is valid. Using this condition in Eq. (3.33), we have

$$\mathscr{E} = -\frac{D_n - D_p}{n_n\mu_n + p_n\mu_p}\frac{\partial p'_n}{\partial x} \tag{3.34}$$

Thus, we see that

$$\frac{J_p(\text{drift})}{J_p(\text{diff})} = \frac{\mu_p p_n \mathscr{E}}{-D_p\frac{\partial p'_n}{\partial x}} = \frac{p_n(D_n - D_p)}{D_n n_n + D_p p_n} \tag{3.35}$$

Since $n_n >> p_n$ and $D_n > D_p$, from Eq. (3.35), we have

$$\frac{J_p(\text{drift})}{J_p(\text{diff})} << 1$$

In other words, the drift of minority carriers can be neglected since the internal electric field created due to the charge imbalance is small. However, the drift current for the majority carriers (electrons in this case) is appreciable since the drift current is proportional to the product of the (small) electric field and (large) majority carrier concentration.

Neglecting the drift of holes for the n-type semiconductor, we can rewrite the continuity equation (3.31b) as

$$\frac{\partial p_n}{\partial t} = D_p\frac{\partial^2 p_n}{\partial x^2} + G_p - \frac{p_n - p_{n0}}{\tau_p} \tag{3.36}$$

At the steady state, $\partial p_n/\partial t = 0$. Also, $G_p = 0$ towards the right of $x = 0$ as shown in Fig. 3.5(a). Hence Eq. (3.36) can be further simplified to

$$D_p\frac{\partial^2 p_n}{\partial x^2} = \frac{p_n - p_{n0}}{\tau_p} \tag{3.37}$$

Noting that $\frac{\partial^2 p_n}{\partial x^2} = \frac{\partial^2 (p_n - p_{n0})}{\partial x^2}$, the solution of Eq. (3.37) is of the form

$$p_n - p_{n0} = A\exp\left(\frac{x}{\sqrt{D_p\tau_p}}\right) + B\exp\left(-\frac{x}{\sqrt{D_p\tau_p}}\right) \tag{3.38}$$

where A and B are constants that can be evaluated from the two boundary conditions given by

At $x = 0$, $\quad p_n = p_n(0)$

At $x = \infty$, $\quad p_n = p_{n0}$

Here, $p_n(0)$ = concentration of holes at the illuminated surface as shown in Fig. 3.5. Also, as the generated carriers diffuse and move deeper into the bulk, they recombine resulting in a decrease in the carrier concentration. At an arbitrarily large distance from the surface at which the carriers are generated, the carrier concentration reduces to the thermal equilibrium value. Since p_n is a finite value, the substitution of these boundary conditions in Eq. (3.38) implies that at $x = 0$ and $B = (p_n(0) - p_{n0})$. Substituting these values of A and B in Eq. (3.38), we can write

$$p_n - p_{n0} = (p_n(0) - p_{n0})\exp\left(-\frac{x}{L_p}\right) \tag{3.39}$$

where $L_p = \sqrt{D_p\tau_p}$ and is referred to as the *diffusion length*.

Substituting Eq. (3.39) into Eq. (3.20) and noting that $p_n - p_{n0} = p'_n$ is actually the excess hole concentration, we obtain an expression for the hole diffusion current in this case as

$$J_{pdiff} = -qD_p\frac{\partial p'_n}{\partial x} = \frac{qD_p}{L_p}(p_n(0) - p_{n0})\exp\left(-\frac{x}{L_p}\right) = \frac{qD_p p'_n}{L_p} \tag{3.40}$$

As shown in Fig. 3.5 with increasing x, the excess hole concentration dies out exponentially due to recombination. The hole diffusion current, which is proportional to the excess hole concentration also decays at the same rate. The variable L_p represents the distance at which the excess hole concentration falls to 1/e of its value at the point of injection. It should be noted that Fig. 3.5 shows a steady state profile, that is the variation does not change with time. This implies that the recombination of carriers in any given region of the semiconductor is compensated by a constant inflow of carriers

$$I \exp\left(-\frac{x}{L_p}\right)$$

Considering a small slice of thickness dx at a distance x from the injecting surface (as in Fig. 3.4), the number of carriers which have recombined in the slice can be given by

$$I \exp\left(-\frac{x}{L_p}\right) - I \exp\left(-\frac{x+dx}{L_p}\right) = I \exp\left(-\frac{x}{L_p}\right)\left[1 - \exp\left(-\frac{dx}{L_p}\right)\right] \approx \frac{I\,dx}{L_p} \exp\left(-\frac{x}{L_p}\right)$$

Dividing this expression by I which is the number of carriers which were injected at $x = 0$, we get the probability of a carrier recombining between x and $x + dx$ as

$$\frac{dx}{L_p} \exp\left(-\frac{x}{L_p}\right)$$

Using the usual method of averaging, the average distance travelled by a carrier before recombining can then be expressed as

$$x_{av} = \int_0^\infty \frac{x}{L_p} \exp\left(-\frac{x}{L_p}\right) dx = L_p$$

We have thus shown that L_p is the average distance travelled by the carriers before they recombine.

PROBLEMS

P3.1 In a uniformly doped silicon sample, the hole component of current is hundred times the electron component in an applied electric field. Assuming $\mu_n = 3\mu_p$, calculate the equilibrium electron and hole concentrations, the net doping, and the sample resistivity at 300 K.

P3.2 A silicon sample is doped with 10^{14} phosphorus atoms per cm^3 and 1.8×10^{16} boron atoms per cm^3. Find out the resistivity of the sample at 300 K assuming all impurities are ionized. If this sample is now uniformly illuminated with light that generates 10^{18} excess electron-hole pairs (EHP) per cm^3 per second, calculate the change in resistivity of the sample caused by the light in the steady state. Assume $\tau_n = \tau_p = 10^{-6}$ s.

P3.3 A sample of intrinsic semiconductor has a resistance of 30 Ω at 375 K and 300 Ω at 300 K. Assuming that mobilities are almost constant at this temperature range, calculate the band gap of the semiconductor.

P3.4 The resistivity of a silicon sample (ρ_0) is measured at 300 K. The sample is then re-melted and doped with an additional 5×10^{14} arsenic atoms per cm^3. A new crystal is grown that has a resistivity of 0.1 Ω cm and is n-type. Determine the type and concentration of dopant in the original sample and value of ρ_0.

P3.5 A silicon sample is doped with 10^{15} donor atoms per cm^3. Calculate the excess electron and hole concentration required to increase the sample conductivity by 20%. What is the carrier generation rate required to maintain these concentrations? Assume $\tau_p = 10^{-6}$ s and $T = 300$ K.
[Note: Use $n_i = 1.5 \times 10^{10}$ cm^{-3}, $\mu_n = 1350$ cm^2/Vs, and $\mu_p = 450$ cm^2/Vs for silicon at 300 K.]

P3.6 Consider an n-type silicon sample with $N_D = 10^{15}$ per cm^3. Plot the variation of conductivity with temperature for this sample. The temperature range should be from very low temperatures to very high temperatures. (Hint: Refer to Figs. 1.14 and 1.19)

REFERENCES AND SUGGESTED FURTHER READING

[1] Grove, A.S., *Physics and Technology of Semiconductor Devices*, Wiley, New York, 1967.

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[3] Streetman, B.G. and S. Banerjee, *Solid State Electronic Devices*, 5th ed., Prentice Hall Inc., New Jersey, 2000.

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$$J_n = J_{nN} + J_{nH} = 0 \tag{4.1b}$$

and

Also we know that for any junction at thermal equilibrium, the Fermi level must be constant throughout. The energy band diagram for this abrupt p-n junction can therefore be drawn simply by aligning the Fermi level as shown in Fig. 4.2(c). Far from the junction, the electron and hole concentrations on both sides remain unaffected and hence, the positions of E_i and E_F with respect to the Fermi level also remain the same as they were before the junction was formed. In the space charge region, however, the conduction and valence band edges bend, since the electronic potential (as discussed in section 1.7.4, the electronic potential is the negative of the electrostatic potential), the n-region is at an electric field. Since the energy band diagram reflects the variation of the electrostatic potential, the n-region is at a higher electrostatic potential than the p-region. This difference in E_C (or E_V) from the p-region to the n-region is given by $q\mathcal{V}_0$, where $\mathcal{V}_0$ is the called the *contact potential* or the *built-in potential*. We shall see later that this barrier can be modified by application of bias.

We have seen in Fig. 1.9 that electrons at a higher potential energy can easily "fall down" to a region of lower potential energy in the conduction band. On the other hand, since hole energy in an energy band diagram increases downwards, holes can easily "climb up" a potential energy barrier. However, the situation in Fig. 4.2(c) is quite different. There is a variation in the valence band. However, the situation in Fig. 4.2(c) is quite different. There is a large concentration of electrons in the conduction band in the n-region and holes in the valence band in the p-region. These electrons have to "climb up" while the holes have to "fall down" in order to cross the junction. This is possible only if the carriers have sufficiently large kinetic energy. The potential difference across the space charge layer therefore creates a barrier to the flow of carriers called the **potential energy barrier**. This barrier is equal to $q\mathcal{V}_0$ at thermal equilibrium. We shall see later that this barrier can be modified by application of bias.

4.1.1 Built-in Potential

The built-in potential can be calculated as the difference in the positions of the intrinsic levels in the n- and p-regions of the junction in terms of the doping concentrations. In the p-region, using Eq. (1.30), the position of the Fermi level is given by

$$E_{Fp} = E_{ip} - kT \ln \frac{N_A}{n_i} \tag{4.2a}$$

Similarly, in the n-region

$$E_{Fn} = E_{in} + kT \ln \frac{N_D}{n_i} \tag{4.2b}$$

Since the Fermi level is constant throughout, $E_{Fp} = E_{Fn}$. Therefore, from Eq. (4.2), we have

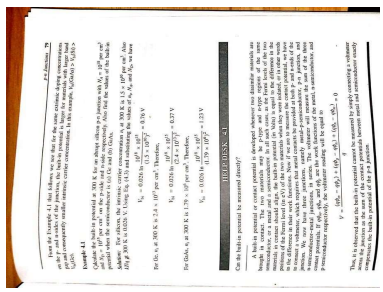
$$E_{ip} - E_{in} = kT \ln \frac{N_A N_D}{n_i^2} = q\mathcal{V}_{bi} \tag{4.3}$$

Assuming complete ionization, we can write $p_{p0} = N_A$, $n_{n0} \approx N_D$, $n_{p0} = n_i^2/p_{p0}$ and $p_{n0} = n_i^2/n_{n0}$. So, the built-in potential can be expressed as

$$\mathcal{V}_{bi} = V_t \ln \frac{p_{p0}}{p_{n0}} = V_t \ln \frac{n_{n0}}{n_{p0}} \tag{4.4}$$

where

p_{p0} and n_{n0} = majority carrier concentrations at thermal equilibrium and
p_{n0} and n_{p0} = minority carrier concentrations at thermal equilibrium.



From the Example 4.1 that follows, we see that in the p-region (doping concentration $N_A = 10^{16}$ per cm^3), the majority carrier concentration $p_{p0} \approx N_A = 10^{16}$ per cm^3 and the minority carrier concentration is $n_{p0} = n_i^2/p_{p0} = (1.5 \times 10^{10})^2/10^{16} = 2.25 \times 10^4$ per cm^3. Similarly, in the n-region, $n_{n0} \approx N_D = 10^{16}$ per cm^3 and $p_{n0} = n_i^2/n_{n0} = 2.25 \times 10^4$ per cm^3. Therefore, $\mathcal{V}_{bi} = V_t \ln \frac{10^{16}}{2.25 \times 10^4} = 0.026 \ln(4.44 \times 10^5) = 0.026 \times 13.0045 = 0.338$ V.

Can the built-in potential be measured externally?

A built-in potential or contact potential is developed whenever two dissimilar materials are brought in contact with each other. For example, when a metal and a semiconductor are in contact, some of the free electrons in the metal will diffuse into the semiconductor leaving behind the positively charged immobile donor ions. This diffusion of the electrons will continue till the electrostatic potential developed by the donor ions is sufficient to stop further diffusion. Thus, a built-in potential is developed across the metal-semiconductor junction. In this case, the semiconductor is usually lightly doped so that the depletion layer penetrates deeply into the semiconductor region. The metal-semiconductor junction is known as a Schottky barrier diode. Similarly, when two dissimilar metals are in contact, a built-in potential is developed. The magnitude of the built-in potential is usually small (a few tenths of a volt) and cannot be measured externally by a voltmeter.

Thus, if a voltmeter is connected between the two terminals of the p-n junction diode, the voltmeter will read zero. This is because the built-in potential developed at the metal-semiconductor junction of the positive terminal of the voltmeter is exactly cancelled by the built-in potential developed at the metal-semiconductor junction of the negative terminal of the voltmeter. Therefore, the voltmeter reading will be equal to zero.

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4 p-n Junctions

The p-n junction is one of the basic building blocks of integrated circuits. Such a junction can be formed by selective diffusion or ion implantation of n-type (or p-type) dopants into a p-type (or n-type) semiconductor sample. It has already been mentioned in Chapter 2, that the resulting dopant distribution follows a complementary error function (erfc) or a Gaussian distribution as shown in Fig. 4.1(a). However, in order to obtain simplified analytical expressions, the doping profile is generally approximated as a box-like or abrupt junction as shown in Fig. 4.1(b). An abrupt junction is one having constant dopant concentrations in both the n-type and p-type regions. We shall first consider the electric field and potential distribution in such an abrupt p-n junction and later extend the analysis to linearly graded junctions. Finally, the junction capacitances and the flow of current in a p-n junction under applied bias are discussed.

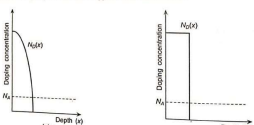

Figure 4.1 (a) Actual doping profile in a p-n junction; and (b) the approximate abrupt doping profile.

4.1 p-n JUNCTION UNDER THERMAL EQUILIBRIUM

We have already seen in Chapter 1 that the Fermi level in a p-type semiconductor lies close to the valence band edge, while in an n-type semiconductor, it lies close to the conduction band edge.

This means that the p-region has a higher concentration of holes (and few electrons) while n-region has a higher concentration of electrons (and few holes). Now, when a p-region and an n-region are brought in close contact (i.e. a junction is formed), this large concentration gradient at the junction causes diffusion of carriers. While holes diffuse from the p-region to the n-region, electrons diffuse from the n-region to the p-region. This process results in some uncompensated donor ions (N_D^+) in the n-region and some uncompensated acceptor ions (N_A^-) in the p-region near the junction. This is schematically shown in Fig. 4.2(a). Consequently, a *negative space charge* builds up in the p-region and a *positive space charge* in the n-region. This creates a built-in electric field directed from positive charge to negative charge (that is, from n-region to p-region) which gives rise to a drift current. The direction of this drift current will be opposite to that of the diffusion current for both electrons and holes as shown in Fig. 4.2(b). An equilibrium condition is reached where there is no net transport of carriers as the diffusion component is balanced by an equal and opposite drift component of current.

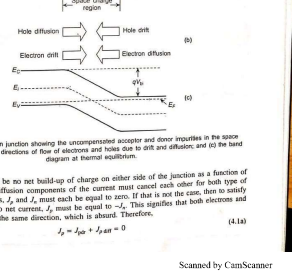

Figure 4.2 (a) A p-n junction showing the uncompensated acceptor and donor impurities in the space charge region; (b) the directions of flow of electron and hole currents due to drift and diffusion; and (c) the band diagram at thermal equilibrium.

Since there can be no net build-up of charge on either side of the junction as a function of time, the drift and diffusion components of the current must cancel each other for both type of charge carriers, that is, J_p and J_n must each be equal to zero. If that is not the case, then to satisfy the requirement of no net current, J_p must be equal to $-J_n$. This signifies that both electrons and holes are moving in the same direction, which is absurd. Therefore,

$$J_p = J_{pD} + J_{pH} = 0 \tag{4.1a}$$

4.1.2 Concept of Space Charge Layer

It has already been pointed out earlier in section 4.1 that due to the concentration gradient existing at the junction, a space charge layer (also called transition layer) is created with uncompensated donor ions in the n-region and acceptor ions in the p-region. Within this space charge region, electrons and holes are in transit from one side to the other. However, it is important to note that, there are much fewer charge carriers than dopants in most of this transition region. So, we can consider that within the space charge region the charge is only due to the immobile acceptor and donor ions. In other words, this region is depleted of mobile carriers and is also referred to as *depletion layer*. In this depletion region, as we move from the p-region to n-region, the difference between the intrinsic level and Fermi level ($E_i - E_F$) at first reduces, becomes zero and then becomes negative as can be seen from Fig. 4.2(c). This actually means that the hole concentration across this region decreases from p_p in the p-region to p_n in the n-region. Similarly, the concentration of electrons decreases from n_n in the n-region to n_p in the p-region. Figure 4.3 shows the variation of the carrier concentrations in the space charge region.

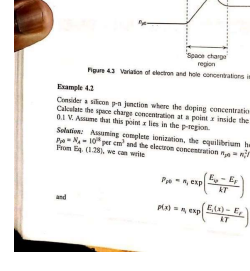

Figure 4.3 Variation of electron and hole concentrations in the space charge region.

Example 4.2

Consider a silicon p-n junction where the doping concentration in the p-region is 10^{18} per cm^3. Calculate the space charge concentration at a point x inside the space charge region where $\psi(x) = 0.1$ V. Assume that this point x lies in the p-region.

Solution: Assuming complete ionization, the equilibrium hole concentration in the p-region $p_{p0} = N_A = 10^{18}$ per cm^3 and the electron concentration $n_{p0} = n_i^2/N_A = 2.25 \times 10^2$ per cm^3 at 300 K. From Eq. (1.28), we can write

$$P_{p0} = n_i \exp\left(\frac{E_{ip} - E_F}{kT}\right)$$

and

$$p(x) = n_i \exp\left(\frac{E_i(x) - E_F}{kT}\right)$$

Therefore, from these equations, we can write

$$p(x) = p_{p0} \exp\left(\frac{E_i(x) - E_{ip}}{kT}\right) = p_{p0} \exp\left(\frac{-q\psi(x)}{kT}\right)$$

Now, at the point x where $\psi(x) = 0.1$ V, the hole concentration $p(x)$ is given by

$$p(x) = 10^{18} \exp\left(\frac{-0.1}{0.026}\right) = 2.13 \times 10^{16} \text{ per cm}^3$$

Following a similar procedure for electrons using Eq. (1.27), it can be shown that

$$n(x) = n_{p0} \exp\left(\frac{q\psi(x)}{kT}\right) = 2.25 \times 10^2 \exp\left(\frac{0.1}{0.026}\right) = 1.05 \times 10^4 \text{ per cm}^3$$

Thus, the total space charge density at the point x is

$$Q(x) = -q\,[N_A - p(x) + n(x)]$$

$$= -q\,[10^{18} - 2.13 \times 10^{16} + 1.05 \times 10^4] = -q\,9.98 \times 10^{17} \approx -q10^{18} = -qN_A$$

Example 4.2 serves as a validation of the *depletion approximation*, according to which the number of mobile charges inside the space charge layer is negligible. It is evident that the hole and electron concentrations are much smaller compared to the ionized dopant concentration and hence, can be ignored. Thus, neglecting mobile carriers within the space charge, the charge density in the p-region is $\rho_p = -qN_A$ and on the n-region, it is $\rho_n = qN_D$.

It must, however, be noted that at the edges of the space charge layer, the mobile carrier concentration approaches the concentration of the dopants, and hence depletion approximation may not be valid. (For example, consider $\psi(x) = 0.026$ V in Example 4.2). However, the thickness of this region is very small compared to the overall space charge layer thickness. Hence, in all further analysis in this book, for the sake of simplicity, we shall assume that the depletion approximation is valid throughout the space charge region. In other words, we assume that *the space charge region has sharp boundaries, where the charge concentration changes abruptly from $\rho_p = -qN_A$ to zero in the p-region and $\rho_n = qN_D$ to zero in the n-region.*

4.1.3 Distribution of Electric Field and Potential within the Space Charge Layer for Abrupt Junctions at Zero Bias

As the electric field lines must begin and end on charges of opposite sign, the total number of uncompensated acceptor ions in the p-region must be equal to that of the uncompensated donors in the n-region of the depletion layer. This causes the space charge layer to extend unequally into the p- and n-regions depending upon the relative doping concentrations of both sides. For example, if the doping concentration in the p-region (N_A) is greater than the doping concentration in the n-region (N_D), the space charge region will extend further into the n-region than into p-region to uncover equal amount of charge. Thus, the total uncompensated charge on each side of the junction Q_0 is given by

$$Q_0 = AqN_A x_p = AqN_D x_n \tag{4.5}$$

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where

A = cross-sectional area of the junction,
x_p = penetration of depletion region into p-region,
x_n = penetration of depletion region into n-region, and
q = electronic charge.

In this book, we shall consider only one-dimensional analysis for simplicity, thereby assuming that there is no variation in the device properties along the cross-section. A p-n junction, where the depletion region extends from $x = -x_p$ to $x = x_n$, is shown schematically in Fig. 4.4(a). The corresponding charge distribution (ρ versus x) is shown in Fig. 4.4(b). Since the number of acceptor impurities in the space charge region in the p-region is equal to the donor impurities in the n-region, the area under the charge distribution curve for the p-region (given by $N_A x_p$) must be equal to that for the n-region (given by $N_D x_n$). From Eq. (4.5) as well as Fig. 4.4(b), a relationship between the relative doping concentration and the penetration of depletion region can be obtained as

$$\frac{x_n}{x_p} = \frac{N_A}{N_D} \tag{4.6}$$

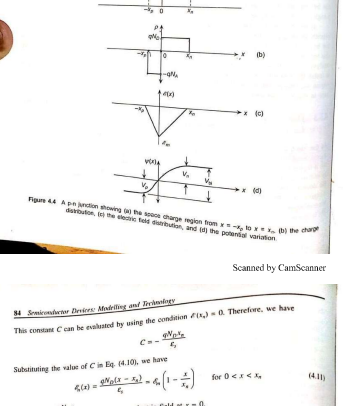


Figure 4.4 A p-n junction showing (a) the space charge region from $x = -x_p$ to $x = x_n$, (b) the charge distribution, (c) the electric field distribution, and (d) the potential variation.

In other words, if the doping concentration in the p-region of the junction is 1000 times the doping concentration in the n-region, the width of the depletion layer in the n-region will be 1000 times that in the p-region. In such cases for all practical purposes, the depletion region exists only on one side of the junction and the penetration on the heavily doped region can be neglected. Such a junction is called a *one-sided abrupt junction* and is used in our analysis later in this chapter.

Let us now again refer to Fig. 4.4 and obtain an expression for the electric field distribution inside the depletion region. To relate the charge distribution with the electrostatic potential ψ, we begin with Poisson's equation, which can be derived from Gauss law. According to Gauss law, the total normal electric flux coming out of a closed surface is equal to the charge enclosed by it, or

$$\nabla \cdot (\varepsilon_s \mathcal{E}) = \rho \tag{4.7a}$$

where

ρ = charge density in the space charge layer,
ε_s = permittivity of the semiconductor, and
$\mathcal{E}$ = electric field given by $\mathcal{E} = -\nabla\psi$.

Considering the existence of electric field only in the x-direction, the one-dimensional Poisson's equation can now be derived as

$$\frac{d^2\psi}{dx^2} = -\frac{\rho}{\varepsilon_s} \tag{4.7b}$$

Neglecting the presence of mobile carriers in the space charge region, Eq. (4.7b) reduces to

$$\frac{d\mathcal{E}}{dx} = \frac{q(N_D - N_A)}{\varepsilon_s} \tag{4.8}$$

Referring to Fig. 4.4(b), we see that

$$\frac{d\mathcal{E}}{dx} = \frac{qN_D}{\varepsilon_s} \quad \text{for } 0 < x < x_n \tag{4.9a}$$

$$\frac{d\mathcal{E}}{dx} = -\frac{qN_A}{\varepsilon_s} \quad \text{for } -x_p < x < 0 \tag{4.9b}$$

Thus, a plot of $\mathcal{E}(x)$ as a function of x within the depletion region has two slopes as seen from Fig. 4.4(c). The slope is positive in the n-region and negative in the p-region. Also, the magnitude of the field is maximum at $x = 0$. The physical interpretation of this fact is that *all* the field lines originating from the positive charges and terminating on the negative charges must pass through $x = 0$. This is not true for any other point in the space charge region. The negative value of the electric field in Fig. 4.4(c) refers to its direction, which is from the n-region to the p-region, that is, in the negative x-direction. Also, at the edge of the space charge region, electric field is assumed to fall to zero value as there are no charges outside the depletion region.

Integrating Eq. (4.9a) with respect to x, we get an expression for the electric field in the n-region as

$$\mathcal{E}(x) = \int \frac{qN_D}{\varepsilon_s} dx = \frac{qN_D x}{\varepsilon_s} + C \tag{4.10}$$

where

C = integration constant.

4.1.4 Distribution of Electric Field and Potential within the Space Charge Layer for Linearly Graded Junctions at Zero Bias

For many practical situations, the approximation of abrupt doping profile may not be valid. Let us now consider the case of a linearly graded junction where the doping concentration is a linear function of x as shown in Fig. 4.5(a). For such a situation

$$N_D - N_A = ax \tag{4.19}$$

where a = constant of proportionality.

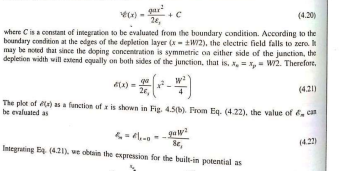


Figure 4.5 (a) The doping concentration distribution and (b) the electric field variation in a linearly graded p-n junction.

Then the solution of Poisson's equation yields an expression for the electric field as

$$\mathcal{E}(x) = \frac{qax^2}{2\varepsilon_s} + C \tag{4.20}$$

where C is a constant of integration to be evaluated from the boundary condition. According to the boundary condition at the edges of the depletion layer ($x = \pm W2$), the electric field falls to zero. It may be noted that since the doping concentration is symmetric on either side of the junction, the depletion width will extend equally on both sides of the junction, that is, $x_n = x_p = W2$. Therefore,

$$\mathcal{E}(x) = \frac{qa}{2\varepsilon_s}\left(x^2 - \frac{W^2}{4}\right) \tag{4.21}$$

The plot of $\mathcal{E}(x)$ as a function of x is shown in Fig. 4.5(b). From Eq. (4.22), the value of $\mathcal{E}_m$ can be evaluated as

$$\mathcal{E}_m = \mathcal{E}|_{x=0} = -\frac{qaW^2}{8\varepsilon_s} \tag{4.22}$$

Integrating Eq. (4.21), we obtain the expression for the built-in potential as

$$V_{bi} = \int_{-x_p}^{x_n} \mathcal{E}(x)\,dx = \frac{qaW^3}{12\varepsilon_s} \tag{4.23}$$

It can be shown that the built-in potential is equal to the area under the $\mathcal{E}(x)$ curve in Fig. 4.5(b). Also, we know that the built-in potential is actually the difference in the Fermi level positions with respect to the intrinsic levels at the two edges of the depletion layer. That is,

$$qV_{bi} = (E_F - E_i(x))|_{x=-W2} - (E_F - E_i(x))|_{x=W2} \tag{4.24}$$

Expressing $(E_F - E_i(x))$ in terms of the doping concentration as in Eqs. (1.29) and (1.30) at the edges of the depletion layer, we have

$$V_{bi} = V_t \ln\left(\frac{(aW2)(aW2)}{n_i^2}\right) = 2V_t \ln\left(\frac{aW}{2n_i}\right) \tag{4.25}$$

Solving Eqs. (4.23) and (4.25) simultaneously, it is possible to obtain numerical values for the depletion layer width and built-in potential for a particular case of linearly graded junction. However, this cannot be done analytically and numerical techniques have to be used.

4.2 THE p-n JUNCTION UNDER APPLIED BIAS

When an external voltage is applied to the p-n junction, the electron and hole concentrations deviate from their equilibrium values. The diode current is closely related to these variations. Since the space charge region is devoid of mobile carriers, its resistance is much greater than that of the neutral p- and n-regions. Therefore, practically all the applied voltage is dropped across this space charge layer and is superimposed on the built-in potential. In other words, the potential difference across the depletion layer deviates from its equilibrium value of V_0 by the amount of applied bias. When the p-n junction is forward biased by V_F, that is, the positive terminal of the battery is connected to the p-region and the negative terminal to the n-region, the potential difference across the space charge region is reduced to $(V_{bi} - V_F)$ as shown in Fig. 4.6(a). On the other hand, when the p-n junction is reverse biased by V_R, that is, the positive terminal of the battery is connected to the n-region and the negative terminal to the p-region, the potential difference is increased to $(V_{bi} + V_R)$ as shown in Fig. 4.6(b). Figure 4.6 also shows the variation in quasi-Fermi levels, which is discussed later in section 4.3.2.

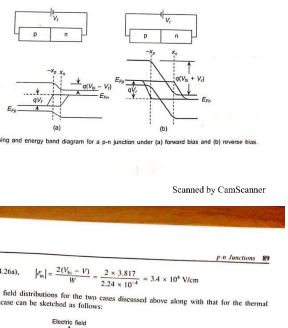


Figure 4.6 Biasing and energy band diagram for a p-n junction under (a) forward bias and (b) reverse bias.

This constant C can be evaluated by using the condition $\mathcal{E}(x_n) = 0$. Therefore, we have

$$C = -\frac{qN_D x_n}{\varepsilon_s}$$

Substituting the value of C in Eq. (4.10), we have

$$\mathcal{E}_n(x) = \frac{qN_D(x - x_n)}{\varepsilon_s} = \mathcal{E}_m\left(1 - \frac{x}{x_n}\right) \quad \text{for } 0 < x < x_n \tag{4.11}$$

where $\mathcal{E}_m = -\frac{qN_D x_n}{\varepsilon_s}$ is the maximum electric field at $x = 0$.

Following a similar analysis for the p-region, we have

$$\mathcal{E}_p(x) = -\frac{qN_A(x + x_p)}{\varepsilon_s} = \mathcal{E}_m\left(1 + \frac{x}{x_p}\right) \quad \text{for } -x_p < x < 0 \tag{4.12}$$

where $\mathcal{E}_m = -\frac{qN_A x_p}{\varepsilon_s}$ is the maximum electric field at $x = 0$.

Using Eq. (4.5), we can show that the expressions for $\mathcal{E}_m$ obtained from the n- and p-regions are indeed equivalent, or

$$\mathcal{E}_m = -\frac{qN_D x_n}{\varepsilon_s} = -\frac{qN_A x_p}{\varepsilon_s} \tag{4.13}$$

It may be noted that $\mathcal{E}_m$ is proportional to the area under the charge concentration plots in Fig. 4.4(b). Now, integrating Eqs. (4.11) and (4.12) over the entire space charge region, we obtain the built-in potential as

$$V_{bi} = -\int_{-x_p}^{x_n} \mathcal{E}\,dx = -\int_0^{x_n} \mathcal{E}_n\,dx - \int_{-x_p}^{0} \mathcal{E}_p\,dx = \frac{q(N_A x_p^2 + N_D x_n^2)}{2\varepsilon_s} = \frac{|\mathcal{E}_m|W}{2} \tag{4.14}$$

where $W (= x_p + x_n)$ is the total width of the space charge layer.

It can be easily verified from Fig. 4.4(c) that the built-in potential, that is, the potential dropped across the depletion layer is given by the area of the triangle covered by the $\mathcal{E}(x)$ versus x plot. This area can also be viewed to be the sum of two triangles having the same altitude ($= |\mathcal{E}_m|$) but different base lengths (x_n and x_p). The areas of these two triangles signify the potential dropped across the n-region of the depletion layer (V_n) and that dropped across the p-region of the depletion layer (V_p). The values of V_n and V_p calculated geometrically from the areas of the triangles agree perfectly with Eq. (4.14), which has been derived analytically. In fact, the geometric method of obtaining the electric field distribution and subsequently the potential variation from the doping concentration curve is often more convenient than analytical methods.

We know that V_n and V_p are related by the following expression

$$\frac{V_n}{V_p} = \frac{N_D}{N_A} = \frac{x_p}{x_n} \tag{4.15}$$

From Eq. (4.15), we can write

$$x_p = \frac{N_D W}{N_A + N_D}$$

and

$$x_n = \frac{N_A W}{N_A + N_D}$$

Substituting these values of x_p and x_n in Eq. (4.14), we get

$$W = \sqrt{\frac{2\varepsilon_s V_{bi}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)}{q}} = \sqrt{\frac{2\varepsilon_s V_{bi}}{qN_{eff}}} \tag{4.16}$$

where N_{eff} can be considered to be an effective doping concentration given by

$$\frac{1}{N_{eff}} = \frac{1}{N_A} + \frac{1}{N_D} \tag{4.17}$$

For a one-sided abrupt junction, if $N_A >> N_D$, $N_{eff} \approx N_D$. Also Eq. (4.17) can be approximated as

$$W = \sqrt{\frac{2\varepsilon_s V_{bi}}{qN_D}} \tag{4.18}$$

This also implies that virtually the entire built-in potential is dropped across the lightly doped side.

Example 4.3

Calculate the maximum electric field and the width of the depletion region at zero bias for an abrupt silicon p-n junction with $N_A = 10^{18}$ per cm^3 and $N_D = 10^{16}$ per cm^3 at room temperature. (Use $n_i = 1.5 \times 10^{10}$ per cm^3, $\varepsilon_r = 11.9$ for Si, $\varepsilon_0 = 8.85 \times 10^{-14}$ F/cm)

Solution: Using Eq. (4.3), we have

$$V_{bi} = 0.026 \ln \frac{10^{18} \times 10^{16}}{(1.5 \times 10^{10})^2} = 0.817 \text{ V}$$

In this problem, since $N_D/N_D = 10^4$, we consider the junction to be a one-sided abrupt junction. Now, from Eq. (4.18),

$$W = \sqrt{\frac{2 \times 11.9 \times 8.85 \times 10^{-14} \times 0.817}{1.6 \times 10^{-19} \times 10^{16}}} = 1.037 \times 10^{-4} \text{ cm} = 1.037 \text{ μm}$$

From Eq. (4.14),

$$|\mathcal{E}_m| = \frac{2V_{bi}}{W} = \frac{2 \times 0.817}{1.037 \times 10^{-4}} = 1.58 \times 10^4 \text{ V/cm}$$

It is left as an exercise for the student to verify that the depletion layer width is nearly the same, if a one-sided junction is not considered and Eq. (4.16) is used.

When a forward voltage is applied, the direction of the applied field is opposite to that of the built-in field. Consequently, the effective field across the junction reduces with forward bias. Conversely, when a reverse bias is applied, the effective field is increased. This change in the electric field also affects the depletion layer width. For forward voltages, the electric flux lines reduce, leading to fewer uncompensated charges and smaller W. Since the flux lines and therefore a larger W. Since the by the same logic, results in increase in electric flux lines, the new values of W and voltage drop across the junction is now $(V_{bi} - V)$, for an abrupt junction, the new values of W and $\mathcal{E}_m$ can be obtained by replacing V_{bi} with $(V_{bi} - V)$ in Eqs. (4.14) and (4.16), that is,

$$|\mathcal{E}_m| = \frac{2(V_{bi} - V)}{W} \tag{4.26a}$$

$$W = \sqrt{\frac{2\varepsilon_s(V_{bi} - V)}{qN_{eff}}} \tag{4.26b}$$

where

V = applied voltage
$V = V_F$ for forward bias
$V = -V_R$ for reverse bias

Example 4.4

For the p-n junction of Example 4.3, obtain the maximum electric field and depletion layer width when (a) a forward voltage of 0.3 V and (b) a reverse voltage of 3 V is applied. Also sketch the electric field distribution for these two cases along with that in thermal equilibrium as obtained in Example 4.3.

Solution: In Example 4.3, we have calculated $V_{bi} = 0.817$ V

Case (a): When $V_F = 0.3$ V,

$$V_{bi} - V_F = 0.817 - 0.3 = 0.517 \text{ V}$$

Now, from Eq. (4.26b),

$$W = \sqrt{\frac{2 \times 11.9 \times 8.85 \times 10^{-14} \times 0.517}{1.6 \times 10^{-19} \times 10^{16}}} = 8.25 \times 10^{-5} \text{ cm} = 0.825 \text{ μm}$$

From Eq. (4.26a),

$$|\mathcal{E}_m| = \frac{2(V_{bi} - V_F)}{W} = \frac{2 \times 0.517}{8.25 \times 10^{-5}} = 1.25 \times 10^4 \text{ V/cm}$$

Case (b): When $V_R = 3$ V,

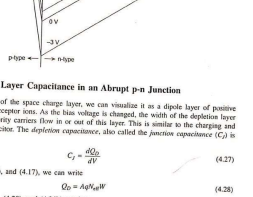
$$V_{bi} + V_R = 0.817 + 3.0 = 3.817 \text{ V}$$

Now, from Eq. (4.26b),

$$W = \sqrt{\frac{2 \times 11.9 \times 8.85 \times 10^{-14} \times 3.817}{1.6 \times 10^{-19} \times 10^{16}}} = 2.24 \times 10^{-4} \text{ cm} = 2.24 \text{ μm}$$

From Eq. (4.26a), $|\mathcal{E}_m| = \frac{2(V_{bi} - V_R)}{W} = \frac{2 \times 3.817}{2.24 \times 10^{-4}} = 3.4 \times 10^4$ V/cm

The electric field distributions for the two cases discussed above along with that for the thermal equilibrium case can be sketched as follows:



4.2.1 Depletion Layer Capacitance in an Abrupt p-n Junction

From our discussion of the space charge layer, we can visualize it as a dipole layer of positive donor and negative acceptor ions. As the bias voltage is changed, the width of the depletion layer also changes, as majority carriers flow in or out of this layer. This is similar to the charging and discharging of a capacitor. The *depletion capacitance*, also called the *junction capacitance* (C_j) is thus expressed as

$$C_j = \frac{dQ_0}{dV} \tag{4.27}$$

Using Eqs. (4.5), (4.6), and (4.17), we can write

$$Q_0 = AqN_{eff}W \tag{4.28}$$

Now from Eqs. (4.27), (4.28), and (4.26b), we have

$$C_j = \frac{dQ_0}{dV} = \frac{dQ_0}{dW}\frac{dW}{dV}$$

$$= AqN_{eff}\frac{\varepsilon_s}{2qN_{eff}(V_{bi} - V)} = A\varepsilon_s\sqrt{\frac{qN_{eff}}{2\varepsilon_s(V_{bi} - V)}} \tag{4.29}$$

Substituting Eq. (4.26b) in Eq. (4.29), we now have

$$C_j = \frac{A\varepsilon_s}{W} \tag{4.30}$$

Equation (4.30) is the well known expression for a parallel plate capacitor where the two plates are separated by a distance W. In other words, the expression for the depletion capacitance is identical to that of a parallel plate capacitor. However, unlike a parallel plate capacitor, W here varies

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nonlinearly with voltage. Consequently, C_j also varies with a change in the applied bias. This change in C_j can be measured to extract the doping concentration of the substrate as well as the built-in potential, especially in case of a one-sided abrupt p-n junction. For such a junction (p^+ n), $N_{aa} \gg N_d$. Hence from Eq. (4.29), we can write

$$C_j = A\sqrt{\frac{q\varepsilon_s N_{dn}}{2(V_{bi} - V)}} = C_{j0}\left(1 - \frac{V}{V_{bi}}\right)^{-1/2} \tag{4.31}$$

where $C_j = C_{j0}$ for $V = 0$.

From Eq. (4.31), we can write

$$\frac{1}{C_j^2} = \frac{2}{A^2 q\varepsilon_s N_{dn}}(V_{bi} - V) \tag{4.32}$$

So we find that if we change the bias voltage across the p-n junction diode, measure the capacitance C_j at each bias point and finally plot $(1/C_j^2)$ versus the applied voltage, we get a straight line with a slope equal to $-2/A^2 q\varepsilon_s N_{dn}$ as shown in Fig. 4.7. Now, since q and ε_s are well known constants, if A is known we can find out the value of N_{dn} from the slope of the straight line. It must be pointed out that the measurements must be made with the diode in the reverse biased condition (that is $V < 0$), since in the forward biased condition the stored charge capacitance, which we shall discuss in section 4.4.3, dominates and the measured capacitance does not equal C_j. From Eq. (4.32), we find that when $V = V_{bi}$, the reciprocal of C_j^2 is zero. Therefore, extrapolating the $(1/C_j^2)$ versus V curve for positive values of V, we get the value of V_{bi} from the intercept of the curve with the V-axis.

Thus, we see that the reverse bias C–V measurement is an important tool for obtaining both the doping concentration and the built-in potential of a p-n junction.

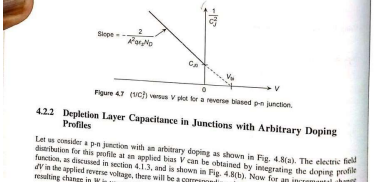

Figure 4.7 $(1/C_j^2)$ versus V plot for a reverse biased p-n junction.

4.2.2 Depletion Layer Capacitance in Junctions with Arbitrary Doping Profiles

Let us consider a p-n junction with an arbitrary doping as shown in Fig. 4.8(a). The electric field distribution for this profile at an applied bias V can be obtained by integrating the doping profile function, as discussed in section 4.1.3, and is shown in Fig. 4.8(b). Now for an incremental change dV in the applied reverse voltage, there will be a corresponding change in the electric field ($d\mathcal{E}$). If the resulting change in W is very small, that is, $dW \ll W$, we can see from the figure that $|A'| = |a|d\mathcal{E}$.

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where dV is given by the area of the shaded region in Fig. 4.8(b). According to Gauss law, the total electric flux coming out of a closed surface equals the charge enclosed by the surface. Therefore, $A\varepsilon_s d\mathcal{E} = dQ_D$, where $A\varepsilon_s d\mathcal{E}$ is the incremental total flux over the area A corresponding to the incremental depletion charge dQ_D. Therefore, substituting these values in Eq. (4.27), we obtain

$$C_j = \frac{dQ_D}{dV} = \frac{dQ_D}{Wd\mathcal{E}} = \frac{dQ_D}{W\left(\frac{dQ_D}{A\varepsilon_s}\right)} = \frac{A\varepsilon_s}{W} \tag{4.33}$$

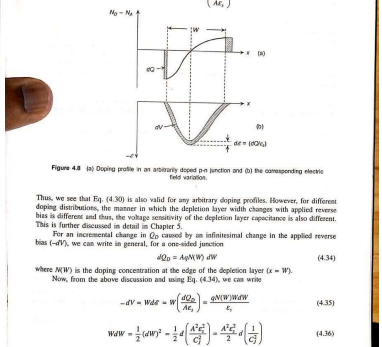

Figure 4.8 (a) Doping profile in an arbitrarily doped p-n junction and (b) the corresponding electric field variation.

Thus, we see that Eq. (4.30) is also valid for any arbitrary doping profiles. However, for different doping distributions, the manner in which the depletion layer width changes with applied reverse bias is different and thus, the voltage sensitivity of the depletion layer capacitance is also different. This is further discussed in detail in Chapter 5.

For an incremental change in Q_D caused by an infinitesimal change in the applied reverse bias ($-dV$), we can write in general, for a one-sided junction

$$dQ_D = Aq\mathcal{N}(W)\,dW \tag{4.34}$$

where $\mathcal{N}(W)$ is the doping concentration at the edge of the depletion layer ($x = W$).

Now, from the above discussion and using Eq. (4.34), we can write

$$-dV = Wd\mathcal{E} = W\left(\frac{dQ_D}{A\varepsilon_s}\right) = \frac{qW\mathcal{N}(W)dW}{\varepsilon_s} \tag{4.35}$$

$$WdW = \frac{1}{2}(dW)^2 = \frac{1}{2}d\left(\frac{A^2C_j^2}{C_j^2}\right) = \frac{A^2}{2}d\left(\frac{1}{C_j^2}\right) \tag{4.36}$$

Now substituting Eq. (4.36) in Eq. (4.35), we have

$$-dV = \frac{qN(W)A^2}{2\varepsilon_s}d\left(\frac{1}{C_j^2}\right)$$

or

$$\frac{d\left(\frac{1}{C_j^2}\right)}{dV} = -\frac{2}{qN(W)A^2\varepsilon_s} \tag{4.37}$$

Equation (4.37) shows that the C–V characteristics of p-n junctions can also be used to obtain the doping profile for non-uniformly doped regions. The initial process is the same as that for uniformly doped junctions, that is after C–V measurements, the $(1/C_j^2)$ versus V curve is plotted. Using Eq. (4.37), we see that from the slope of this curve, the doping concentration at the edge of the depletion region $\mathcal{N}(W)$ corresponding to a particular C_j can be obtained. Equation (4.33) can be used to obtain W for each value of C_j, and therefore the $\mathcal{N}(W)$ versus W plot can be drawn. It can be easily verified that for uniform doping concentration (that is, $\mathcal{N}(W) = \text{constant}$), Eq. (4.37) reduces to Eq. (4.32).

4.3 STATIC CURRENT-VOLTAGE CHARACTERISTICS OF p-n JUNCTIONS

At zero bias, there is no net transfer of carriers across the junction. With the application of forward bias, the potential barrier is lowered and many more electrons from n-region are able to diffuse to the p-region as they now have sufficient energy to cross the barrier. Similar situation prevails for the holes and thus, the diffusion current in a forward-biased junction is quite large. An opposite situation is encountered in a reverse-biased junction, where the potential barrier is so large that virtually no electrons from the n-region and no holes from the p-region can diffuse across the junction. The current is now due to electrons from the p-region and holes from the n-region crossing the junction. Since these are minority carriers and are therefore few in number, the reverse current is extremely small. In this section, we shall derive a relation for the steady state current-voltage characteristics of p-n junction diodes in terms of the various device parameters.

4.3.1 Current–Voltage Relationship in an Infinitely Long Diode

After the qualitative discussion regarding the p-n junction under the biased conditions as presented in the previous section, let us now derive analytical expression for the current–voltage relation in an infinitely long diode. In doing so, the following assumptions are made.

(a) The current flow across the junction is one-dimensional, that is, only in the x-direction.

(b) All the applied voltage is assumed to drop across the space charge region. As the electric field in the neutral p- and n-regions is negligibly small, the minority carrier drift current can be neglected as shown in section 3.7.

(c) There is no net generation or recombination in the space charge layer and the current in this region remains constant throughout.

(d) Low level injection condition prevails. This is valid for low

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(e) Drift and diffusion currents almost balance each other in the depletion region even when a bias is applied.

While the other assumptions are easy to justify, the last one (assumption e) needs some explanation. Actually, the drift current is equal to the diffusion current only under zero bias (thermal equilibrium) condition. The hole diffusion current is given by

$$J_{pdiff} = qD_p\frac{dp}{dx}$$

Assuming the depletion width to be 1 μm, and the hole concentration differences to be about 10^{18} cm^{-3} across the depletion region, $J_{pdiff} \approx 10^4$ A/cm^2, which is a large current density. This is the junction is the difference between the hole drift current and the hole diffusion current. Since the balanced by an equally large hole drift current. Under bias conditions, the hole current flowing across individual components under low injection levels, even under bias conditions we can write

$$q\mu_p\mathcal{E}\cdot n \approx qD_p\frac{dn}{dx} \tag{4.38a}$$

Similarly, for electrons,

$$q\mu_n\mathcal{E}\cdot n \approx qD_n\frac{dn}{dx} \tag{4.38b}$$

Since $\mathcal{E} = -d\phi/dx$, from Eq. (4.38a) and using Einstein's relation, we have

$$\frac{dp}{p} = -\frac{V_T}{D_p}\frac{d\phi}{dx} \qquad \because\ \frac{D_p}{\mu_p} = V_T$$

Integrating between the edges of the depletion region, we can write

$$\int_0^{V_{bi}-V} d\phi = -V_T\int_{p_{po}}^{p_{no}}\frac{dp}{p} \tag{4.40}$$

where

p_{po} = hole concentration at the edge of depletion region in the p-region and

p_{no} = hole concentration at the edge of depletion region in the n-region as shown in Fig. 4.9.

From Eq. (4.40), we have

$$p_{no} = p_{po}\exp\left(-\frac{V_{bi}-V}{V_T}\right) \tag{4.41}$$

Substituting for V_{bi} from Eq. (4.4) into Eq. (4.41), we get

$$p_{no} = \frac{p_{no}p_{a0}}{p_{p0}}\exp\left(\frac{V}{V_T}\right) \tag{4.42}$$

According to assumption (d), low level injection condition prevails. This means that the majority carrier concentration does not deviate much from its equilibrium value. Hence, $p_{po} \approx p_{p0}$. Equation (4.42) can thus be rewritten as

$$p_{no} = p_{n0}\exp\left(\frac{V}{V_T}\right) \tag{4.43a}$$

By analogy, for electrons in the p-type semiconductor, the same relation can be represented as

$$n_{po} = n_{p0}\exp\left(\frac{V}{V_T}\right) \tag{4.43b}$$

We now take a critical look at Fig. 4.9. It can be seen that when a forward bias is applied, the minority carrier concentrations at the edges of the depletion region (p_{no} and n_{po}) increase exponentially by a factor $\exp(V/V_T)$ from their equilibrium values. In other words, there is *minority carrier injection* due to forward bias. On the other hand, when a reverse bias is applied, p_{no} and n_{po} decrease exponentially as $\exp(-V/V_T)$. This is because the minority carriers which are close to the space charge region, diffuse to the space charge region and are then swept across by the strong electric field to the other side of the junction, resulting in *minority carrier extraction*. It may be mentioned here that along with the change in the minority carrier concentration, there is a corresponding change in the majority carrier concentration in the neutral regions, thereby maintaining charge neutrality. However, under low level injection condition, the change in the majority carrier concentration is negligibly small.

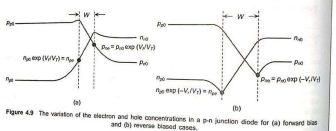

Figure 4.9 The variation of the electron and hole concentrations in a p-n junction diode for (a) forward bias and (b) reverse biased cases.

The situation in the n-region is quite similar to the illustrative example in section 3.7, where the semiconductor was illuminated to raise the minority carrier concentration at the surface. In this case, in a forward biased junction, instead of illumination, minority carriers are injected from the p-region to increase the hole concentration at the surface adjacent to the space charge region. Similar to the case detailed in section 3.7, the excess carriers diffuse into the bulk due to the concentration gradient. Neglecting minority carrier drift current (assumption b), the steady state continuity equation in the n-region is given by [refer to Eq. (3.37)].

$$\frac{d^2p_n}{dx^2} - \frac{(p_n - p_{n0})}{L_p^2} = 0 \tag{4.44}$$

Since the hole concentration at $x = x_n$ is given by Eq. (4.43a) and the hole concentration reduces due to recombination to the thermal equilibrium value far away from the junction, the boundary conditions for an infinitely long diode are:

(a) $p_n(x_n) = p_{no} = p_{n0}\exp\left(\frac{V}{V_T}\right)$ at $x = x_n$

(b) $p_n(\infty) = p_{n0}$ at $x = \infty$

Now solving Eq. (4.44) following the same procedure as in Chapter 3 section 3.7, we obtain

$$p_n - p_{n0} = p_{n0}\left[\exp\left(\frac{V}{V_T}\right) - 1\right]\exp\left(-\frac{x - x_n}{L_p}\right) \tag{4.45a}$$

From Eq. (4.45a) we see that the hole concentration in the n-region ($x > x_n$) decays exponentially as we move away from the junction. Similarly, the electron concentration in the p-region ($x < -x_p$) also decays exponentially and this can be represented as

$$n_p - n_{p0} = n_{p0}\left[\exp\left(\frac{V}{V_T}\right) - 1\right]\exp\left(\frac{x + x_p}{L_n}\right) \tag{4.45b}$$

Figure 4.9 shows the variation of the electron and hole concentrations in the diode for both forward and reverse biased conditions. Now the hole diffusion current at the edge of the depletion layer is given by

$$I_p(x_n) = -AqD_p\left.\frac{dp_n}{dx}\right|_{x=x_n} = \frac{AqD_pp_{n0}}{L_p}\left[\exp\left(\frac{V}{V_T}\right) - 1\right] \tag{4.46a}$$

Similarly, the electron diffusion current at $x = -x_p$ can be obtained as

$$I_n(-x_p) = AqD_n\left.\frac{dn_p}{dx}\right|_{x=-x_p} = \frac{AqD_nn_{p0}}{L_n}\left[\exp\left(\frac{V}{V_T}\right) - 1\right] \tag{4.46b}$$

According to our assumption (c), there is no generation or recombination in the depletion region. Therefore, the two current components as derived in Eq. (4.46a) and (4.46b) remain constant throughout the depletion layer. The sum of these two components gives the total current flowing through the p-n junction as

$$I = I_p(x_n) + I_n(-x_p) = Aq\left(\frac{D_pp_{n0}}{L_p} + \frac{D_nn_{p0}}{L_n}\right)\left[\exp\left(\frac{V}{V_T}\right) - 1\right] \tag{4.47}$$

Therefore, the diode current can be expressed as

$$I = I_0\left[\exp\left(\frac{V}{V_T}\right) - 1\right] \tag{4.48}$$

where I_0 is called the *reverse saturation current* of the diode and it is given by

$$I_0 = Aq\left(\frac{D_pp_{n0}}{L_p} + \frac{D_nn_{p0}}{L_n}\right) = Aqn_i^2\left(\frac{D_p}{L_pN_D} + \frac{D_n}{L_nN_A}\right) \tag{4.49}$$

Figure 4.10 plots the current–voltage characteristics of the diode as given by Eq. (4.48). Equation (4.48) shows that when the diode is forward biased (that is, $V > 0$), the current increases exponentially with applied voltage while when it is reverse biased (that is, $V < 0$), the current is limited to $I \approx -I_0$. This property of the diode is used in several applications.

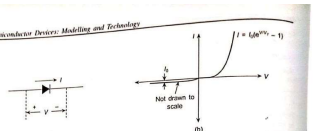

Figure 4.10 Current–voltage characteristics of a p-n junction diode.

For a one-sided abrupt p$^+$n junction $p_{n0} \gg n_{p0}$, and hence in the expression of I_0 given by Eq. (4.49), only the first term dominates. Thus for such a situation, the reverse saturation current can be approximated as

$$I_0 = \frac{AqD_pp_{n0}}{L_p} = \frac{AqD_pn_i^2}{L_pN_D} \tag{4.50}$$

An interesting point to be noted is that although the current in a p$^+$n junction is primarily due to holes injected from the heavily doped p-region into the n-region, the actual magnitude of the current is independent of the doping concentration in the p-region but depends on the doping concentration of the lightly doped n-region. Decreasing the donor concentration in the n-region increases reverse saturation current I_0 in such a junction.

Figure 4.11 plots the hole and electron currents flowing through a forward biased diode as a function of x. Using Eq. (4.45), we can write the expressions for the minority carrier diffusion current in the neutral n-region ($x > x_n$) as

$$I_p(x) = -AqD_p\frac{dp_n}{dx} = I_p(x_n)\exp\left(-\frac{x - x_n}{L_p}\right) \tag{4.51a}$$

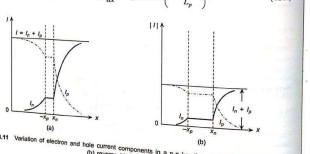

Figure 4.11 Variation of electron and hole current components in a p-n junction under (a) forward bias and (b) reverse bias conditions.

Similarly, the minority carrier diffusion current in the neutral p-region ($x < -x_p$) is expressed as

$$I_n(x) = AqD_n\frac{dn_p}{dx} = I_n(-x_p)\exp\left(-\frac{x + x_p}{L_n}\right) \tag{4.51b}$$

From the above two equations, it is obvious that the components of the minority carrier diffusion current decay exponentially away from the junction. However, since the total current flowing in a diode must remain constant, the decay in the minority carrier current is compensated by a corresponding increase in the majority carrier current as shown in Fig. 4.11(a). Thus, for away from the junction, in the n-region, I_p falls to zero and the current is carried entirely by electrons flowing from the n-contact towards the junction. To reiterate, let us trace the flow of electrons from the battery move towards the junction as a drift current with the aid of a small electric field in the neutral n-region. The field required to cause this flow is very small as electrons are the majority carriers. It has therefore been neglected in our analysis as mentioned in assumption (b). As the electrons flow towards the junction, they partially recombine with the holes moving in the opposite direction resulting in a decrease in electron current. (Since electrons and holes recombine in pairs, the decrease in electron current is equal to the decrease in hole current in the n-region. This ensures that the sum of the electron and hole currents is always constant). The electrons, which reach the junction, are injected into the p-region as a minority carrier diffusion current. Far away from the junction, all the electrons recombine, reducing the electron current to zero. Similarly, at the p-contact, the entire current is carried by holes, and this current ultimately reduces to zero as the holes move across the junction towards the n-contact.

Figure 4.11(b) shows the variation of electron and hole current densities for a reverse biased diode. In this case, since the minority carrier concentrations in the neutral regions are reduced below the thermal equilibrium values, there is excess generation of carriers. The electrons, which are generated in the p-region of the junction, diffuse towards the junction and are swept across it by the electric field in the space charge regions. The electron current increases further in the n-region due to generation, as these carriers flow towards the positive terminal of the battery connected to the n-contact. Similarly, the holes flow from the n-region to the p-region.

Physically, one can now understand why the forward current is much larger than the reverse bias current. Under forward bias conditions, the hole current injected from the p-region (where holes are majority carriers) into the n-region is proportional to $p_{n0}[\exp(V/V_T) - 1]$, which can be very large. On the other hand, under reverse bias conditions, holes are injected from the n-region (where they are minority carriers) to the p-region. The current, in this case, depends on the generation rate of carriers, which is proportional to p_{n0}. This explains why the reverse current is almost independent of the reverse bias voltage and is equal to I_0, while the forward bias current is higher by a factor $[\exp(V/V_T) - 1]$.

HELP DESK 4.2

Although the excess holes in the neutral n-region are continuously recombining, how is it that the steady state minority carrier concentration distribution curve shown in Fig. 4.9 does not change with time?

Let us first calculate the total minority carrier recombination rate, that is, the number of holes recombining per unit time, in the neutral n-region. Let (Q_p/τ_p) be the total excess hole concentration

in the neutral n-region. Now since τ_p is the average lifetime of these holes, the total recombination rate R is given by (Q_p/τ_p). That is,

$$R = \frac{Q_p}{\tau_p} = \frac{A}{\tau_p}\int_{x_n}^{\infty}(p_n - p_{n0})\,dx = \frac{A}{\tau_p}\int_{x_n}^{\infty}p_{n0}\left[\exp\left(\frac{V}{V_T}\right) - 1\right]\exp\left(-\frac{x - x_n}{L_p}\right)dx$$

$$= \frac{Ap_{n0}L_p}{\tau_p}\left[\exp\left(\frac{V}{V_T}\right) - 1\right]$$

On the other hand, the injected hole current is given by

$$I_p(x_n) = \frac{AqD_pp_{n0}}{L_p}\left[\exp\left(\frac{V}{V_T}\right) - 1\right] = \frac{AqD_pL_pp_{n0}}{\tau_p}\left[\exp\left(\frac{V}{V_T}\right) - 1\right] = \frac{Aqp_{n0}L_p}{\tau_p}\left[\exp\left(\frac{V}{V_T}\right) - 1\right]$$

Thus, we see that the number of holes injected into the n-region per unit time given by $I_p(x_n)/q$ is exactly equal to the number of holes recombining per unit time, thereby maintaining the steady state profile. A similar analysis can also be carried out for electrons in the p-region.

4.3.2 Quasi-Fermi Levels Under Bias Condition

Let us now refer to Fig. 4.6 and take another look at the energy band diagram with application of bias. As there is no change in the majority carrier concentration, the quasi-Fermi level for holes E_{Fp} deep inside the neutral p-region and the quasi-Fermi level for electrons E_{Fn} deep inside the neutral n-region are essentially at the equilibrium value and remain constant. However, at the edges of the depletion layer, the minority carrier concentration deviates from its equilibrium value by a factor $\exp(V/V_T)$ so that at the edges of the depletion region

$$pn = n_i^2\exp\left(\frac{V}{V_T}\right)$$

Referring to Eq. (1.51), this implies a separation between E_{Fp} and E_{Fn} on either side of the junction by an amount qV, as shown in Fig. 4.6(a) and (b). It can be easily seen that under forward bias, the quasi-Fermi level for electrons (E_{Fn}) is above that for holes (E_{Fp}) by a value qV_p and under reverse bias E_{Fn} is higher than E_{Fp} by qV_r. As the minority carriers decay away from the depletion layer edge towards the bulk and approach their equilibrium values, E_{Fp} and E_{Fn} come closer and ultimately merge. Also, in the absence of significant generation–recombination, E_{Fp} and E_{Fn} remain constant throughout the depletion region.

4.3.3 Current–Voltage Relation in Practical Diodes Having Finite Length

For a practical diode with finite length, the boundary conditions change and the equations as derived in the previous section have to be modified. Also the continuity equation given by Eq. (4.44) now needs to be solved subject to these new boundary conditions. While the first boundary condition specifying the carrier concentration at the edge of the depletion region remains the same, the second boundary condition has to be changed to take into account the finite length of the neutral regions. We have already discussed the role of surface recombination in section 1.4.4. The surface recombination velocities at the metal–semiconductor contacts are very

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high resulting in high recombination rates. Consequently, the excess carrier concentration reduces to zero at these contacts. The boundary conditions to find out the solution of Eq. (4.44) in the case of a diode with an n-region of finite length are therefore,

(a) $p_n(x_n) = p_{no} = p_{n0}\exp\left(\frac{V}{V_T}\right)$ at $x = x_n$

(b) $p_n(W_n) = p_{n0}$ at $x = W_1$

where $W_1 = x_n + W_n$. W_n being the width of the neutral n-region.

The solution of Eq. (4.44) now has the form

$$p_n - p_{n0} = p_{n0}\left[\exp\left(\frac{V}{V_T}\right) - 1\right]\frac{\sinh\left(\frac{W_1 - x}{L_p}\right)}{\sinh\left(\frac{W_1 - x_n}{L_p}\right)} \tag{4.52}$$

Equation (4.52) shows the variation of the minority carrier distribution in the n-region. From Eq. (4.52), it can be shown that when $(W_1/L_p) \gg 1$, that is the width of the n-layer is much larger than the diffusion length, the expression for $(p_n - p_{n0})$ reduces to Eq. (4.45) for an infinitely long n-region. This indicates an exponential decay of carriers away from the junction. Another very interesting situation is encountered for a short diode. For $y \ll 1$, $\sinh(y) \approx y$. So for $W_n \ll L_p$, Eq. (4.52) reduces to

$$p_n - p_{n0} = \frac{p_{n0}(W_1 - x)}{W_n}\left[\exp\left(\frac{V}{V_T}\right) - 1\right] \tag{4.53}$$

In other words, the minority carrier concentration approaches a linear distribution instead of an exponential distribution. The nature of variation of the minority carrier concentration for various (W_n/L_p) values is shown in Fig. 4.12.

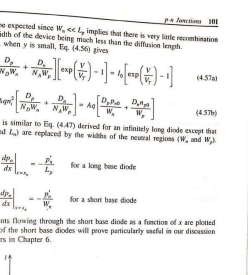

Figure 4.12 Hole concentration variation in the neutral n-region of a p-n junction diode for various (W_n/L_p) ratios.

The hole diffusion current in the neutral n-region can now be expressed as

$$I_p(x) = -AqD_p\frac{dp_n}{dx} = \frac{AqD_pp_{n0}}{L_p}\left[\exp\left(\frac{V}{V_T}\right) - 1\right]\frac{\cosh\left(\frac{W_1 - x}{L_p}\right)}{\sinh\left(\frac{W_1 - x_n}{L_p}\right)} \tag{4.54}$$

So, the hole diffusion current at the edge of the depletion layer in the n-region is given by

$$I_p(x_n) = -AqD_p\left.\frac{dp_n}{dx}\right|_{x=x_n} = \frac{AqD_pp_{n0}}{L_p}\left[\exp\left(\frac{V}{V_T}\right) - 1\right]\coth\left(\frac{W_n}{L_p}\right) \tag{4.55a}$$

Similarly, if the p-region is also of finite length, the electron diffusion current at the edge of the depletion layer in the p-region is given by

$$I_n(-x_p) = \frac{AqD_nn_{p0}}{L_n}\left[\exp\left(\frac{V}{V_T}\right) - 1\right]\coth\left(\frac{W_p}{L_n}\right) \tag{4.55b}$$

where W_p = width of the neutral p-region.

The total diode current is now given by adding these two components as

$$I = Aq\left[\frac{D_pp_{n0}\coth\left(\frac{W_n}{L_p}\right)}{L_p} + \frac{D_nn_{p0}\coth\left(\frac{W_p}{L_n}\right)}{L_n}\right]\left[\exp\left(\frac{V}{V_T}\right) - 1\right]$$

$$= Aqn_i^2\left[\frac{D_p\coth\left(\frac{W_n}{L_p}\right)}{N_DL_p} + \frac{D_n\coth\left(\frac{W_p}{L_n}\right)}{N_AL_n}\right]\left[\exp\left(\frac{V}{V_T}\right) - 1\right] \tag{4.56}$$

Now let us look at two extreme cases:

Case 1

Long diode (also called *long base diode*): For $y > 4$, $\coth(y) \approx 1$. Therefore if $W_n > 4L_p$ and $W_p > 4L_n$, Eq. (4.56) reduces to the ideal diode equation given by Eq. (4.47). So the ideal diode equation is said to be valid for a long diode satisfying the above restrictions regarding the widths of the n- and p-regions.

Case 2

Short diode (also called *short base diode*): We have already seen that the minority carrier concentration distribution in the neutral region of a short diode shows a linear variation. Since the slope of this curve is constant, the minority carrier diffusion current in the neutral region also remains constant. Since $\cosh(y) \approx 1$ when y is small, it can be seen from Eq. (4.54) that $I_p(x)$ is a constant

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when $W_n \ll L_p$. This is to be expected since $W_n \ll L_p$ implies that there is very little recombination in the neutral region, the width of the device being much less than the diffusion length.

Since $\coth(y) = (1/y)$, when y is small, Eq. (4.56) gives

$$I = Aqn_i^2\left[\frac{D_p}{N_DW_n} + \frac{D_n}{N_AW_p}\right]\left[\exp\left(\frac{V}{V_T}\right) - 1\right] = I_0\left[\exp\left(\frac{V}{V_T}\right) - 1\right] \tag{4.57a}$$

where

$$I_0 = Aqn_i^2\left[\frac{D_p}{N_DW_n} + \frac{D_n}{N_AW_p}\right] = Aq\left[\frac{D_pp_{n0}}{W_n} + \frac{D_nn_{p0}}{W_p}\right] \tag{4.57b}$$

We also see that Eq. (4.57) is similar to Eq. (4.47) derived for an infinitely long diode except that the diffusion lengths (L_p and L_n) are replaced by the widths of the neutral regions (W_n and W_p). This is because

$$\left.\frac{dp_n}{dx}\right|_{x=x_n} = -\frac{p_n'}{L_p} \qquad \text{for a long base diode}$$

while

$$\left.\frac{dp_n}{dx}\right|_{x=x_n} = -\frac{p_n'}{W_n} \qquad \text{for a short base diode}$$

The electron and hole currents flowing through the short base diode as a function of x are plotted in Fig. 4.13. This property of the short base diodes will prove particularly useful in our discussion of bipolar junction transistors in Chapter 6.

Figure 4.13 Electron and hole current variation in a p-n junction diode with short p- and n-regions.

HELP DESK 4.3

We have seen (refer HELP DESK 4.2) that $I_p(x = x_n) = Q_p/\tau_p$ in the neutral n-region of a long base diode. Is this also valid for a short base diode?

For a long base diode, all the holes injected into the n-region recombine before reaching the n-contact, and Q_p/τ_p represents the recombination current in the neutral base region. However in the case of a short base diode, a hole current exists even at the n-contact. Since the concentration of holes together with the majority carrier electrons at the n-contact. Since the surface recombination velocities are extremely large, there will be a large recombination current at the n-contact surface. Since the holes recombining at the surface, in addition to those recombining in the bulk, given by Q_p/τ_p, are supplied by the injected hole current, I_p ($x = x_n$) will not be equal to Q_p/τ_p in a short base diode. It has been left as a problem at the end of the chapter to show that $I_p/\tau_p > Q_p/\tau_p$ for a short base diode.

Example 4.5

For an abrupt p$^+$n silicon diode, the doping concentration in the n-region is 10^{16} per cm^3. The width of the n-region is 2 μm. Assuming this width is much smaller than the hole diffusion length in the n-region, calculate the reverse saturation current at (i) 300 K and (ii) 400 K if the area of the diode is 100 μm × 100 μm. Assume that the hole mobility in the n-region is 350 cm^2/Vs at both the temperatures.

Solution: From Eq. (4.57), we know that for a short diode

$$I_0 = Aqn_i^2\left[\frac{D_p}{N_DW_n} + \frac{D_n}{N_AW_p}\right]$$

In a p$^+$n diode, since $N_A \gg N_D$,

$$I_0 \approx Aqn_i^2\left[\frac{D_p}{N_DW_n}\right]$$

Also we know from Einstein's relation, $D_p = \mu_pV_T$

At 300 K,

$D_p = 0.026 \times 350$ cm^2/s $= 9.1$ cm^2/s and $n_i = 1.5 \times 10^{10}$ per cm^3

Therefore,

$$I_0 = \frac{(100 \times 10^{-4})^2 \times 1.6 \times 10^{-19} \times 9.1 \times (1.5 \times 10^{10})^2}{2 \times 10^{-4} \times 10^{16}} = 1.638 \times 10^{-14}\text{ A}$$

At 400 K,

$D_p = 0.035 \times 350$ cm^2/s $= 12.25$ cm^2/s and $n_i = 5.185 \times 10^{12}$ per cm^3 (see Ex. 1.1)

Therefore,

$$I_0 = \frac{(100 \times 10^{-4})^2 \times 1.6 \times 10^{-19} \times 12.25 \times (5.185 \times 10^{12})^2}{2 \times 10^{-4} \times 10^{16}} = 2.635 \times 10^{-8}\text{ A}$$

So, we see that for 100°C rise in temperature, the reverse saturation current increases by five orders of magnitude.

HELP DESK 4.4

Compare the plots of the forward current–voltage characteristics of Ge, Si, and GaAs p$^+$n diodes having identical structures and doping concentrations.

For two materials, say material 1 and material 2, having band gap energies E_{g1} and E_{g2} respectively, the ratio of their intrinsic carrier concentrations n_{i1} and n_{i2} is given by [from Eq. (1.24)]

$$\frac{n_{i1}}{n_{i2}} = \frac{\sqrt{N_{C1}N_{V1}}\exp\left[-\left(\frac{E_{g1}}{2kT}\right)\right]}{\sqrt{N_{C2}N_{V2}}\exp\left[-\left(\frac{E_{g2}}{2kT}\right)\right]} = \sqrt{\frac{N_{C1}N_{V1}}{N_{C2}N_{V2}}}\exp\left(\frac{E_{g2} - E_{g1}}{2kT}\right)$$

Assuming that the differences in the values of N_C and N_V are small,

$$\frac{n_{i1}^2}{n_{i2}^2} = \exp\left(\frac{E_{g2} - E_{g1}}{kT}\right)$$

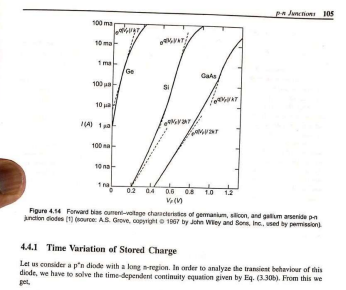
Considering the band gaps given in Table 1.3,

$$\frac{(n_i^2)_{Ge}}{(n_i^2)_{Si}} \approx \exp\left(\frac{1.12 - 0.66}{0.026}\right) = 4.83 \times 10^7$$

Similarly,

$$\frac{(n_i^2)_{Si}}{(n_i^2)_{GaAs}} \approx \exp\left(\frac{1.42 - 1.12}{0.026}\right) = 1.03 \times 10^5$$

Since other terms in the expression for I_0 [Eq. (4.57b)] have much smaller dependence on the material, we can say that $(I_0)_{Ge} >> (I_0)_{Si} >> (I_0)_{GaAs}$ and this gives the following I–V characteristics.



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4.3.4 Ideality Factor of a p-n Junction Diode

We have seen in the preceding section that the current in a p-n junction diode can be expressed by Eq. (4.48). For sufficiently large forward bias voltage ($V >> V_T$), the diode current can be further approximated as

$$I \approx I_0\exp\left(\frac{V}{V_T}\right) \tag{4.58a}$$

In practice, however, the characteristics of a diode follow a slightly modified equation which is given by

$$I = I_0\exp\left(\frac{V}{mV_T}\right) \tag{4.58b}$$

where m is called the *ideality factor* of the diode (its value lies between 1 and 2).

This difference between a practical diode and an ideal diode is present because we have neglected the recombination within the space charge layer (assumption (c)) in our analysis. In a practical diode, because of recombination within the depletion layer, there is another component of current given by[1]

$$I_{rec} = \frac{Aqn_iW}{2\tau}\exp\left(\frac{V}{2V_T}\right) \tag{4.59}$$

Thus, for a p$^+$n junction under forward bias, the diode current is actually the sum of I_{rec} (given by Eq. (4.59)) and I (given by Eq. (4.58a)). The ratio I_{rec}/I signifying the relative dominance of these two currents then determines the value of m. For an abrupt p$^+$n junction, from Eqs. (4.58a), (4.50), and (4.59) we have

$$\frac{I_{rec}}{I} = \frac{WN_D}{2L_pn_i}\exp\left(-\frac{V}{2V_T}\right) \tag{4.60}$$

We see from Eq. (4.60) that for large forward biases, that is $V >> (2V_T)$, $I_{rec} << I$, and therefore $m \approx 1$. However, when the forward bias is reduced, the semiconductor material properties as well as the operating temperature play an important role in determining this ratio. For example, in a germanium diode n_i is large (2.4×10^{13} per cm^3 at 300 K) and hence I_{rec}/I is small. The diode current is therefore dominated by the diffusion current and $m \approx 1$. On the other hand, n_i is small for GaAs (1.79×10^6 per cm^3 at 300 K) and therefore I_{rec}/I is large and $m \approx 2$. For silicon diodes, at room temperature, up to a moderate forward bias of 0.4–0.5 V,

Physically, Eq. (4.62) states that the hole current injected across the p⁺n junction (and therefore, approximately the total diode current) is related to two charge storage effects. The first is the usual recombination term Q_p/τ_p. The second term is $\partial Q_p/\partial t$, which represents the fact that the stored recombination charge Q_p can either increase or decrease with time. In steady state, when distribution of excess charge can either increase or decrease with time. However, injection of holes may result in an increase in the excess hole charges stored in the neutral n-region, if the rate of influx (the injected current) is greater than the recombination rate. On the other hand, there is a reduction in the stored charge (that is, $\partial Q_p/\partial t$ is negative), if the rate of inflow is less than the recombination rate. Also, the rate of increase of charge is equal to difference of the rate at which charges flow in and the rate at which the carriers recombine. More simply, if X be the rate at which carriers flow in and Y be the rate at which carriers recombine per unit time, then $X-Y$ gives the rate of increase in the number of carriers.

$Q_p(t) = 0$, the junction voltage has to fall to zero, τ_p must be small. This can be achieved by deliberately introducing gold or platinum into the silicon diode as discussed in Section 1.4.3. Gold and platinum create deep levels in the band gap which act as recombination centres and effectively kill τ_p. Alternatively for fast switching diodes, the device geometry may also be modified by designing a p⁺n diode with a very narrow base (n-region). If the n-region width is less than the hole diffusion length, very few charges can be stored and hence, a very short time is required to switch the diode off and on.

Now, for a given current transient, the stored charge can be obtained as a function of time using Eq. (4.62). The following example clarifies the concept to a greater extent. Let us assume that a constant current I is flowing through a diode. Since the diode is at a steady state, $\partial Q_p/\partial t = 0$, and from Eq. (4.62), we have $Q_p = I\tau_p$. At time $t = 0$, the connection to the battery is open-circuited so that the current suddenly becomes zero as shown in Fig. 4.15(a). Since the excess holes in the n-region must die out by recombination, some time is needed for Q_p to become zero. Equation (4.62) now reduces to

$$\frac{\partial Q_p}{\partial t} = -\frac{Q_p(t)}{\tau_p} \tag{4.63}$$

Solving Eq. (4.63) using the initial condition $Q_p(0) = I\tau_p$, we get

$$Q_p(t) = I\tau_p \exp\left(-\frac{t}{\tau_p}\right) \tag{4.64}$$

In other words, as shown in Fig. 4.15(b), the stored charge dies exponentially with a time constant τ_p that is, the lifetime of a hole in the n-region.

An important point to be noted in the above example is that although the diode current has suddenly fallen to zero, the voltage across the junction does not immediately do so. Since the excess hole concentration at the edge of the depletion region p'_{no} is related to the junction voltage, we can write using Eq. (4.43a).

$$p'_{no} = p_{n0}\left[\exp\left(\frac{V}{V_T}\right) - 1\right] \tag{4.65}$$

Therefore from Eq. (4.65), we have

$$V(t) = V_T \ln\left[\frac{p'_{no}(t)}{p_{n0}} + 1\right] \tag{4.66}$$

If we know $p'_{no}(t)$, we can find the junction voltage as a function of time. Unfortunately, it is not so simple. Since $I = 0$, the slope of $p(x)$ must be zero at $x = x_n$ for $t > 0$ (refer Eq. (4.46a)) and x. However, neglecting this minor variation and expressing the stored charge at any instant as given by Eq. (4.45), we have

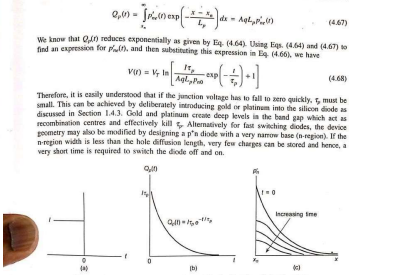

Figure 4.15 (a) The time variation of current flowing through a p⁺n junction diode (b) the resultant stored charge variation, and (c) charge distribution variation in the neutral n-region.

4.4.2 Reverse Recovery of a Diode

In most switching applications, the diode has to be switched from forward conduction to the reverse-biased state and back. As we shall see, in such a situation, a reverse current much larger than the reverse saturation current (I_0) can flow through the diode for a short time. Let us consider the diode circuit shown in Fig. 4.16(a) and assume that the input voltage V_a varies as shown in Fig. 4.16(b). At $t < t_1$, $V_a = V_F$ and in steady state, the current I_F flowing through the circuit is limited by the resistor R. Neglecting the small forward voltage drop V_D of the diode, $I = I_F = (V_F - V_D)/R \approx V_F/R$. At $t > t_1$, the applied voltage becomes $-V_R$. However, as already pointed out, the stored charge and hence the junction voltage V_D cannot change immediately as shown in Figs. 4.16(c) and 4.16(e). So just as the voltage is reversed, the diode junction voltage remains at the same small forward bias value. Consequently, a large reverse current $I = -I_F = -V_R/R \approx -V_R/R$ flows temporarily as shown in Fig. 4.16(d). As the stored charge is depleted with time, the junction voltage reduces. So long as Q_p is positive, the junction voltage has a small

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$$Q_p(t) = \int_{x_n}^{\infty} p'_{no}(t)\exp\left(-\frac{x - x_n}{L_p}\right)dx = AqL_p p'_{no}(t) \tag{4.67}$$

We know that $Q_p(t)$ reduces exponentially as given by Eq. (4.64). Using Eqs. (4.64) and (4.67) to find an expression for $p'_{no}(t)$, and then substituting this expression in Eq. (4.66), we have

$$V(t) = V_T \ln\left[\frac{I\tau_p}{AqL_p p_{n0}}\exp\left(-\frac{t}{\tau_p}\right) + 1\right] \tag{4.68}$$

Therefore, it is easily understood that if the junction voltage has to fall to zero quickly, τ_p must be small. This can be achieved by deliberately introducing gold or platinum into the silicon diode as discussed in Section 1.4.3. Gold and platinum create deep levels in the band gap which act as recombination centres and effectively kill τ_p. Alternatively for fast switching diodes, the device geometry may also be modified by designing a p⁺n diode with a very narrow base (n-region). If the n-region width is less than the hole diffusion length, very few charges can be stored and hence, a very short time is required to switch the diode off and on.

positive value. The current remains approximately constant at $I = -V_R/R$ till Q_p falls to zero. However, once Q_p becomes negative, the junction voltage also becomes negative. As the source voltage begins to be divided between the diode and the resistor, the magnitude of the current reduces. Finally, almost all the applied voltage is dropped across the diode and the current in the circuit becomes equal to I_R. The time required for the rate of removal of stored charge depends on the carrier lifetime.

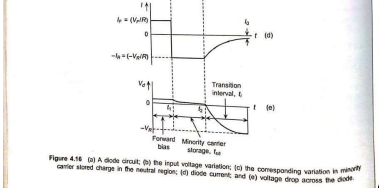


Figure 4.16 (a) A diode circuit; (b) the input voltage variation; (c) the corresponding variation in minority carrier stored charge in the neutral region; (d) diode current; and (e) voltage drop across the diode.

We now proceed to obtain an expression for t_{sd} in terms of τ_p and the diode current by solving Eq. (4.62) for this particular situation. We note that at $t = t_1$, the diode current becomes $-I_R$ and remains constant till $t = t_1 + t_{sd}$. Substituting in Eq. (4.62), for $t_1 < t < t_1 + t_{sd}$, we have

$$-I_R = \frac{Q_p(t)}{\tau_p} + \frac{dQ_p(t)}{dt} \tag{4.69}$$

The solution of this equation [Eq. (4.69)], is of the form

$$Q_p(t) = C_1\exp\left(-\frac{t}{\tau_p}\right) - I_R\tau_p$$

where C_1 is the constant of integration (evaluated from the initial boundary condition).

We know that since a constant current I_F was flowing through the diode till the time $t = t_1$, the stored charge Q_p at $t = t_1$ is $I_F\tau_p$. Thus, we can write the solution of Eq. (4.69) as

$$Q_p(t) = I_F\tau_p\exp\left(\frac{t - t_1}{\tau_p}\right) + I_R\tau_p\left[\exp\left(-\frac{t - t_1}{\tau_p}\right) - 1\right] \tag{4.70}$$

Now, substituting Eq. (4.67) in Eq. (4.70), we have

$$p'_{no}(t) = \frac{1}{AqL_p}\left[(I_F + I_R)\tau_p\exp\left(\frac{t - t_1}{\tau_p}\right) - I_R\tau_p\right] \tag{4.71}$$

By the definition of the storage delay time, $p'_{no}(t) = 0$ at $t = t_1 + t_{sd}$. Therefore, from Eq. (4.71), we get

$$t_{sd} = \tau_p\ln\left(1 + \frac{I_F}{I_R}\right) \tag{4.72}$$

Thus, the presence of a large reverse current reduces t_{sd}. This is to be as expected since the reverse current helps to remove the excess stored charge and I_R is the rate at which excess holes are removed. Also, t_{sd} is higher for higher I_F, as in that case the initial stored charge itself is more.

4.4.3 Charge Storage Capacitance

There are essentially two capacitances associated with a junction. One of them is due to the charge dipole at the depletion layer (depletion capacitance) and has already been discussed in sections 4.2.1 and 4.2.2. In addition, as pointed out already, the junction voltage lags behind the current due to charge storage and therefore we have another capacitance effect. This is termed the *charge storage capacitance* or the *diffusion capacitance* C_d. While the depletion capacitance dominates when the diode is in reverse biased condition, the diffusion capacitance becomes important when the junction is forward biased. To calculate this diffusion capacitance due to charge storage effects, let us consider a p⁺n junction with a long n-region at forward bias with a steady current I flowing through it. Assuming that the junction voltage $V > 3V_T$, from Eq. (4.67) the stored charge can be given as

$$Q_p = I\tau_p = \int_{x_n}^{\infty} p'_{no}\exp\left(-\frac{x - x_n}{L_p}\right)dx = AqL_p p'_{no} = AqL_p p_{n0}\exp\left(\frac{V}{V_T}\right) \tag{4.73}$$

The diffusion capacitance C_d is therefore given by

$$C_d = \frac{dQ}{dV} = \frac{AqL_p p_{n0}}{V_T}\exp\left(\frac{V}{V_T}\right) = \frac{I\tau_p}{V_T} \tag{4.74}$$

The small signal conductance of the diode ($G = dI/dV$), calculated by differentiating the ideal diode current equation (Eq. (4.48)), is given by

$$G = \frac{I}{V_T} \tag{4.75}$$

The diffusion capacitance can therefore also be expressed as

$$C_d = G\tau_p \tag{4.76}$$

An important point to note here is that C_d is directly proportional to the diode current.

The above equations are valid for long base diode. Let us now consider a p⁺n diode with a short n-region. We have already seen that for such a diode, the minority carrier concentration in the n-region shows a linear variation, as shown in Fig. 4.17. From this figure (Fig. 4.17), the excess minority carrier charge can be obtained from the shaded area under the curve as

$$Q_p = \frac{AqW_n p'_{no}}{2} \tag{4.77}$$

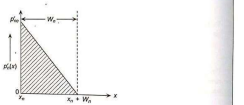


Figure 4.17 Variation of the minority carrier stored charge in the neutral n-region of a short base p⁺n diode.

while the current $I_p(x_n)$ is obtained from the slope of the concentration profile. This is found to be

$$I_p(x_n) = -AqD_p\left.\frac{dp_n}{dx}\right|_{x=x_n} = \frac{AqD_p p'_{no}}{W_n} \tag{4.78}$$

Now from Eqs. (4.77) and (4.78), we can write for a p⁺n diode

$$I \approx I_p(x_n) = \frac{Q_p}{\tau_t} \tag{4.79}$$

where $\tau_t = \frac{W_n^2}{2D_p}$ is called the *transit time* of the diode.

Using $Q_p = I\tau_t$ (instead of $Q_p = I\tau_p$) in Eq. (4.74), for a short base diode we obtain

$$C_d = \frac{I\tau_t}{V_T} = G\tau_t \tag{4.80}$$

In fact, Eq. (4.80) is the equation for the diffusion capacitance for a diode where $\tau_t = \tau_p$ for a long base diode and $\tau_t = W_n^2/2D_n$ for a short base diode.

HELP DESK 4.5

Why is τ_t called the transit time of the diode?

Let us consider the minority carrier profile in Fig. 4.17. If the time taken for the charges to move from the edge of the depletion region ($x = x_n$) to the n-contact ($x = W_n + x_n$) is τ, the total charge which reaches the contact in time τ is Q_p. Therefore, the charge reaching the contact per unit time (which is the current I), is Q_p/τ. Comparing with Eq. (4.79), we see that $\tau = \tau_t$. That is, τ_t can be considered to be the time required by the carriers to cross the n-region, and hence the name.

The small signal equivalent circuit of a diode is shown in Fig. 4.18. In this figure, the circuit components are G, C_j, and C_d given by Eqs. (4.75), (4.31), and (4.74) or (4.80) respectively.

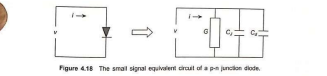


Figure 4.18 The small signal equivalent circuit of a p-n junction diode.

4.5 BREAKDOWN MECHANISMS

When a sufficiently large reverse voltage is applied to the diode, the p-n junction breaks down and conducts a large current. The breakdown process is not inherently destructive, that is, when the applied voltage is reduced the diode again behaves normally. However, the maximum current must be limited by connecting an external resistance R as shown in Fig. 4.19(a) to avoid excessive heating. If the breakdown voltage of the diode is V_{BR} as shown in Fig. 4.19(b), the maximum reverse current that can flow is

$$I_R = \frac{V - V_{BR}}{R} \tag{4.81}$$

It must be ensured that the product $I_R V_{BR}$ is less than the maximum power rating of the diode to avoid burn up. The reverse breakdown encountered in a p-n junction can occur by two mechanisms, each of which needs a critical electric field in the space charge region. One is called *zener breakdown*, which usually occurs at low reverse voltages (a few volts) and the other mechanism called *avalanche breakdown* is responsible for breakdown occurring at higher voltages.

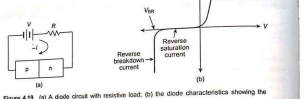


Figure 4.19 (a) A diode circuit with resistive load; (b) the diode characteristics showing the breakdown region.

4.5.1 Zener Breakdown

When two heavily doped p- and n-regions form a junction, then on application of reverse bias, the valence band of the p⁺ region can be aligned opposite to the conduction band of the n⁺ region as shown in Fig. 4.20. In other words, the conduction and valence band edges on either side of the junction may cross even at relatively low voltages. So in the p-region, there is a large density of filled states and in the n-region there are empty states in the conduction band at the same energy. If the barrier separating the p- and n-regions is narrow (the width of the space charge layer is small), tunnelling of electrons can occur from p-region to n-region causing an appreciable reverse current to flow. This is called the *zener effect*. Since the tunnelling probability depends on the width of the barrier, it is important that the barrier is abrupt and doping concentrations on both sides are high (> 5×10^{18} per cm³) for the zener effect to take place.

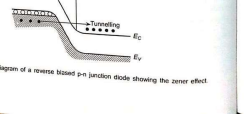


Figure 4.20 Band diagram of a reverse biased p-n junction diode showing the zener effect.

The zener effect can be explained as the field ionization of the host atoms at the junction. The applied reverse bias across a very narrow depletion region results in a large electric field inside the space charge region. The bonds are broken at the critical field strength as the valence electrons forming the covalent bonds are torn out by the field. The electrons and holes which are created in this process, move in opposite directions under the influence of the strong electric field in the space charge region, resulting in a large increase in current. The critical electric field for zener breakdown is of the order of 10^6 V/cm. Diodes, which are deliberately designed for particular zener breakdown voltages, are referred to as *zener diodes*.

4.5.2 Avalanche Breakdown

For lightly doped junctions, electron tunnelling is negligible. Instead, the breakdown in such cases involves impact ionization of atoms by energetic carriers. For example, if the electric field $\mathscr{E}$ inside the depletion region is large, an electron entering from the p-region may acquire sufficiently high energy to cause an ionizing collision with the lattice atom creating an EHP. A single such event results in carrier multiplication; the original electron as well as the generated (secondary) electron are swept to the n-region, while the generated hole is swept to the p-region as shown in Fig. 4.21. The degree of multiplication can become very high if the carriers generated within the depletion region also have ionizing collisions with the lattice. That is, an incoming carrier undergoes ionizing collision producing an EHP; each of these carriers creates a new EHP; again each new carrier produces an EHP, and the process continues. This is called an *avalanche* process since each incoming carrier can create a large number of new carriers resulting in an avalanche of carriers.

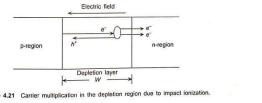


Figure 4.21 Carrier multiplication in the depletion region due to impact ionization.

We shall now make an approximate analysis of this avalanche multiplication of carriers. Let us assume that a carrier (electron or hole), while being accelerated through a depletion region of width W, has a probability P of creating an EHP by undergoing ionizing collision with the lattice. Then for n_0 incoming electrons entering the depletion region from the p-region, Pn_0 secondary EHPs will be generated. Now these generated electrons move to the n-region while the generated holes travel to the p-region under the electric field. As the total distance traversed by this EHP is still W, they in turn generate new EHPs with the same probability and consequently there are $(Pn_0)P$ tertiary electrons. In all, assuming no recombination, the total number of electrons coming out of the depletion region in the n-region is obtained as

$$n_{out} = n_0(1 + P + P^2 + P^3 + \cdots) \tag{4.82}$$

Therefore, the multiplication factor is given by

$$M = \frac{n_{out}}{n_{in}} = (1 + P + P^2 + P^3 + \cdots) = \frac{1}{1 - P} \tag{4.83}$$

The probability of an ionizing collision as a carrier travels through the depletion region can also be expressed as

$$P = \int_0^W \alpha\,dx \tag{4.84}$$

where α = ionization coefficient.

For the avalanche process to be self-sustaining, M should be infinite. That is,

$$P = \int_0^W \alpha\,dx = 1 \tag{4.85}$$

Now, α is a function of the electric field and can be expressed as

$$\alpha = a_0\exp\left[-\left(\frac{b}{\mathscr{E}}\right)^m\right] \tag{4.86}$$

where the constants a_0, b, and m are characteristics of the particular semiconductor.

It must be pointed out that our analysis is oversimplified and actually the ionization probability is related to the junction parameters in a much more complicated manner. Qualitatively, we can expect the ionization probability to increase with increasing electric field and therefore on the applied reverse bias. A widely used empirical relation between the multiplication factor M and the applied reverse voltage (V_R) near breakdown is given by

$$M = \frac{1}{\left[1 - \left(\frac{V_R}{V_{BR}}\right)^n\right]} \tag{4.87}$$

where n varies between 3 and 6 depending on the semiconductor material. In general, the critical reverse voltage for breakdown increases with increasing values of band gap, since more energy is required for ionization for larger bandgap materials.

From the breakdown conditions described so far in this section and the field dependence of the ionization coefficient, the critical electric field ($\mathscr{E}_c$) at which the avalanche process takes place and breakdown occurs, may be calculated as shown in Fig. 4.22. As can be seen from this figure, the value of the critical electric field is only weakly dependent on the doping concentration till tunnelling becomes the dominant breakdown mechanism. Now from Eq. (4.26a), noting that $|\mathscr{E}_m| = \mathscr{E}_c$ when $V = -V_{BR}$, the breakdown voltage for one-sided abrupt p⁺n junction is given by

$$V_{BR} = \frac{\mathscr{E}_c W}{2} = \frac{\mathscr{E}_c\varepsilon_s\mathscr{E}_c}{2qN_D} \tag{4.88}$$

From Eq. (4.88), we see that for junctions where breakdown takes place due to the avalanche process, the breakdown voltage increases with reduction in the doping concentration. The

avalanche breakdown voltage is also found to increase with temperature. This is because as the temperature increases, the scattering of the carriers increases. This energy loss obviously does not contribute to the ionization process. That means a larger voltage now has to be applied across the junction to achieve the condition for M to become infinite.

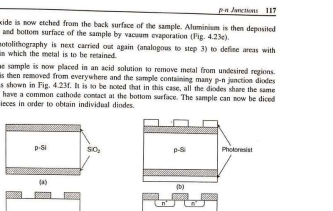


Figure 4.22 Critical field at breakdown versus doping concentration in Si and GaAs [2]

Example 4.6

In a p⁺nn⁺ junction diode, the doping concentration of the n-region is 2×10^{15} cm^{-3}. If the critical field at avalanche breakdown is $\approx 3 \times 10^5$ V/cm, find out

(a) The breakdown voltage if the width of the n-region is 10 μm.
(b) The breakdown voltage if the width of the n-region is 1 μm.
(c) Sketch the electric field distribution for both cases.

Solution: (a) For this structure, the depletion region exists essentially only in the n-region. At breakdown the peak electric field is $|\mathscr{E}_m| = \mathscr{E}_c \approx 2 \times 10^5$ V/cm. Substituting this value in Eq. (4.13) and assuming $x_n = W$, we get

$$W = \frac{2 \times 10^5 \times 11.9 \times 8.854 \times 10^{-14}}{1.6 \times 10^{-19} \times 2 \times 10^{15}}\ \text{cm} = 6.585\ \mu\text{m}$$

The breakdown voltage is now obtained using Eq. (4.88) as

$$V_{BR} = \frac{2 \times 10^5 \times 6.585 \times 10^{-4}}{2} = 65.85\ \text{V}$$

(b) Since, the width of the n-region is now 1 μm, the depletion region width is virtually restricted to 1 μm. The extension of depletion region in the heavily doped p⁺ and n⁺ regions will be negligible and hence can be ignored. As the slope of the electric field distribution in n-region is

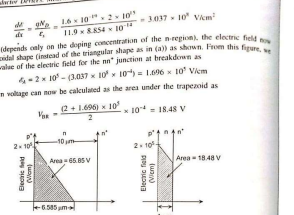
$$\frac{d\mathscr{E}}{dx} = \frac{qN_D}{\varepsilon_s} = \frac{1.6 \times 10^{-19} \times 2 \times 10^{15}}{11.9 \times 8.854 \times 10^{-14}} = 3.037 \times 10^9\ \text{V/cm}^2$$

is a constant (depends only on the doping concentration of the n-region), the electric field now takes a trapezoidal shape (instead of the triangular shape as in (a)) as shown. From this figure, we calculate the value of the electric field for the nn⁺ junction at breakdown as

$$\mathscr{E}_c = 2 \times 10^5 - (3.037 \times 10^9 \times 10^{-4}) = 1.696 \times 10^5\ \text{V/cm}$$

The breakdown voltage can now be calculated as the area under the trapezoid as

$$V_{BR} = \frac{(2 + 1.696) \times 10^5}{2} \times 10^{-4} = 18.48\ \text{V}$$



4.6 FABRICATION OF DISCRETE PLANAR p-n JUNCTION DIODES

In the previous sections, we have discussed various aspects of the p-n junction diodes. Let us now take a look at the fabrication of this device. The basic steps involved in the fabrication of a p-n junction diode using planar technology are outlined in Fig. 4.23. In this case, p-type silicon is taken as substrate.

Step 1: The wafer is cleaned to remove organic and/or metallic impurities as well as any trace of native oxide that may be present on the surface. Thermal oxidation is then carried out in order to grow a layer of SiO_2 (500–800 nm) on both top and bottom surfaces (Fig. 4.23a).

Step 2: Next, the photoresist is spin-coated and photolithography is carried out in order to realize a mask pattern on the top surface. The bottom surface is fully coated with photoresist (Fig. 4.23b).

Step 3: The sample is now placed in a solution containing HF. HF etches SiO_2 through the openings in the photoresist on the top surface. However, SiO_2 is not affected in the regions protected by the photoresist. Photoresist is then removed from everywhere in an organic solvent (acetone in case of positive photoresist) and the structure has "oxide windows" in the top surface as shown in Fig. 4.23c.

Step 4: Phosphorus diffusion is next carried out to produce n⁺-regions in the windows. As the diffusion coefficient of phosphorus in SiO_2 is small, phosphorus diffuses only through the windows, that is, in the regions no longer protected by the oxide. The p-n junction thus realized is shown in Fig. 4.23d.

Step 5: Oxide is now etched from the back surface of the sample. Aluminium is then deposited on both top and bottom surface of the sample by vacuum evaporation (Fig. 4.23e).

Step 6: Photolithography is next carried out again (analogous to step 3) to define areas with photoresist in which the metal is to be retained.

Step 7: The sample is now placed in an acid solution to remove metal from undesired regions. Photoresist is then removed from everywhere and the sample containing many p-n junction diodes is realized as shown in Fig. 4.23f. It is to be noted that in this case, all the diodes share the same cathode and have a common cathode contact at the bottom surface. The sample can now be diced into small pieces in order to obtain individual diodes.

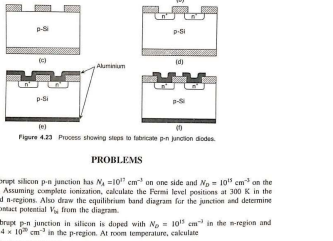


Figure 4.23 Process showing steps to fabricate p-n junction diodes.

PROBLEMS

P4.1 An abrupt silicon p-n junction has $N_A = 10^{17}$ cm^{-3} on one side and $N_D = 10^{15}$ cm^{-3} on the other. Assuming complete ionization, calculate the Fermi level positions at 300 K in the p- and n-regions. Also draw the equilibrium band diagram for the junction and determine the contact potential V_{bi} from the diagram.

P4.2 An abrupt p-n junction in silicon is doped with $N_D = 10^{15}$ cm^{-3} in the n-region and $N_A = 4 \times 10^{20}$ cm^{-3} in the p-region. At room temperature, calculate

(a) the built-in potential.
(b) the depletion layer width and the maximum field at zero bias.

(c) the depletion layer width and the maximum field at a reverse bias of 5 V.
(d) the depletion layer width and the maximum field at a forward bias of 0.5 V.

P4.3 The peak electric field at the junction of a silicon p⁺n junction of arbitrary n-doping profile is 1.5×10^5 volts/cm. Determine

(a) the total charge per unit area in the depletion region on the n-side.
(b) the depletion layer width on the p⁺-side of the junction if the doping on this side is uniform and equal to 10^{19} cm^{-3}.
(c) the voltage across the depletion layer in the p⁺-region.

P4.4 In Eq. 4.2(c), the point in the space charge layer at which $E_F = E_i$ is referred to as the intrinsic point. At this point, $n = p = n_i$. Show that the intrinsic point lies on the side of the space charge layer with the lower doping concentration.

P4.5 A p⁺n silicon diode is doped with $N_D = 10^{16}$ cm^{-3} in the n-region, where $D_p = 10$ cm^2/s and $\tau_p = 0.1$ μs. The junction area is 10^{-4} cm^2. Calculate the reverse saturation current and the forward current when $V = 0.5$ volts.

P4.6 The hote injection efficiency of a junction is defined as I_p/I where I is the total diode current. Assuming that the junction follows the long diode equations, show that $I_p/I = 1/[1 + (L_n\sigma_n/L_p\sigma_p)]$, where L_n is the diffusion length of electrons in the p-region, L_p is the diffusion length of holes in the n-region, σ_n is the conductivity in the n-region, and σ_p is the conductivity of the p-region.

P4.7 Determine N_D and area of a long base abrupt silicon p⁺n junction which provides a forward current of 1 mA when $V_F = 0.6$ V and has a peak electric field of 20 volts/micron at a reverse bias voltage of 150 volts. Assume that $\tau_p = 0.1$ μs in the n-region and $\mu_p = 450$ cm^2/Vs.

P4.8 A p⁺n junction with a reverse saturation current I_R is carrying a steady forward current I_F for $t < 0$. At $t = 0$, the diode current is abruptly increased to I_2. Assuming that the n-region is very long and $I_R = 1$ μA, calculate the stored charge and the diode voltage at $t = 0.5$ μs if $I_F = 1$ mA, $I_2 = 10$ mA, $\tau_p = 1$ μs, and $T = 300$ K.

P4.9 For a p⁺n diode, $\mu_p = 450$ cm^2/Vs and $\tau_p = 1$ μs in the n-region. Calculate the widths of the n-region for which (a) $W_n \le 0.1L_p$ (i.e. a short base diode) and (b) $W_n \ge 4L_p$ (i.e. a long base diode).

P4.10 How is Eq. (4.62) to be modified for a short base diode?

P4.11 Show that $I_p(x = x_n) > \langle Q_p/\tau_p \rangle$ for a diode with a short n-region.

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[3] Streetman, B.G. and S. Banerjee, *Solid State Electronic Devices*, 5th ed., Prentice Hall, New Jersey, 2000.
[4] Tyagi, M.S., *Introduction to Semiconductor Materials and Devices*, John Wiley & Sons, New York, 1991.

Zener diode is designed to have a specific breakdown voltage, it is called a concentration. When a diode is designed to have a specific breakdown voltage, it is called a *breakdown diode* or a *zener diode*. The term zener diode, however, is a misnomer in many situations since the actual breakdown is usually due to avalanche multiplication of carriers. Zener effect dominates only when both sides of the junction are heavily doped.

Such a diode with a tailored breakdown voltage of V_{BR} can be used as a *voltage regulator* as shown in the circuit of Fig. 5.1. Then, so long as the input voltage is greater than V_{BR}, the output voltage across the diode remains fixed at V_{BR} even if there is a change in the value of the input voltage. In other words, the voltage is regulated. This fixed breakdown voltage can also be used as a reference voltage in circuits that require a specific known value of voltage.

Figure 5.1 A voltage regulator circuit and the regulated output voltage signal.

5.3 VARIABLE CAPACITOR (VARACTOR)

Varactor is actually an abbreviated form of VARiable reACTOR. One property of a p-n junction that the width of the junction depletion region (and hence the depletion capacitance) is a function of the applied voltage, is utilized in this application. As already pointed out in section 4.2.2, the manner in which the depletion layer width changes with the applied reverse voltage depends on the actual doping profile of the junction.

In general, let us consider a p⁺n junction where the doping profile in the n-region is given by

$$N_D(x) = N_{D0}\left(\frac{x}{x_0}\right)^n \tag{5.1}$$

where N_{D0}, x_0, and n are constants for a particular doping profile and x is the distance from the p-n junction in the n-region.

Then, following the usual procedure of integrating Poisson's equation twice using appropriate boundary conditions, the depletion region width is obtained as a function of the applied reverse bias V_j as a function of the applied reverse bias V_R as

$$W \propto (V_{bi} + V_j)^{1/(n+2)} \tag{5.2}$$

Consequently, using Eq. (4.33) and neglecting V_{bi} when a large reverse bias is applied, the depletion capacitance is given by

$$C_j = \frac{A\varepsilon_s}{W} \propto (V_j)^{-1/(n+2)} \tag{5.3}$$

For an abrupt junction, $n = 0$, and therefore the depletion capacitance C_j is proportional to $(V_j)^{-1/2}$ as seen from Eq. (4.31). Similarly using Eq. (5.3), it can be shown that for a linearly graded junction ($n = 1$), C_j will be proportional to $(V_j)^{-1/3}$. So the *voltage sensitivity* (dC_j/dV) is found to be greater for an abrupt junction.

The sensitivity can be further enhanced by using a *hyper-abrupt junction*, where the doping profile is tailored specifically to obtain a value of $n < 0$. A special case is seen for $n = -3/2$ as shown in Fig. 5.2. In this case, using Eq. (5.3), the depletion capacitance variation can be expressed as $C_j \propto V_j^{-2}$. If this varactor is used with an inductance L in a resonant circuit, the resonant frequency of the circuit is given by

$$f_r = \frac{1}{2\pi\sqrt{LC_j}} \propto V_j \tag{5.4}$$

In other words, the resonant frequency of the circuit can be varied linearly by changing the applied reverse voltage. This property is widely used for tuning in the radio/TV receivers.

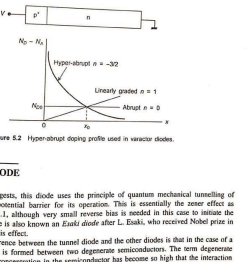

Figure 5.2 Hyper-abrupt doping profile used in varactor diodes.

5.4 TUNNEL DIODE

As the name itself suggests, this diode uses the principle of quantum mechanical tunnelling of electrons through the potential barrier for its operation. This is essentially the zener effect as discussed in section 4.5.1, although very small reverse bias is needed in this case to initiate the tunnel action. This device is also known as *Esaki diode* after L. Esaki, who received Nobel prize in 1973 for his work on this effect.

A significant difference between the tunnel diode and the other diodes is that in the case of a tunnel diode, a junction is formed between two degenerate semiconductors. The term degenerate implies that the doping concentration in the semiconductor has become so high that the interaction between the dopant atoms themselves can no longer be ignored. In such cases, the impurity levels cease to be discrete energy states and form a band instead. This band may overlap with the bottom

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5 Applications of p-n Junctions

5.1 INTRODUCTION

In the previous chapter, we have discussed about the basic properties of p-n junction diodes. From this discussion, it is evident that such a junction allows substantial current to flow through it only when it is biased in the forward direction. When reverse-biased, a very small current can pass through this diode. Thus, ideally, a p-n junction diode can be thought of as short circuit in forward bias and open circuit in reverse bias. When an ac sinusoidal voltage is applied as input to a circuit consisting of a diode and a load resistance in series, a rectified signal is obtained at the output since the current can flow only in the forward direction. That is, the output signal across the load contains only the positive half cycles of the original sine wave and hence, has a positive average value. By suitably filtering this signal, ac to dc conversion can be achieved. This is perhaps the most common application of diodes.

Also, this property of a diode of conducting only in the forward direction has been made use of in various switching applications. The time response of the diode may be a critical criterion for such applications. As already pointed out in section 4.4.1, if a diode has to be switched on and off rapidly, the minority carrier lifetime must be small and therefore either deep levels have to be deliberately introduced during fabrication of the device or the device geometry has to be suitably adjusted so as to allow minimum charge storage.

Apart from the use of diodes as switching devices or rectifiers, specially designed p-n junction diodes are widely used in various other applications such as voltage regulator, variable capacitor, negative resistance devices, and optoelectronic transducers. All these applications are based on the fundamental junction properties. In this chapter, we shall discuss how the p-n junction can be suitably designed for some specific application.

5.2 VOLTAGE REGULATOR

We have discussed the breakdown mechanisms of diodes in section 4.5

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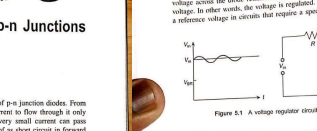

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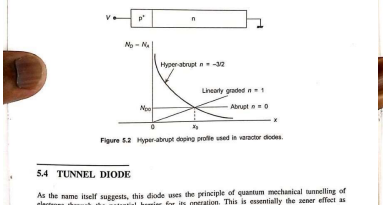

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of the conduction band for donors and with the top of the valence band for acceptors. If the electron concentration in the conduction band is more than the effective density of states N_C, the Fermi level no longer lies within the energy gap but moves inside the conduction band. Similarly, if the hole concentration becomes greater than N_V, the Fermi level moves inside the valence band. The energy band diagram of a p-n junction at equilibrium where both p- and n-regions are degenerately doped is shown in Fig. 5.3(a). The Fermi level E_F is constant at equilibrium and lies below E_V in the p-region and above E_C in the n-region. When a small reverse voltage is applied, the quasi-Fermi level for holes (E_{Fp}) moves above that for electrons (E_{Fn}), as shown in Fig. 5.3(b). As there are filled states below E_{Fp} and empty states above E_{Fn} at the same energy and the barrier width is very small, electrons can tunnel from the p-region to the n-region. This is akin to zener effect except that a very small reverse voltage is required to initiate the tunnelling phenomenon. As the reverse bias is increased, E_{Fp} moves further up with respect to E_{Fn}, creating more filled states in the p-region and empty states in the n-region at the same energy. Thus, the tunnelling of electrons from p-region to n-region increases with increasing reverse bias.

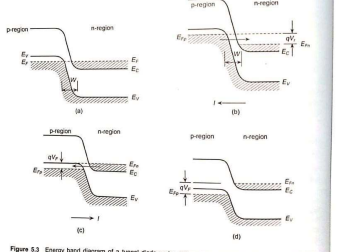

Figure 5.3 Energy band diagram of a tunnel diode under different bias conditions: (a) thermal equilibrium, (b) reverse bias, (c) small forward bias resulting in tunnelling of carriers, and (d) larger forward bias showing negative resistance effect.

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On the other hand, when a small forward voltage is applied across this junction, a situation depicted in Fig. 5.3(c) is reached. Now E_{Fn} moves up with respect to E_{Fp}, placing empty states in p-region and the tunnelling increases with the increase in applied voltage.

However, with increasing forward voltage, a point is reached where the conduction band edge in the n-region approaches the valence band edge in the p-region as shown in Fig. 5.3(d). The electrons in the n-region now see fewer accessible empty states in the p-region and hence, tunnelling reduces with further increase in the forward voltage. This is known as the *negative resistance effect* since the current reduces with increase in applied voltage. Once the conduction band edge in the n-region moves above the valence band edge in the p-region, tunnelling stops and the device behaves like a conventional diode. The current–voltage characteristics of a tunnel diode are shown in Fig. 5.4. It is observed that the current decreases rapidly beyond a voltage V_p, termed as the *peak voltage* of the tunnel diode. The ratio of the *peak current* I_p to the *valley current* I_v is often used as a figure of merit for this diode. The negative resistance region can be used for various applications such as oscillation, amplification and switching to name a few. Also, as tunnelling does not have the time delays involved in drift and diffusion, tunnel diodes are very useful for high-speed circuits. However, the applications of tunnel diode are limited by its comparative low current drive.

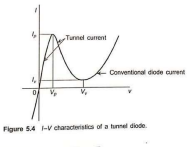

Figure 5.4 I–V characteristics of a tunnel diode.

5.5 SOLAR CELLS AND PHOTODIODES

Optoelectronic devices convert optical energy into electrical energy and vice versa. The phenomenon of converting optical radiation into electrical energy is called the *photovoltaic effect*. Two important devices based on the photovoltaic effect are solar cell and photodetector. The inverse of the photovoltaic effect is termed *electroluminescence*, that is, the phenomenon of light emission when subjected to electrical signal. The LEDs and Lasers form good examples of this effect.

5.5.1 Photovoltaic Effect

It has already been discussed in section 1.1.3 that if a photon with energy $h\nu$ which is greater than or equal to the band gap E_g is incident on a semiconductor, absorption of light can take

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place with consequent generation of an EHP. In other words, in order to generate EHPs in a semiconductor, the wavelength of the incident radiation must be lower than a particular wavelength, the *cut-off wavelength* λ_c. The ability of the material to absorb light is also dependent on another parameter, called the *absorption coefficient* α. α has the characteristic value for the material, called the *effective absorption length*, is essentially a unit of inverse of length while $(1/\alpha)$, called the *effective absorption length*, is essentially a measure of the material thickness within which the incident photon is absorbed by the semiconductor. In fact, the fraction of the incident radiation absorbed within a distance x from the surface is given by $[1 - \exp(-\alpha x)]$. Thus, the fraction of the incident radiation absorbed within a distance $1/\alpha$ is then evaluated as $[1 - \exp(-1)] = 0.632$. An absorption coefficient of 10^4 cm^{-1} corresponds to an absorption length of 1 μm, which means that 63.2% of the incident radiation is absorbed within 1 μm from the surface. The absorption coefficients for different semiconductors are plotted as a function of the photon energy in Fig. 5.5. For indirect band gap semiconductors, the value of α at a given photon energy is usually lower for materials with higher E_g, as can be seen from the plots for Ge and Si. However, direct band gap materials (such as GaAs) have much larger values of α even though their E_g may be larger than indirect band gap materials, as seen from the plots for GaAs and Si. This means that for a given photon energy, a much thinner layer of GaAs (than silicon) is needed to absorb the light.

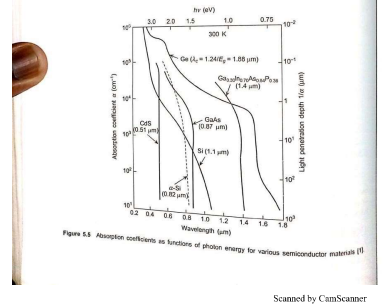

Figure 5.5 Absorption coefficients as functions of photon energy for various semiconductor materials [1]

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When a p-n junction is illuminated with light of photon energy greater than E_g, photons are absorbed in the semiconductor and EHPs are generated both in the n-region and the p-region of the junction. For the EHPs to contribute towards current in the external circuit, the generated electrons and holes must be separated before they recombine. This is achieved if the EHPs are generated in the depletion layer, where the electric field sweeps away the electrons and holes in opposite directions. The photo-generated minority carriers, which are generated within one diffusion length from the depletion layer edge, can also diffuse to the depletion region without recombining. They are then swept across the junction due to the electric field present in the depletion region as shown in Fig. 5.6. Due to the direction of the electric field being from the n-region to the p-region, the holes flow towards the p-region and electrons to the n-region. Since the direction of this photo-generated current I_L is opposite to that in a forward-biased diode, we can write an expression for the total current in an illuminated diode as

$$I = I_0\left[\exp\left(\frac{V}{V_T}\right) - 1\right] - I_L \tag{5.5}$$

In other words, the total current of the diode is lowered by an amount I_L when illuminated.

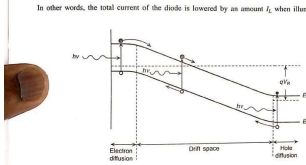

Figure 5.6 An illuminated p-n junction showing the generation and subsequent separation of EHPs.

The *I–V* characteristics of an illuminated p-n junction diode are depicted in Fig. 5.7. An interesting point to be noted is that the *I–V* characteristics pass through the first, third, and fourth quadrants. Depending on the intended application, the diode can be made to operate so that either power is delivered to the device (operating in the first or the third quadrant) or supplied by the device to the external circuit (operating in fourth quadrant). When the diode is used as a solar cell, it is made to operate in the fourth quadrant and work as a battery. On the other hand, to make use of the diode as a photodetector, it is usually operated in the third quadrant. However, though the underlying principle is the same, the actual structures are very different for these two devices. We shall now discuss the operation of these two devices in the next two sections.

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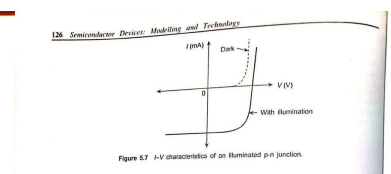

Figure 5.7 I–V characteristics of an illuminated p-n junction.

Example 5.1

The band gap of silicon and GaAs are $E_g = 1.12$ eV and $E_g = 1.424$ eV respectively. What are the cut-off wavelengths λ_c in these materials? (Take Planck's constant $h = 6.62 \times 10^{-34}$ Js and velocity of light $c = 3 \times 10^{08}$ cm/s.)

Solution: At the cut-off wavelength λ_c,

$$h\nu = E_g$$

or

$$\frac{hc}{\lambda_c} = E_g$$

or

$$\lambda_c = \frac{hc}{E_g}$$

Since 1 eV = 1.6×10^{-19} J,

$$hc = \frac{6.62 \times 10^{-34} \times 3 \times 10^{08}}{1.6 \times 10^{-19}} = 1.24 \times 10^{-4} \text{ eVcm} = 1.24 \text{ eV}\mu\text{m}$$

Therefore, when λ_c is expressed in μm and E_g in eV, we have the simple relation

$$\lambda_c = \frac{1.24}{E_g}$$

Therefore, for silicon,

$$\lambda_c = \frac{1.24}{1.12} = 1.11 \text{ } \mu\text{m}$$

and for GaAs,

$$\lambda_c = \frac{1.24}{1.424} = 0.87 \text{ } \mu\text{m}$$

5.5.2 Solar Cell

A solar cell is basically a p-n junction, which can deliver power to the external circuit when illuminated. Arrays of such junctions are used to supply electrical power to space satellites and also

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for remote and rural areas. Solar cells are preferred over conventional batteries for these applications as they are environment-friendly and provide maintenance-free services for long time. However, the high cost of solar cells has so far curtailed its widespread use for terrestrial applications.

Solar cells are usually large area devices so that a greater amount of optical energy is absorbed and output power is high. In order to increase the light-generated current I_L, a large number of the minority carriers generated outside the depletion region have to reach the edge of the depletion region. This requires the diffusion lengths L_n and L_p to be large. As seen from the discussion on fabrication of diodes in section 4.6, the top layer is produced by diffusion or ion-implantation with a high impurity concentration. High impurity concentrations create lattice damage which introduces recombination centres. This makes the lifetime of carriers very short in the top layer and consequently the diffusion length is small. Hence, in a practical solar cell, the topmost layer is kept very thin so as to allow more light to reach the base. On the other hand, the diffusion length in the substrate (or base) is large and the carriers generated in this region mostly contribute to useful photocurrent. Also, n^+p structures, with the n^+ layer at the top, are preferred. This is because the diffusion constant of electrons in the p-base (D_n) is larger than that of holes in the n-base (D_p) of a p^+n junction. Larger diffusion constant means larger diffusion length and therefore an improvement in I_L, as more carriers generated outside the depletion region can diffuse to this high field region. Usually, in a practical solar cell the surface concentration of the n-region is kept below 10^{20} cm^{-3} with a junction depth less than 0.2 μm. The surface of the solar cell is coated with *antireflection coating* in order to minimize reflection losses. In general, the antireflection materials used are SiO_2, TiO_2, and Ta_2O_5. The series resistance of the solar cell has to be made very small so that power is not lost due to losses in the cell itself. To achieve this, thin contact fingers are provided on the n-region. Figure 5.8(a) shows the structure of a solar cell.

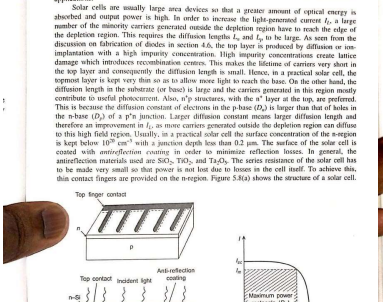

Figure 5.8 (a) Structure of a solar cell; (b) *I–V* characteristics of an illuminated solar cell showing the point of maximum power.

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Solar cells are usually calibrated at AM1 or AM0 illumination. The AM1 spectrum represents sunlight incident at the sea level on a clear day with the sun at the zenith and the incident optical power in this case is about 92.5 mW/cm^2. On the other hand, the incident optical power for AM0 is 135 mW/cm^2 and is considered to be the intensity of the solar spectrum just outside the earth's atmosphere. While AM0 calibration is generally used for solar cells in space satellites, AM1 is usually used for cells on terrestrial applications.

The fourth quadrant *I–V* characteristics of a solar cell is shown in Fig. 5.8(b). The three most important parameters of a solar cell are the open-circuit voltage V_{oc}, the *short circuit current* I_L and the *fill factor FF*. Since in a solar cell, current flows from the n-region to the p-region, that is, in the direction opposite to that of a conventional diode, putting $V = 0$ in Eq. (5.5) we obtain the short circuit current I_{sc} as

$$I_{sc} = -I_L \tag{5.6}$$

Again for $I = 0$, we have the open circuit voltage V_{oc} from Eq. (5.5) as

$$V_{oc} = V_T \ln\left(1 + \frac{I_L}{I_0}\right) \approx V_T \ln\left(\frac{I_L}{I_0}\right) \tag{5.7}$$

However, no power is delivered when the solar cell is operating at either of these two points. So, the operating point of the cell should be selected such that the output power becomes maximum. The output power is given by

$$P = VI = V\left[I_L - I_0\left\{\exp\left(\frac{V}{V_T}\right) - 1\right\}\right] \tag{5.8}$$

The condition for maximum power is obtained from $dP/dV = 0$. Hence, differentiating Eq. (5.8) and equating it to zero, we get the expressions for the voltage and the current at the point of maximum power as

$$V_m = V_T \ln\frac{1 + \frac{I_L}{I_0}}{1 + \frac{V_m}{V_T}} = V_{oc} - V_T \ln\left(1 + \frac{V_m}{V_T}\right) \tag{5.9a}$$

$$I_m = (I_0 + I_L)\left[\frac{V_m}{V_m + V_T}\right] \approx I_L\left[1 - \frac{V_T}{V_m}\right] \tag{5.9b}$$

The optimum load resistance R_m, for maximum power is thus given by rearranging Eq. (5.9b) as

$$R_m = \frac{V_m}{I_m} = \frac{V_m + V_T}{I_0 + I_L} \approx \frac{V_m}{I_L} \tag{5.10}$$

The fill factor of the solar cell is defined as

$$FF = \frac{V_m I_m}{V_{oc} I_L} = \frac{P_m}{V_{oc} I_L} \tag{5.11}$$

where $P_m = V_m I_m$ is the maximum power output of the solar cell. For a well-designed cell, FF usually lies in the range 0.7–0.8. The efficiency of the solar cell is given by

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$$\eta = \frac{V_{oc} I_m}{P_{in}} = FF\frac{V_{oc} I_L}{P_{in}} \tag{5.12}$$

where P_{in} = incident optical power.

Thus, it is evident that to realize a solar cell with high efficiency, it is not only necessary to have high V_{oc} and I_L, but also a high FF. Solar cells with $\eta = 15\%$ under AM1 illumination are commercially available.

To improve the efficiency of solar cells, I_L has to be increased. This can be achieved by increasing the collection of photo-generated carriers. One way of improving the carrier collection is to heavily dope the back of the cell so as to realize an n^+pp^+ structure. The potential barrier at the back pp^+ junction forces the minority carriers to be confined to the p-region. The decrease in the availability of minority carriers reduces recombination at the back contact and enhances the chance of collection of the generated carriers. Other techniques employed to increase I_L include surface texturing, use of solar concentrators, and so on.

The efficiency of a solar cell depends, to some extent, on the material used. For a material with larger band gap, V_{oc} is usually higher as I_0 reduces drastically with increase in E_g [see Eq. (5.7)]. On the other hand, with an increase in E_g, some part of the incident light for which $h\nu$ is now less than E_g, no longer contributes to I_L, thereby reducing I_L [see Eq. (5.6)]. Therefore, there is an optimum band gap for the highest possible efficiency. GaAs solar cells have higher efficiency than their silicon counterparts, and are therefore preferred for space applications, where cost is not of much concern. On the other hand, silicon is still preferred for terrestrial applications, for their lower cost. Very high efficiency cascade solar cells, consisting of a series of junctions of materials with different bandgaps, which efficiently absorb different portions of the solar spectrum, have also been demonstrated primarily for space applications.

Example 5.2

A solar cell has an area of 1 cm × 1 cm. The cell has an n^+p structure, where the doping concentration of the p-region is 10^{15} per cm^3. Considering $\mu_n = 1350$ cm^2/Vs, $\tau_n = 1$ μs in the p-region, and $L_I = 40$ mA, calculate I_{sc}, V_{oc}, and maximum power output (P_m) if Fill Factor (FF) is 0.7. If the incident optical power is 90 mW/cm^2, what is the efficiency (η) of the solar cell?

Solution: In the p-region,

$$L_n = \sqrt{D_n \tau_n} = \sqrt{\mu_n V_T \tau_n} = \sqrt{1350 \times 0.026 \times 10^{-6}} = 5.925 \times 10^{-3} \text{ cm}$$

Since a solar cell is usually a long base diode,

$$I_0 = \frac{A q D_n n_i^2}{L_n N_A} = \frac{1.6 \times 10^{-19} \times (1350 \times 0.026) \times (1.5 \times 10^{10})^2}{5.925 \times 10^{-3} \times 10^{15}} = 2.13 \times 10^{-10} \text{ A}$$

Now, from Eq. (5.7),

$$V_{oc} = V_T \ln\left(\frac{I_L}{I_0}\right) = 0.026 \ln\left(\frac{40 \times 10^{-3}}{2.13 \times 10^{-10}}\right) = 0.495 \text{ V}$$

From Eq. (5.6),

$$I_{sc} = I_L = 40 \text{ mA}$$

Using Eq. (5.11),

$$P_m = FF\, I_m\, V_m = 0.7 \times (40 \times 10^{-3}) \times 0.495 = 13.86 \text{ mW}$$

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From Eq. (5.12),

$$\eta = \frac{P_m}{P_{in}} = \frac{13.86}{90} = 0.154$$

Therefore, the solar cell has an efficiency of 15.4%.

5.5.3 Photodiode

In a photodiode, the incident optical signal generates EHPs that gives rise to a photocurrent. The diode is reverse biased as less than its breakdown voltage ($V_r < V_{BR}$), so that under dark conditions, diode is reverse biased as less than its breakdown voltage ($V_r < V_{BR}$), so that under dark conditions, only a very small current flows, usually referred to as the *dark current* of the photodiode. Now, when light is incident on the diode, a large photocurrent flows. This change in current allows the photodiode to detect the presence of light and hence the photodiode is also termed a *photodetector*. The dark current must be made small and the photocurrent large as much as possible so that the photodiode can detect weak signals and the signal-to-noise ratio (SNR) is large.

Most photodiodes have a thin p^+ layer at the top unlike solar cells where the n^+ layer is at the top. This is because a diode with the p^+ layer at the top has a faster time response compared to the diode with the n^+ layer at the top (In a solar cell, the response time is not important). For a well designed photodiode, most EHPs are generated within the depletion region. Due to the electric field, the electrons move towards the n-region and the holes towards the p-region. Now the mobility of holes is much less compared to that of electrons. So with a p-layer at the top, the slower carriers (holes) have to move a much shorter distance to the p-contact.

An important parameter of a photodiode is the *quantum efficiency* η, which is defined as the number of useful EHPs generated per incident photon. The term 'useful EHPs', refers to those EHPs which do not recombine and therefore contribute to current in the external circuit. The value of η may be less than 1 if all the light is not absorbed within the depletion layer or a diffusion length from it. Another related parameter of a photodiode is the *responsivity* R. It is the ratio of the photocurrent to the incident optical power and is expressed in units of Amperes/Watt. If $P_{optical}$ is the incident optical power at a wavelength λ corresponding to a photon energy $h\nu$, and I_p is the photo-generated current then η and R are expressed as

$$\eta = \frac{\text{Rate of generation of useful EHPs}}{\text{No. of incident photons/unit time}}$$

$$= \frac{\left(\frac{I_p}{q}\right)}{\left(\frac{P_{optical}}{h\nu}\right)} = \exp(-\alpha W_p) - \exp\{-\alpha(W + L_p)\} \tag{5.13a}$$

and

$$R = \frac{I_p}{P_{optical}} = \frac{\eta q}{h\nu} \tag{5.13b}$$

In Eq. (5.13a),

α = absorption coefficient
W_p = thickness of the top p-layer,
W = depletion width, and
L_p = minority carrier diffusion length in the underlying n-layer.

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It has been assumed that only those EHPs which are generated in the depletion layer and one diffusion length away from it in the n-region contribute to the photocurrent. From Eq. (5.13a), we see that for high efficiency, W_p must be small and the depletion layer width W must be large. Physically this means that most of the incident photons are absorbed within the depletion region rather than in the neutral p- and n-regions. As there exists a strong electric field in the depletion region, as soon as an EHP is generated within this region, electrons are swept into the n-region and holes into the p-region eventually contributing to photocurrent. Also a large W results in a small capacitance and improves the speed of response of the diode. However, if W is too large, the transit time delay of the carriers dominates and hence, a compromise has to be made.

HELP DESK 5.1

Equation (5.13b) suggests that for a given quantum efficiency, the responsivity increases with wavelength. What is the physical reason for this?

Let us consider two ideal photodiodes, P_1 and P_2, having quantum efficiency η equal to 1. Let us assume that the same number of photons is incident on P_1 and P_2 per unit time. Therefore, it follows that the photo-generated current is also the same in P_1 and P_2, that is, $I_{L1} = I_{L2}$. Let λ_1 and λ_2 be the incident wavelengths on P_1 and P_2 respectively, and $\lambda_1 > \lambda_2$. Since each photon incident on P_2 has higher energy than those incident on P_1, the incident optical power on P_2 ($P_{optical\,2}$) is greater than the incident power on P_1 ($P_{optical\,1}$).

Now, since

$$P_{optical\,2} > P_{optical\,1}$$

and

$$I_{L2} = I_{L1}$$

it follows that

$$R_1 > R_2$$

where R_1 and R_2 are the responsivities of diodes P_1 and P_2 respectively. Therefore for two photodiodes having the same η, responsivity increases with increase in wavelength.

Also note that one photon can generate only one EHP. If the energy of the photon $h\nu > E_g$, the excess energy ($h\nu - E_g$) does not contribute towards I_L but is lost in the substrate. Therefore, reduction in wavelength below the cut-off wavelength only increases the input power, without increasing output current, thereby reducing responsivity.

A convenient way of tailoring the depletion layer width is to use a *p-i-n photodiode* rather than a simple p-n junction. Most of the reverse bias appears across the 'intrinsic' layer, which need not be truly intrinsic, as long as the doping is low. As the doping is low, the carrier lifetime is high in this region and most of the photo-generated carriers are collected by the n- and p-regions. Figure 5.9(a) shows the cross-section of a p-i-n photodiode. The curvature at the junction edges leads to higher electric field intensity and a smaller radius of curvature results in a smaller breakdown voltage. A guard ring of larger junction depth is usually employed in p-i-n photodiodes as shown in Fig. 5.9(a) to increase the breakdown voltage and reduce the dark current.

Photodiodes, which are operated close to the avalanche breakdown voltage to enable avalanche multiplication of photo-generated carriers, are called *avalanche photodiodes*. The

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avalanche multiplication process increases the photocurrent and hence the responsivity of the photodiode. However, since the avalanche multiplication process is statistical in nature, there is a fluctuation in the responsivity due to the avalanche multiplication noise.

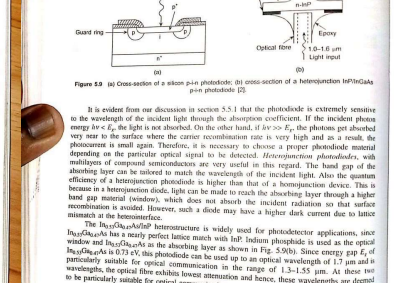

Figure 5.9 (a) Cross-section of a silicon p-i-n photodiode; (b) cross-section of a heterojunction InP/InGaAs p-i-n photodiode [2].

It is evident from our discussion in section 5.5.1 that the photodiode is extremely sensitive to the wavelength of the incident light through the absorption coefficient. If the incident photon energy $h\nu < E_g$, the light is not absorbed. On the other hand, if $h\nu >> E_g$, the photons get absorbed very near to the surface where the carrier recombination rate is very high and as a result, the photocurrent is small again. Therefore, it is necessary to choose a proper photodiode material depending on the particular optical signal to be detected. *Heterojunction photodiodes*, with multilayers of compound semiconductors are very useful in this regard. The band gap of the absorbing layer can be tailored to match the wavelength of the incident light. Also the quantum efficiency of a heterojunction photodiode is higher than that of a homojunction device. This is because in a heterojunction diode, light can be made to reach the absorbing layer through a higher band gap material (window), which does not absorb the incident radiation so that surface recombination is avoided. However, such a diode may have a higher dark current due to lattice mismatch at the heterointerface.

The $In_{0.53}Ga_{0.47}As/InP$ heterostructure is widely used for photodetector applications, since $In_{0.53}Ga_{0.47}As$ has a nearly perfect lattice match with InP. Indium phosphide is used as the optical window and $In_{0.53}Ga_{0.47}As$ as the absorbing layer as shown in Fig. 5.9(b). Since energy gap E_g of $In_{0.53}Ga_{0.47}As$ is 0.73 eV, this photodiode can be used up to an optical wavelength of 1.7 μm and is particularly suitable for optical communication in the range of 1.3–1.55 μm. At these two wavelengths, the optical fibre exhibits lowest attenuation and hence, these wavelengths are deemed to be particularly suitable for optical communication.

5.6 LIGHT EMITTING DIODES (LEDs) AND LASERS

In this section we discuss those p-n junction devices, which convert electrical energy to optical energy. This effect is called *injection electroluminescence*. When forward biased, lasers emit *coherent* light in much narrower wavelength bands than LEDs

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and are more directional. This is because the dominant operating process for LEDs is *spontaneous emission* and for lasers it is *stimulated emission*. We shall discuss these two processes in the following subsection.

5.6.1 Spontaneous and Stimulated Emission

We have already seen that a photon of appropriate energy can be absorbed by a semiconductor, creating an EHP in the process. This is called *optical absorption*. Let us consider Fig. 5.10(a) which depicts two energy levels in a semiconductor E_1 and E_2, where E_1 corresponds to the ground state and E_2 to the excited state. At room temperature, most of the electrons are in ground state. When a photon of energy greater than or equal to $h\nu_{12} = E_2 - E_1$ is incident on the system, an electron in the ground state absorbs it and goes to the excited state. However, the excited state is unstable. So, after a short time, without any external stimulus, the electron comes back to the ground state emitting a photon of energy $h\nu_{12}$. This process is referred to as *spontaneous emission* and is schematically represented in Fig. 5.10(b). On the other hand, if a photon of energy $h\nu_{12}$ impinges on the electron while it is in the excited state [as shown in Fig. 5.10(c)], the electron can be stimulated back to the ground state with emission of a photon having an energy $h\nu_{12}$, in phase with the incident radiation. This is called *stimulated emission*. The radiation emitted from stimulated radiation is monochromatic because each photon has energy precisely equal to $h\nu_{12}$ and coherent because they are all in phase.

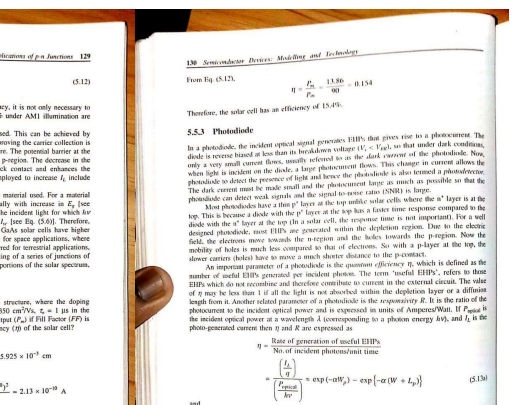

Figure 5.10 Schematic showing the three basic processes: (a) photon absorption; (b) spontaneous emission; and (c) stimulated emission.

5.6.2 Light Emitting Diodes

LEDs are p-n junctions that can emit spontaneous radiation in ultraviolet, visible, or infrared regions. Visible LEDs are widely used as visual information links between electronic instruments and their

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users. Thus, it is a common sight to see them blinking on the control panel of almost all electronic gadgets. On the other hand, infrared LEDs are used for fibre-optic communication systems where silica fibres are used to guide the optical signal over long distances. An important application of infrared LED is in opto-isolators where an input electrical signal is applied to the LED. Light is generated and subsequently detected by a photodiode and converted back to an electrical signal as a current flowing through a load resistor. Opto-isolators allow signal transmission at the speed of light and are electrically isolated.

$GaAs_{1-y}P_y$ is the most preferred material for visible LED. The phosphorous mole fraction in this ternary compound is denoted by y, that is, as y is increased, more and more phosphorus atoms replace arsenic in the crystal lattice. For $0 < y < 0.45$, the band gap of the material is direct and increases from 1.424 eV (at $y = 0$) to 1.977 eV (at $y = 0.45$). As already pointed out in section 1.1.3, the probability of direct (radiative) transition is high in direct band gap semiconductors. Hence, $GaAs_{1-y}P_y$ ($y < 0.45$) is used for light emission in the wavelength range of 627–870 nm. For $y > 0.45$, the material has an indirect band gap. So, special recombination centres have to be introduced to facilitate radiative recombination. Incorporation of nitrogen results in formation of such a recombination

5.6.3 Semiconductor Laser

Virtually all the semiconductor lasers use direct band gap materials. The dominant process for semiconductor laser operation is stimulated emission. If the stimulated emission is to dominate over the spontaneous emission, the photon field energy density must be very high and therefore an optical resonant cavity is to be used. In addition, if stimulated emission of photons has to dominate over the absorption of photons, we must have more electrons in the excited state than in the ground state. This is called *population inversion*, since under equilibrium the reverse is true.

Population inversion in a laser can be achieved in the following manner. Let us consider a p-n junction formed between two degenerate semiconductors. The energy band diagram of such a junction under thermal equilibrium is shown in Fig. 5.11(a). When a forward voltage is applied, electrons are injected from the n-region and holes from the p-region into the space charge region. When a sufficiently large forward voltage is applied, high injection occurs, that is, there is a large concentration of electrons in the conduction band and holes in the valence band in the space charge region. This is the condition of population inversion as shown in Fig. 5.11(b). For band-to-band transition, the minimum energy required is E_g. Therefore, we can write the condition for population inversion as $qV_F = E_{Fn} - E_{Fp} > E_g$.

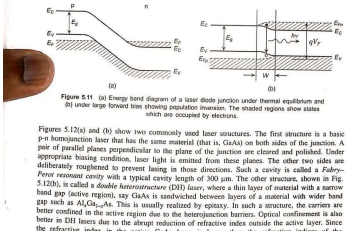

Figure 5.11 (a) Energy band diagram of a laser diode junction under thermal equilibrium and (b) under large forward bias showing population inversion. The shaded regions show states which are occupied by electrons.

Figures 5.12(a) and (b) show two commonly used laser structures. The first structure is a basic p-n homojunction laser that has the same material (that is, GaAs) on both sides of the junction. A pair of parallel planes perpendicular to the plane of the junction are cleaved and polished. Under appropriate biasing condition, laser light is emitted from these planes. The other two sides are deliberately roughened to prevent lasing in those directions. Such a cavity is called a *Fabry–Perot resonant cavity* with a typical cavity length of 300 μm. The other structure, shown in Fig. 5.12(b), is called a *double heterostructure* (DH) *laser*, where a thin layer of material with a narrow band gap (active region), say GaAs is sandwiched between layers of a material with wider band gap such as $Al_xGa_{1-x}As$. This is usually realized by epitaxy. In such a structure, the carriers are better confined in the active region due to the heterojunction barriers. Optical confinement is also better in DH lasers due to the abrupt reduction of refractive index outside the active layer. Since the refractive index in the active GaAs layer is larger than the refractive indices of the surrounding $Al_xGa_{1-x}As$ layers, the propagation of the electromagnetic radiation is confined in a direction parallel to the layer interfaces. The *threshold current density* J_{th} defined as the minimum

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current density required for lasing is lower for DH lasers compared to homojunction lasers. With increase in temperature, J_{th} for DH lasers increases at a slower rate than that for homojunction lasers and hence DH lasers are preferred for continuous operation at room temperature.

At low currents where the spontaneous emission dominates, the laser spectrum has broad spectral distribution, and the full width at half maximum intensity may be about 10–50 nm. As the current approaches threshold, the spectral distribution becomes narrower and above J_{th}, the spectral width is usually of the order of 0.1–1 nm.

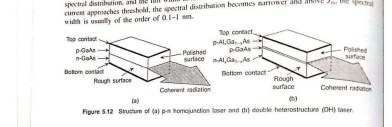

Figure 5.12 Structure of (a) p-n homojunction laser and (b) double heterostructure (DH) laser.

PROBLEMS

P5.1 At a particular wavelength of incident radiation, the absorption coefficients of Si and GaAs are 4×10^3 per cm and 3×10^4 per cm respectively. What are the thicknesses of Si and GaAs necessary to absorb 80% of the incident radiation?

P5.2 A 5 μm thick silicon sample is illuminated with monochromatic light having $h\nu = 2$ eV. The incident power is 10 mW. (a) What is the total energy absorbed in the sample per second? (b) How much energy per second is dissipated as heat? (c) How many EHPs are generated in the sample per second?

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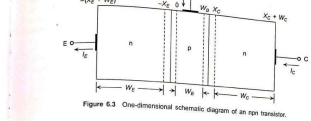

Figure 6.3 One-dimensional schematic diagram of an npn transistor.

In order to obtain expressions for the terminal currents, we shall use the minority carrier distributions and currents in the neutral regions. This is because, for minority carriers the drift current can be neglected, as we assume that all the applied voltages appear across the space charge region and there is negligible electric field in the neutral regions. The continuity equations are therefore simplified. We have already seen in Chapter 4 that under bias, the minority carrier concentration at the edge of the depletion region gets modified by a factor $\exp(V/V_T)$, where $V = V_F$ for a forward biased junction and $V = -V_r$ for a reverse biased junction. Also, at the emitter and collector contacts, the large surface recombination velocity reduces the minority carrier concentration to the thermal equilibrium values. Using these considerations, the distribution of minority carrier concentration in the BJT for the different modes of operation is as shown in Fig. 6.4. The polarities of V_{BE} and V_{BC} for each mode of operation are also shown.

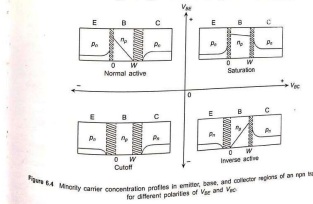

Figure 6.4 Minority carrier concentration profiles in emitter, base, and collector regions of an npn transistor for different polarities of V_{BE} and V_{BC}.

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Due to the concentration gradients of the minority carriers in the neutral regions, there is flow of diffusion currents. The minority carrier electron and hole current components at the EB junction are denoted by I_{nE} and I_{pE} respectively. When the EB junction is forward biased ($V_{BE} > 0$), holes are injected from the base to the emitter and I_{pE} is negative (that is, in the negative x-direction). On the other hand, when $V_{BE} < 0$, the hole concentration in the emitter reduces below the thermal equilibrium value as the minority carrier holes are swept to the base, giving rise to a positive I_{pE}. When $V_{BE} > V_{BC}$, the concentration gradient of electrons in the base causes the injection of electrons from the emitter to the base, giving rise to a negative I_{nE}. (Note that the direction of current is opposite to the direction of electron flow). However, when $V_{BE} < V_{BC}$, the direction of the concentration gradient changes and electrons may be injected from base to emitter resulting in a positive I_{nE}. (There is a special case when $V_{BE} = V_{BC}$, which is discussed later in section 6.6.2). For the CB junction also, there exist electron and hole current components denoted by I_{nC} and I_{pC} respectively. Using similar arguments as in the case of the EB junction, I_{pC} is seen to be positive when $V_{BC} > 0$ and negative when $V_{BC} < 0$. The current component I_{nC} is in general negative when $V_{BE} > V_{BC}$, and positive when $V_{BE} < V_{BC}$. (Note that positive or negative values of the current components only refer to their directions of flow. For positive hole current, the flow of holes is in the positive x-direction while for positive electron current, the flow is in the negative x-direction).

We have already seen in section 6.2 that for proper action and operation of a transistor, most of the electrons injected into the base at one of the junctions should reach the other junction. This implies that there is dependence between I_{nE} and I_{nC}. However, $I_{nE} \neq I_{nC}$. For example, when electrons injected from the EB junction flow towards the CB junction, they partly recombine with the majority carriers (holes) in the base, and therefore $I_{nE} \neq I_{nC}$. The difference, $I_{BB} = -(I_{nE} - I_{nC})$ is referred to as the *base recombination current*. An equivalent hole current flows in from the base terminal to compensate for the holes lost during the base recombination process. Therefore, the base current consists of the following components: hole current at the EB junction (I_{pE}), hole current at the CB junction (I_{pC}), and base recombination current (I_{BB}). The emitter current is due to the hole current at the EB junction (I_{pE}) and the electron current at the EB junction (I_{nE}), while the collector current is due to the hole current at the CB junction (I_{pC}) and the electron current at the CB junction (I_{nC}). Hence, considering that I_E and I_C actually flow in the negative x-direction (refer to Fig. 6.3), we can write

$$\begin{aligned} I_E &= -(I_{nE} + I_{pE}) \\ I_B &= I_{pC} - I_{pE} + I_{BB} = I_{pC} - I_{pE} - (I_{nE} - I_{nC}) \\ I_C &= -(I_{nC} + I_{pC}) \end{aligned} \tag{6.1}$$

Rearranging Eq. (6.1) we get

$$I_E = I_B + I_C \tag{6.2}$$

Equation (6.2) states that the total current flowing into the transistor is equal to the current flowing out, and is only the Kirchoff's Current Law as applied to BJTs. Figure 6.5 shows the various current components and the direction of flow of carriers in the normal active mode of operation, that is, when $V_{BE} > 0$ and $V_{BC} < 0$.

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6

Bipolar Junction Transistors

6.1 INTRODUCTION

The Bipolar Junction Transistor (BJT) is one of the most important solid-state devices. It is a two-junction, three-terminal device in which the current flow properties of one p-n junction can be modulated by another p-n junction. The structure can be either p-n-p or n-p-n. The three terminals are referred to as emitter, base, and collector. The circuit symbols of the two types of BJT are shown in Fig. 6.1. The arrow on the emitter signifies the direction of the current flowing at the emitter terminal. This chapter focuses on the n-p-n transistor, which is the device predominantly used in integrated circuits. However, the same analysis is valid for the p-n-p transistor as well, with appropriate changes made for carrier types and directions of currents.

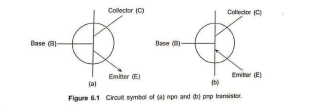

Figure 6.1 Circuit symbol of (a) npn and (b) pnp transistor.

6.2 PRINCIPLE OF OPERATION

In order to understand the interaction of the two p-n junctions in a BJT, let us first consider the structure shown in Fig. 6.2(a) in which a region of p-silicon is sandwiched between two n-regions (n_1 and n_2). Let us assume that the thickness of the intermediate p-layer is much larger than the

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diffusion length of electrons (L_n). If one of the two junctions (J_1) is now forward biased and the other junction (J_2) is reverse biased, practically all the electrons injected by the forward biased junction into the p-region will recombine before they reach the second junction. At J_2, the electron current, which is proportional to the slope of the electron concentration profile, is almost zero. Thus, the two junctions are independent of each other. No appreciable current flows at the n_2 terminal and the structure is equivalent to two diodes connected back to back.

However, as shown in Fig. 6.2(b), if the width of the p-region (base) is very small (less than L_n), most of the electrons injected into the p-region from the n_1-region (emitter) reach J_2. These electrons are then swept across the depletion region of this reverse biased junction by the large electric field present and are finally collected in the n_2-region (*collector*). So, even though J_2 is reverse biased, a large current flows at the n_2 terminal due to the proximity of the two junctions. This is the principal feature of transistor action. The n_1-region which emits electrons when J_1 is forward biased, is called the *emitter* and the narrow p-region is called the *base* of the transistor.

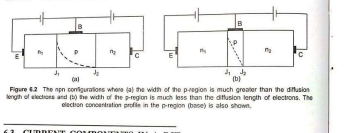

Figure 6.2 The npn configurations where (a) the width of the p-region is much greater than the diffusion length of electrons and (b) the width of the p-region is much less than the diffusion length of electrons. The electron concentration profile in the p-region (base) is also shown.

6.3 CURRENT COMPONENTS IN A BJT

Figure 6.3 shows a one-dimensional n-p-n transistor structure. As shown in the figure, the emitter, base, and collector widths are W_E, W_B, and W_C respectively. The neutral base extends from $x = 0$ to $x = W_B$, while the neutral emitter and collector lies at $-(x_E + W_E) < x < -x_E$ and $x_C < x < (x_C + W_C)$ respectively. The doping concentrations in the emitter, base, and collector are N_{DE}, N_A, and N_{DC} respectively. The transistor has a cross-sectional area A, and is uniform along the cross-section. The applied voltage across the emitter–base junction is denoted by V_{BE}, while the applied voltages across the collector–base junction is V_{BC}. The assumed directions of flow of the terminal currents, I_E, I_B and I_C are also shown in the figure.

Since the transistor has two junctions, namely emitter–base (EB) and collector–base (CB), and each of these junctions can be either forward or reverse biased, we can have four different modes of operation for the bipolar transistor. They are:

(a) *Normal active mode*—The emitter–base junction is forward biased and CB junction is reverse biased.
(b) *Saturation mode*—Both EB and CB junctions are forward biased.
(c) *Inverse active mode*—The emitter–base junction is reverse biased and CB junction is forward biased.
(d) *Cut-off mode*—Both EB and CB junctions are reverse biased.

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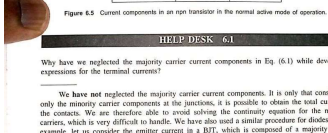

Figure 6.5 Current components in an npn transistor in the normal active mode of operation.

HELP DESK 6.1

Why have we neglected the majority carrier current components in Eq. (6.1) while developing expressions for the terminal currents?

We have not neglected the majority carrier current components. It is only that considering only the minority carrier components at the junctions, it is possible to obtain the total current at the contacts. We are therefore able to avoid solving the continuity equation for the majority carriers, which is very difficult to handle. We have also used a similar procedure for diodes. As an example, let us consider the emitter current in a BJT, which is composed of a majority carrier (electron) current ($I_{nE(maj)}$) and a minority carrier (hole) current ($I_{pE(min)}$) at the emitter contact. The electrons injected at the emitter contact, while flowing towards the EB junction, partly recombine in the bulk emitter. Since I_{nE} is the electron current at the EB junction, which is injected as a minority carrier current into the base, we can write $I_{nE} = I_{nE(maj)} - I_{RE}$, where I_{RE} is the recombination current in the bulk emitter. Similarly, the holes injected into the emitter partly recombine, while flowing towards the emitter contact. Since an identical number of holes and electrons are involved in recombination, we can write $I_{pE(min)} = I_{pE} - I_{RE}$, where I_{pE} is the hole current at the junction. Therefore, we see that $I_{nE(maj)} + I_{pE(min)} = I_{nE} + I_{pE}$, or that *the sum of the majority and minority carrier components at the contact equals the sum of the minority carrier current components at the junction.*

Having defined the various current components in a BJT and their relationships with the external currents, we shall now derive expressions for these components in terms of the various geometrical and doping parameters of the device. The procedure followed is similar to that used for diodes in Chapter 4. We shall first obtain the minority carrier relationships by solving the continuity equations in the emitter, base, and collector regions. The diffusion currents at the EB and CB junctions are then obtained from the slopes of these relations.

Let us first consider the base region. Under steady state condition, that is, $\partial n_B/\partial t = 0$, the continuity equation in the neutral base can be written using Eq. (3.32a) as

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$$D_n \frac{\partial^2 n_p}{\partial x^2} = \frac{n_p - n_{p0}}{\tau_n} \tag{6.3}$$

The solution of this differential equation is of the form

$$n_p - n_{p0} = A \exp\left(\frac{x}{L_n}\right) + B \exp\left(-\frac{x}{L_n}\right) \tag{6.4}$$

where

A and B = constants to be evaluated from two boundary conditions and

$L_n = \sqrt{D_n \tau_n}$ is the minority carrier diffusion length in the base.

The two boundary conditions are the values of n_p at the edges of the depletion region on either side of the base given by

(i) $n_p(x = 0) = n_{pE} = n_{p0} \exp\left(\frac{V_{BE}}{V_T}\right)$

and

(ii) $n_p(x = W_B) = n_{pC} = n_{p0} \exp\left(\frac{V_{BC}}{V_T}\right)$

Substituting these two boundary conditions in Eq. (6.4), we have

$$n_p - n_{p0} = n'_{pE} \left[\frac{\sinh\left(\frac{W_B - x}{L_n}\right)}{\sinh\left(\frac{W_B}{L_n}\right)}\right] + n'_{pC} \left[\frac{\sinh\left(\frac{x}{L_n}\right)}{\sinh\left(\frac{W_B}{L_n}\right)}\right] \tag{6.5}$$

where

n'_{pE} and n'_{pC} = excess minority carrier concentrations in the base at the edges of the EB and CB depletion regions respectively given by

$$n'_{pE} = \left(n_p - n_{p0}\right)_{x=0} = n_{p0}\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] = \frac{n_i^2}{N_A}\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] \tag{6.6}$$

and

$$n'_{pC} = \left(n_p - n_{p0}\right)_{x=W_B} = n_{p0}\left[\exp\left(\frac{V_{BC}}{V_T}\right) - 1\right] = \frac{n_i^2}{N_A}\left[\exp\left(\frac{V_{BC}}{V_T}\right) - 1\right] \tag{6.7}$$

where

N_A = doping concentration in the base.

The electron current injected by the emitter into the base can now be obtained as

$$I_{nE} = \left(A q D_n \frac{dn_p}{dx}\right)_{x=0} = -\frac{A q D_n}{L_n}\left[n'_{pE} \coth\left(\frac{W_B}{L_n}\right) - n'_{pC} \operatorname{cosech}\left(\frac{W_B}{L_n}\right)\right] \tag{6.8}$$

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And similarly the electron current at the CB junction can be written as

$$I_{nC} = \left(A q D_n \frac{dn_p}{dx}\right)_{x=W_B} = -\frac{A q D_n}{L_n}\left[n'_{pE} \operatorname{cosech}\left(\frac{W_B}{L_n}\right) - n'_{pC} \coth\left(\frac{W_B}{L_n}\right)\right] \tag{6.9}$$

Following a similar procedure, that is, solving the continuity equation for holes in the emitter and collector regions using appropriate boundary conditions (say, $p_n = p_{n0}$ at the emitter and collector contacts), we get

$$I_{pE} = \left(-A q D_{pE} \frac{dp_n}{dx}\right)_{x=-x_E} = -\frac{A q D_{pE} p'_{nE}}{L_{pE}} \coth\left(\frac{W_E}{L_{pE}}\right) \tag{6.10}$$

and

$$I_{pC} = \left(-A q D_{pC} \frac{dp_n}{dx}\right)_{x=x_C} = \frac{A q D_{pC} p'_{nC}}{L_{pC}} \coth\left(\frac{W_C}{L_{pC}}\right) \tag{6.11}$$

where

D_{pE} and D_{pC} = the minority carrier diffusion constants in the emitter and collector respectively,

L_{pE} and L_{pC} = the minority carrier diffusion lengths in the emitter and collector respectively, and

p'_{nE} and p'_{nC} = the excess minority carrier concentrations at the edges of the depletion region in the emitter and collector respectively. These are given by

$$p'_{nE} = \left(p_n - p_{n0}\right)_{x=-x_E} = \left(p_n\right)_{x=-x_E}\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] = \frac{n_i^2}{N_{DE}}\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] \tag{6.12}$$

and

$$p'_{nC} = \left(p_n - p_{n0}\right)_{x=x_C} = \left(p_n\right)_{x=x_C}\left[\exp\left(\frac{V_{BC}}{V_T}\right) - 1\right] = \frac{n_i^2}{N_{DC}}\left[\exp\left(\frac{V_{BC}}{V_T}\right) - 1\right] \tag{6.13}$$

The various current components given by Eqs. (6.8)–(6.11) can be substituted in Eq. (6.1) to obtain expressions for the terminal currents I_E, I_B, and I_C as

$$I_E = a_{11}\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] - a_{12}\left[\exp\left(\frac{V_{BC}}{V_T}\right) - 1\right] \tag{6.14}$$

$$I_C = a_{21}\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] - a_{22}\left[\exp\left(\frac{V_{BC}}{V_T}\right) - 1\right] \tag{6.15}$$

$$I_B = (a_{11} - a_{21})\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] - (a_{12} - a_{22})\left[\exp\left(\frac{V_{BC}}{V_T}\right) - 1\right] \tag{6.16}$$

where

$$a_{11} = A q n_i^2 \left[\frac{D_n}{L_n N_A} \coth\left(\frac{W_B}{L_n}\right) + \frac{D_{pE}}{L_{pE} N_{DE}} \coth\left(\frac{W_E}{L_{pE}}\right)\right]$$

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$$a_{12} = a_{21} = A q n_i^2 \frac{D_n}{L_n N_A} \operatorname{cosech}\left(\frac{W_B}{L_n}\right)$$

$$a_{22} = A q n_i^2 \left[\frac{D_n}{L_n N_A} \coth\left(\frac{W_B}{L_n}\right) + \frac{D_{pC}}{L_{pC} N_{DC}} \coth\left(\frac{W_C}{L_{pC}}\right)\right] \tag{6.17}$$

The terminal currents can then be calculated from Eqs. (6.14)–(6.16) for given values of V_{BE} and V_{BC}, provided the values of the required device parameters are known. We can also see that both I_E and I_C increase with increase in V_{BE}, but reduce with increase in V_{BC}. The implications of this relationship are discussed in detail in subsequent sections.

Although the equations developed in this section appear complex and difficult to handle, they can be simplified if the particular mode of operation is known. Also, if the widths of the emitter, base and collector are small compared to the respective minority carrier diffusion lengths, as in the BJTs in present day ICs, further simplification is possible. We shall now demonstrate this in the next section for the normal active mode of operation.

6.4 APPROXIMATE EXPRESSIONS FOR CURRENTS IN NORMAL ACTIVE MODE OF OPERATION

The BJT is biased in the normal active mode in most analog applications. The expressions for the different current components in a practical transistor operating under normal active bias can be simplified by making certain approximations. For a reverse bias applied across the CB junction, the minority carrier concentration at $x = W_B$ can be approximated to be zero. On the other hand, for a forward biased EB junction, the minority carrier concentration at $x = 0$ will be extremely large. Therefore, in Eqs. (6.8) and (6.9), the term involving n'_{pC} can be neglected with respect to the term involving n'_{pE}. Hence, we can write an approximate expression for the emitter current due to injected electrons as

$$I_{nE} \approx -\frac{A q D_n n'_{pE}}{L_n} \coth\left(\frac{W_B}{L_n}\right) \tag{6.18}$$

Similarly, the electron current at the CB junction can be approximated as

$$I_{nC} \approx -\frac{A q D_n n'_{pE}}{L_n} \operatorname{cosech}\left(\frac{W_B}{L_n}\right) \tag{6.19}$$

For well-designed BJTs, the base width W_B is much smaller than the electron diffusion length L_n in the base, or $W_B << L_n$. This condition can be used to further simplify Eqs. (6.18) and (6.19). Since for $x << 1$, $\cosh(x) \approx 1$ and $\operatorname{cosech}(x) \approx 1/x$, we have

$$I_{nE} \approx I_{nC} \approx -\frac{A q D_n n'_{pE}}{W_B} \tag{6.20}$$

Physically interpreting, this means that the recombination in the base is negligible ($I_{BB} \approx I_{nC}$). This is what is expected because the base width is much less than the diffusion length

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and almost all the electrons injected into the base from the emitter reach the collector junction without recombining. Also, it signifies that the electron concentration in the base falls linearly as shown in Fig. 6.6 and

$$\frac{dn}{dx} \approx -\frac{n'_{pE}}{W_B}$$

throughout the base as in a short base diode (refer to section 4.3.3).

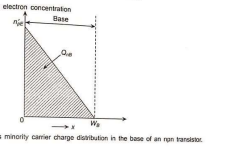

Figure 6.6 Excess minority carrier charge distribution in the base of an npn transistor.

HELP DESK 6.2

Why does the electron concentration fall almost linearly from the emitter to the collector side of the base, although there is negligible recombination?

We know from the continuity equation that the current flowing out from a particular region is equal to the difference between the current flowing in and the recombination current in the region. If recombination is negligible, the current outflow is almost equal to the current inflow. If the current is a diffusion current, it further implies that the slope of the carrier concentration curve at the input is almost equal to that at the output. Therefore in the base region, in the absence of any drift current component, the electron concentration falls linearly so that the current is almost a constant from the emitter end to the collector end.

The approximate expression for the base recombination current I_{BB} is given by

$$I_{BB} = -(I_{nE} - I_{nC})$$

can be obtained from Eqs. (6.18) and (6.19) as

$$I_{BB} = \frac{A q D_n n'_{pE}}{L_n}\left[\coth\left(\frac{W_B}{L_n}\right) - \operatorname{cosech}\left(\frac{W_B}{L_n}\right)\right] = \frac{A q D_n n'_{pE}}{L_n \sinh\left(\frac{W_B}{L_n}\right)}\left[\cosh\left(\frac{W_B}{L_n}\right) - 1\right] \tag{6.21}$$

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Now, for $x << 1$, $\cosh(x) \approx 1 + (x^2/2)$ and $\sinh(x) \approx x$. Assuming $(W_B/L_n) << 1$, from Eq. (6.21), we can write

$$I_{BB} \approx \frac{A q D_n n'_{pE} W_B}{2 L_n} = \frac{A q n'_{pE} W_B}{2 \tau_n} \tag{6.22}$$

Let us calculate I_{BB} using an alternate and perhaps a simpler method. In Eq. (6.5), neglecting the term involving n'_{pC} and taking $(W_B/L_n) << 1$, we have a simplified expression

$$n_p - n_{p0} \approx n'_{pE}\frac{W_B - x}{W_B} = n'_{pE}\left(1 - \frac{x}{W_B}\right) \tag{6.23}$$

Figure 6.6 plots the excess carrier concentration distribution in the base given by Eq. (6.23). The plot is similar to Fig. 4.17, which is for a short base diode. The excess minority carrier charge in the base (Q_{nB}) is obtained from the shaded area under the concentration curve in Fig. 6.6 as

$$Q_{nB} = \frac{A q n'_{pE} W_B}{2} \tag{6.24}$$

Since the base recombination current can be expressed as the minority carrier stored charge in the base divided by the lifetime, we have

$$I_{BB} = \frac{Q_{nB}}{\tau_n} = \frac{A q n'_{pE} W_B}{2 \tau_n} \tag{6.25}$$

We can see that Eq. (6.25) is identical to Eq. (6.22).

Now the terminal currents are obtained by substituting the values of I_{nE}, I_{pE}, I_{nC}, I_{pC}, and I_{BB} from Eqs. (6.20), (6.10), (6.11) and (6.22) in Eq. (6.1) as

$$\begin{aligned} I_E &= -(I_{nE} + I_{pE}) = \frac{A q D_n n'_{pE}}{W_B} + \frac{A q D_{pE} p'_{nE}}{L_{pE}} \coth\left(\frac{W_E}{L_{pE}}\right) \\ &= A q n_i^2 \left[\frac{D_n}{N_A W_B} + \frac{D_{pE}}{N_{DE} L_{pE}} \coth\left(\frac{W_E}{L_{pE}}\right)\right]\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] \end{aligned} \tag{6.26}$$

$$\begin{aligned} I_C &= -(I_{nC} + I_{pC}) = \frac{A q D_n n'_{pE}}{W_B} - \frac{A q D_{pC} p'_{nC}}{L_{pC}} \coth\left(\frac{W_C}{L_{pC}}\right) \approx \frac{A q D_n n'_{pE}}{W_B} \\ &= \frac{A q D_n n_i^2}{N_A W_B}\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] \end{aligned} \tag{6.27}$$

$$\begin{aligned} I_B &= I_{pC} - I_{pE} + I_{BB} = \frac{A q D_{pC} p'_{nC}}{L_{pC}} \coth\left(\frac{W_C}{L_{pC}}\right) + \frac{A q D_{pE} p'_{nE}}{L_{pE}} \coth\left(\frac{W_E}{L_{pE}}\right) + \frac{A q n'_{pE} W_B}{2 \tau_n} \\ &= A q n_i^2 \left[\frac{D_{pE}}{N_{DE} L_{pE}} \coth\left(\frac{W_E}{L_{pE}}\right) + \frac{W_B}{2 N_A \tau_n}\right]\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] \end{aligned} \tag{6.28}$$

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In the Eqs. (6.26)–(6.28), all the terms involving p'_{nC} have been neglected as the CB junction is reverse biased and p'_{nC} is a very small negative quantity. An interesting observation is that I_E, I_C, and I_B are all proportional to $\left[\exp\left(V_{BE}/V_T\right) - 1\right]$ and therefore a ratio of any two of these currents is a constant and independent of bias. (This is not true for other modes of operation). Also, from Eqs. (6.24) and (6.27), we can write

$$I_C = \frac{A q D_n n'_{pE}}{W_B} = \frac{A q n'_{pE} W_B}{2\left(\frac{W_B^2}{2 D_n}\right)} = \frac{Q_{nB}}{\frac{W_B^2}{2 D_n}} = \frac{Q_{nB}}{\tau_B} \tag{6.29}$$

where

$\tau_B = (W_B^2/2D_n)$ is called the *base transit time*.

We have already discussed about transit time in a short base diode in section 4.4.3. The similarity between the two expressions for transit time is quite obvious.

HELP DESK 6.3

Why is it that from Eqs. (6.26)–(6.28), $I_E \neq I_C + I_B$?

We have used $I_{BB} = I_{nC} - I_{pE}$ to develop the base current relation, although we have assumed $I_{nE} + I_{pE}$ (that is $I_{BB} = 0$) to develop the relation for I_E. This gives rise to the discrepancy. In a well designed BJT with $W_B << L_n$, I_{BB} is much smaller than I_C and can therefore be neglected while calculating I_C. However, I_{BB} is an important component of I_B and cannot be neglected when I_B is calculated.

6.5 BASIC BJT PARAMETERS

A bipolar transistor can be operated in any of the three configurations shown in Figs. 6.7(a), (b), and (c). In Fig. 6.7(a), the base terminal is common to both the input and the output and is hence termed the common base configuration. Similarly, the configurations in Figs. 6.7(b) and (c), where the emitter and collector are the common terminals, are called common emitter and common collector configurations respectively. Figure 6.7 also shows the input and output currents and voltages for each configuration.

One of the most important parameters of the BJT is its *common base current gain* (α) defined for the normal active region of operation as

$$\alpha = \left|\frac{I_{nC}}{I_E}\right| \tag{6.30}$$

From Eq. (6.1), we can therefore express I_C as

$$I_C = -(I_{nC} + I_{pC}) = \alpha I_E - I_{pC} = \alpha I_E + I_{CBO} \tag{6.31}$$

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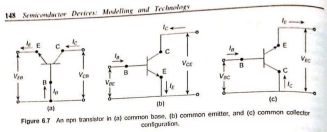

Figure 6.7 An npn transistor in (a) common base, (b) common emitter, and (c) common collector configuration.

where I_{CBO} = small leakage current flowing in the reverse biased CB junction when $I_E = 0$ (that is, the emitter contact is open-circuited).

The first two subscripts (CB) in I_{CBO} refer to the two terminals between which the current is flowing and the third subscript (O) refers to the open-circuited third terminal. Neglecting the small leakage current I_{CBO}, we see that α is the ratio of the output current I_C to the input current I_E in the common base configuration, and is therefore referred to as the *common base current gain*.

The parameter α is dependent on two factors, namely the *emitter efficiency* γ, and the *base transport factor* α_T. The emitter efficiency γ is defined as the ratio of the current due to electron injected from the emitter into the base (I_{nE}) to the total emitter current (I_E). That is,

$$\gamma = \left|\frac{I_{nE}}{I_E}\right| = \frac{1}{1 + \frac{I_{pE}}{I_{nE}}} \tag{6.32}$$

An emitter efficiency close to unity indicates that almost the entire emitter current is due to the "useful" injection of electrons from the emitter to the base. The base transport factor α_T is defined as the ratio of the collector current due to electrons (I_{nC}) to the current due to electron injected from emitter into base (I_{nE}). That is,

$$\alpha_T = \left|\frac{I_{nC}}{I_{nE}}\right| = 1 - \left|\frac{I_{BB}}{I_{nE}}\right| \tag{6.33}$$

If almost all the electrons injected into the base reach the collector junction without recombining, α_T becomes close to unity. From Eqs. (6.30), (6.32), and (6.33) we see that

$$\alpha \approx \gamma \alpha_T \tag{6.34}$$

It may be noted that ideally the maximum values of α_T and γ are unity. However, in all practical situations, they are less than (though very close to) unity. Thus, the value of α is also close to unity in most well designed BJTs.

The other important parameter is the *common emitter current gain* (**usually represented as** β **or** h_{FE}) defined as

$$\beta = \frac{\alpha}{1 - \alpha} \tag{6.35}$$

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Equation (6.39) suggests that the maximum value of γ is unity. In order to obtain γ close to unity, the doping concentration in the base must be made much smaller than that in the emitter (that is $N_A \ll N_D$). Also, the base width must be much smaller than the emitter width (that is $W_B \ll W_E$). In fact, the number of dopants per unit area, called the Gummel number ($GN_E = N_{DE}W_E$) must be much greater than the base Gummel number ($GN_B = N_AW_B$).

The base transport factor α_T can be obtained by substituting the expressions for I_{nC} and I_{nE} given by Eqs. (6.18) and (6.19) into Eq. (6.33) as

$$\alpha_T = \frac{I_{nC}}{I_{nE}} = \frac{\dfrac{AqD_n n'_{pE}}{L_n}\operatorname{cosech}\left(\dfrac{W_B}{L_n}\right)}{\dfrac{AqD_n n'_{pE}}{L_n}\coth\left(\dfrac{W_E}{L_n}\right)} = \frac{1}{\cosh\left(\dfrac{W_B}{L_n}\right)} \tag{6.40}$$

Approaching as before, approximating $\cosh(W_B/L_n)$ using the condition $W_B \ll L_n$, the base transport factor can be approximated as

$$\alpha_T = \frac{1}{1 + \dfrac{W_B^2}{2L_n^2}} \tag{6.41}$$

From Eq. (6.41), we see that in order to have $\alpha_T \approx 1$, W_B must be much less than L_n so that recombination in the base is minimized. Now, substituting Eqs. (6.39) and (6.41) in Eq. (6.34) and making suitable approximations, α can be expressed as

$$\alpha = \frac{1}{\left(1 + \dfrac{D_{pE}N_AW_B}{D_nN_{DE}W_E}\right)\left(1 + \dfrac{W_B^2}{2L_n^2}\right)} = \frac{1}{1 + \dfrac{D_{pE}N_AW_B}{D_nN_{DE}W_E} + \dfrac{W_B^2}{2L_n^2}} \tag{6.42}$$

The common-emitter current gain β can now be expressed by substituting Eq. (6.42) into Eq. (6.35). This gives

$$\beta = \frac{\alpha}{1-\alpha} = \frac{1}{\dfrac{1}{\alpha} - 1} = \frac{1}{\dfrac{D_{pE}N_AW_B}{D_nN_{DE}W_E} + \dfrac{W_B^2}{2L_n^2}} \tag{6.43}$$

In order to obtain a high value of β (and also α close to unity), the denominator in Eq. (6.43) must be made small. Therefore, three conditions must be satisfied, namely (a) $N_A \ll N_{DE}$; (b) $W_B \ll W_E$; (c) $W_B \ll L_n$. The conditions (a) and (b) ensure that the hole injection from the base into the emitter is negligible compared to the forward injection of electrons from the emitter into the base, resulting in $\gamma \approx 1$. Also the base should be very narrow so that almost all the electrons injected by the emitter can reach the collector without recombining in the base and we have $\alpha_T \approx 1$.

Now, considering a transistor with $N_A \ll N_{DE}$, from Eqs. (6.6) and (6.12), we have $n'_{pE} \gg p_{nE}$. Also since the CB junction is reverse biased in the normal active mode, $n'_{pC} \gg p_{nC}$. Therefore, we can simplify Eq. (6.28) by neglecting terms involving p'_{nB} and p'_{nC}, so that

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$$I_B \approx I_{BR} = \frac{Aqn'_{pE}W_B}{2\tau_n} = \frac{Q_{nB}}{\tau_n} \tag{6.44}$$

where Q_{nB} is given by Eq. (6.24).

Now, neglecting I_{CBO} in Eq. (6.36) and from Eqs. (6.29) and (6.44), we have

$$\beta = \frac{I_C}{I_B} = \frac{\tau_n}{\tau_{nB}} \tag{6.45}$$

Thus, we see that the expression for β can be extremely simplified under certain conditions to a ratio of two time constants, namely the electron lifetime in the base and the base transit time.

We have seen in this section that both α and β, which are defined in the normal active mode of operation, are constants depending only on the physical parameters of the BJT. This implies that when a BJT is operating in the normal active mode, its current gain is independent of the bias voltages.

Example 6.1

An npn bipolar junction transistor T_1 has the following parameters:

	Emitter (n-type)	Base (p-type)	Collector (n-type)
Width (μm)	1.0	0.8	3.0
Doping concentration (cm^{-3})	10^{19}	10^{17}	5×10^{15}
Minority carrier diffusion constant (cm^2/s)	2	20	10
Minority carrier lifetime (μs)	0.01	0.1	1

Calculate the emitter efficiency γ, base transport factor α_T, common base current gain α, and common emitter current gain β for this transistor.

Solution: The minority carrier diffusion lengths in the emitter, base, and collector are given by

$$L_{nE} = \sqrt{2 \times 0.01 \times 10^{-6}}\ \text{cm} = 1.4\ \mu\text{m},$$

$$L_{nB} = \sqrt{20 \times 0.1 \times 10^{-6}}\ \text{cm} = 14\ \mu\text{m}, \text{ and}$$

$$L_{nC} = \sqrt{10 \times 10^{-6}}\ \text{cm} = 31.6\ \mu\text{m}.$$

Since W_E is comparable to L_{nE}, we cannot use Eq. (6.39) to evaluate γ. From Eq. (6.32) and Eq. (6.37), and since $W_B \ll L_n$, we can write

$$\gamma = \frac{1}{1 + \dfrac{D_{pE}W_E N_A \coth\left(\dfrac{W_E}{\tau_{pE}}\right)}{D_n L_{nE} N_{DE}}} = \frac{1}{1 + \dfrac{2 \times 0.8 \times 10^{-4} \times 10^{17}\coth\left(\dfrac{1}{1.4}\right)}{20 \times 1.4 \times 10^{-4} \times 10^{19}}} = 0.999$$

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Case 1

$$V_{BE} \neq 0,\ V_{BC} = 0$$

This is the state of the forward operation of the transistor. For $V_{BC} = 0$, Eq. (6.47) gives $I_R = 0$. Therefore from Eqs. (6.48)–(6.50), we get

$$I_C = \alpha_F I_F \tag{6.51}$$

$$I_E = I_F \tag{6.52}$$

$$I_B = I_F - I_C = (1 - \alpha_F)\, I_F \tag{6.53}$$

The **forward common emitter current gain** (β_F), defined as the ratio of collector current to base current when $V_{BC} = 0$, is given by

$$\beta_F = \frac{I_C}{I_B} = \frac{\alpha_F}{1 - \alpha_F} \tag{6.54}$$

Also, we note that for $V_{BC} = 0$, $Q_{BR} = Q_C = 0$.

Under these circumstances, the collector current can be expressed as

$$I_C = \alpha_F I_F = AqD_n \frac{dn}{dx} = AqD_n \frac{n'_{pE}}{W_B}\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] = I_S\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] \tag{6.55}$$

where

$$I_S = \frac{AqD_n n_i^2}{N_A W_B} \tag{6.56}$$

Equation (6.55) is essentially same as Eq. (6.29). Now the base current for this case is the sum of the recombination current I_{BR} and the injected hole current into emitter I_{pE}. Here, I_{pE} can be expressed as

$$I_{pE} = AqD_p \frac{dp}{dx} = \frac{AqD_p n_i^2}{N_{DE}W_E}\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] \tag{6.57}$$

Noting that $Q_{nB} = Q_{BR}$ for this case and using $I_{BR} = (Q_{nB}/\tau_n)$ as given by Eq. (6.25), we have

$$I_B = I_{pE} + I_{BR} = \frac{AqD_p n_i^2}{N_{DE}W_E}\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] + \frac{Aqn_i^2 W_B}{2N_A\tau_n}\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right]$$

$$= Aqn_i^2\left(\frac{D_p}{N_{DE}W_E} + \frac{W_B}{2N_A\tau_n}\right)\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] \tag{6.58}$$

Using Eqs. (6.54), (6.55), and (6.58), we obtain the expression for β_F, which is identical to the expression for the common emitter gain β given by Eq. (6.43).

Case 2

$$V_{BE} = 0,\ V_{BC} \neq 0$$

This is the configuration for reverse operation of the transistor. For $V_{BE} = 0$, from Eq. (6.48), $I_F = 0$. Also, $Q_E = Q_{BF} = 0$. The stored charge in the base is represented by Q_{nB} and the emitter current is due to electrons injected into the base from the collector. Therefore,

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$$I_E = -\alpha_R I_R = -AqD_n \frac{dn}{dx} = -\frac{AqD_n n_i^2}{N_A W_B}\left[\exp\left(\frac{V_{BC}}{V_T}\right) - 1\right] = -I_S\left[\exp\left(\frac{V_{BC}}{V_T}\right) - 1\right] \tag{6.59}$$

It is to be noted that the direction of the emitter current has changed. This is because electrons are now injected from the collector into the base and flow towards the emitter due to the concentration gradient in the base. It is now possible to define a parameter called the reverse common emitter current gain β_R, which is essentially the ratio of the emitter current to the base current when $V_{BE} = 0$. The reverse common emitter current gain is given by

$$\beta_R = \frac{\alpha_R}{1 - \alpha_R}$$

Case 3

$$V_{BC} \neq 0,\ V_{BE} \neq 0$$

In order to analyze this condition we use superposition of the current obtained under the two limiting cases discussed previously. Thus using Eqs. (6.49), (6.55), and (6.59), we can write

$$I_C = \alpha_F I_F - I_R = I_S\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] - \frac{I_S\left[\exp\left(\frac{V_{BC}}{V_T}\right) - 1\right]}{\alpha_R} \tag{6.60}$$

From the expression of β_F, we see that $\alpha_R \approx \beta_R/(1 + \beta_R)$. Thus Eq. (6.60) can be written as

$$I_C = I_S\left[\exp\left(\frac{V_{BE}}{V_T}\right) - \exp\left(\frac{V_{BC}}{V_T}\right)\right] - \frac{I_S\left[\exp\left(\frac{V_{BC}}{V_T}\right) - 1\right]}{\beta_R} \tag{6.61}$$

Similarly the emitter current can be expressed from Eqs. (6.48), (6.55), and (6.59) as

$$I_E = I_F - \alpha_R I_R = \frac{I_S\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right]}{\alpha_F} - I_S\left[\exp\left(\frac{V_{BC}}{V_T}\right) - 1\right] \tag{6.62}$$

Considering $\alpha_F = \beta_F/(1 + \beta_F)$, we can rewrite Eq. (6.62) as

$$I_E = I_S\left[\exp\left(\frac{V_{BE}}{V_T}\right) - \exp\left(\frac{V_{BC}}{V_T}\right)\right] + \frac{I_S\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right]}{\beta_F} \tag{6.63}$$

Therefore the expression for the base current is obtained from $I_B = I_E - I_C$ as

$$I_B = \frac{I_S\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right]}{\beta_F} + \frac{I_S\left[\exp\left(\frac{V_{BC}}{V_T}\right) - 1\right]}{\beta_R} \tag{6.64}$$

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Since $W_B \ll L_n$, from Eq. (6.41), we have

$$\alpha_T = \frac{1}{1 + \dfrac{W_B^2}{2L_n^2}} = \frac{1}{1 + \dfrac{(0.8 \times 10^{-4})^2}{2(14 \times 10^{-4})^2}} = 0.99837$$

From Eq. (6.34),

$$\alpha = \gamma\alpha_T = 0.999 \times 0.99837 = 0.9974$$

From Eq. (6.35),

$$\beta = \frac{\alpha}{1-\alpha} = \frac{0.9974}{1 - 0.9974} = 383.6$$

6.6 THE EBERS–MOLL MODEL

For circuit simulation, it is necessary to have a common set of equations which relate the terminal currents to the terminal voltages, for all modes of operation. Also, the model should have few parameters which can be extracted experimentally. The Ebers–Moll model is ideally suited for this purpose and is included in the popular circuit simulator SPICE.

The basic equivalent circuit used for the Ebers–Moll model of the BJT is shown in Fig. 6.8. It consists of two back-to-back diodes. The currents through the diodes between the EB and CB terminals are denoted by I_F and I_R respectively and are expressed by the usual diode equation. That is,

$$I_F = I_{ES}\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] \tag{6.46}$$

$$I_R = I_{CS}\left[\exp\left(\frac{V_{BC}}{V_T}\right) - 1\right] \tag{6.47}$$

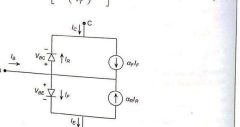

Figure 6.8 Basic Ebers–Moll equivalent circuit of a BJT.

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where

I_{ES} and I_{CS} = the reverse saturation currents of the EB and BC junction diodes respectively. In addition, there are two current sources which stand for the coupling between the two junctions. The current source between the CB terminals is related to the diode current at the EB junction through the forward common base current gain (α_F). Similarly, the current source between the EB terminals is related to the diode current at the CB junction through the reverse common base current gain (α_R). For a symmetrical transistor having identical emitter and collector structure, the gains α_F and α_R are identical. This means that the transistor will behave in exactly the same way if the biasing of the emitter and the collector junction is interchanged. However, practical transistors are not symmetric and hence α_F and α_R have different values.

Now, from Fig. 6.8, the emitter, collector, and base currents in a BJT can be modelled as

$$I_E = I_F - \alpha_R I_R \tag{6.48}$$

$$I_C = \alpha_I I_F - I_R \tag{6.49}$$

$$I_B = I_F - I_C = (1 - \alpha_F)\, I_F + (1 - \alpha_R)\, I_R \tag{6.50}$$

In a modern BJT, all the three regions (that is, emitter, base, and collector) have very small widths. Assuming that their widths are much smaller than the respective diffusion lengths, the minority carrier distributions can be approximated by linear relationships as shown in Fig. 6.9 (refer to HELP DESK 6.2). In this figure, Q_E, Q_{nB}, Q_{BR}, and Q_C refer to the stored charge in the emitter due to holes injected from the base, the stored charge in the base due to electrons injected from the emitter, the stored charge in the base due to electrons injected from the collector, and the stored charge in the collector due to holes injected from the base respectively. Note that the total stored charge in the base is the sum of Q_{BF} and Q_{BR}. With the help of the equivalent circuit and the minority carrier distribution shown in Figs. 6.8 and 6.9, it is now possible to model the currents in a BJT for all possible combinations of V_{BE} and V_{BC}. We first proceed to analyze the two limiting cases when one junction is short-circuited and a bias is applied only across the other junction, and then the general case when non-zero voltages are applied to both the junctions.

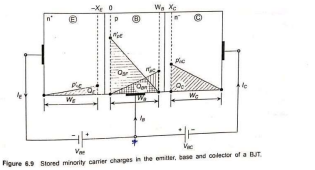

Figure 6.9 Stored minority carrier charges in the emitter, base and collector of a BJT.

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From the symmetry of the expressions for the collector current and the emitter current, it can be seen that the equivalent circuit of a BJT can now be represented by two coupled diodes and only one current source as shown in Fig. 6.10. According to this circuit, the currents flowing through the back-to-back diodes are I_{EC}/β_R and I_{CC}/β_F. The current supplied by the current source is I_{CT}. From this circuit, the collector, emitter, and base currents of a BJT can be expressed as

$$I_C = I_{CT} - \frac{I_{EC}}{\beta_R} \tag{6.65}$$

$$I_E = I_{CT} - \frac{I_{CC}}{\beta_F} \tag{6.66}$$

$$I_B = \frac{I_{CC}}{\beta_F} + \frac{I_{EC}}{\beta_R} \tag{6.67}$$

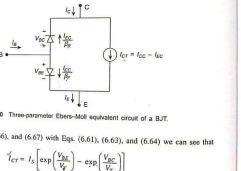

Figure 6.10 Three-parameter Ebers–Moll equivalent circuit of a BJT.

Comparing Eqs. (6.65), (6.66), and (6.67) with Eqs. (6.61), (6.63), and (6.64) we can see that

$$I_{CT} = I_S\left[\exp\left(\frac{V_{BE}}{V_T}\right) - \exp\left(\frac{V_{BC}}{V_T}\right)\right]$$

$$I_{CC} = I_S\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right]$$

$$I_{EC} = I_S\left[\exp\left(\frac{V_{BC}}{V_T}\right) - 1\right]$$

Thus, the Ebers–Moll model helps to obtain the values of the terminal currents for any combination of V_{BE} and V_{BC}, provided the values of only three parameters, namely I_S, β_F, and β_R are known.

The values of these parameters can be extracted very easily through simple experiments. If we short-circuit the collector–base junction (that is, $V_{BC} = 0$) and vary V_{BE}, for $V_{BE} > 3V_T$, we see from Eqs. (6.61) and (6.64) that I_C and I_B vary in the manner given by

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$$I_C \approx I_S \exp\left(\frac{V_{BE}}{V_T}\right) \quad \text{and} \quad I_B \approx \frac{I_C}{\beta_F} \exp\left(\frac{V_{BE}}{V_T}\right)$$

Now, if we plot $\ln(I_C)$ and $\ln(I_B)$ versus V_{BE}, we shall get two parallel straight lines, as shown in Fig. 6.11. (Note that the lines are parallel only in a small range of V_{BE}. The reason for this shall be discussed in Chapter 7.) The intercept of the $\ln(I_C)$ versus V_{BE} curve, when extrapolated, gives $\ln(I_S)$. The ratio of I_C and I_B gives β_F. A similar experiment where $V_{BE} = 0$ and V_{BC} is varied can be used to evaluate β_R.

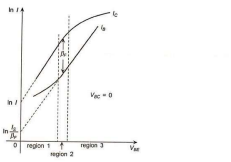

Figure 6.11 $\ln(I_C)$ and $\ln(I_B)$ versus V_{BE} characteristics of a BJT when $V_{BC} = 0$.

HELP DESK 6.5

The EB junction of an npn transistor is forward biased by V_{BE}, while the collector terminal is left floating. (a) What is the open circuit voltage developed across the CB junction? (b) What is the value of V_{CE} at this point? (c) What is the mode of operation of this BJT?

(a) From Eq. (6.65), when $I_C = 0$, $I_{CT} = I_{EC}/\beta_R$. Substituting the expressions for I_{CT} and I_{EC}, we have

$$\left(1 + \frac{1}{\beta_R}\right)\exp\left(\frac{V_{BC}}{V_T}\right) = \exp\left(\frac{V_{BE}}{V_T}\right) + \frac{1}{\beta_R}$$

Therefore,

$$V_{BC} = V_T \ln\left[\frac{\beta_R \exp\left(\dfrac{V_{BE}}{V_T}\right) + 1}{\beta_R + 1}\right] = V_T \ln\left[\alpha_R\left\{\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right\} + 1\right] = V_T \ln\left[\alpha_R \exp\left(\frac{V_{BE}}{V_T}\right)\right]$$

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(b) The above equation gives

$$\exp\left(\frac{V_{BE} - V_{BC}}{V_T}\right) = \alpha_R \exp\left(\frac{V_{BE}}{V_T}\right)$$

or

$$\exp\left(\frac{V_{BE} - V_{BC}}{V_T}\right) = \exp\left(\frac{V_{CE}}{V_T}\right) = \frac{1}{\alpha_R}$$

Therefore,

$$V_{CE} = V_T \ln\left(\frac{1}{\alpha_R}\right)$$

(c) From the solution of (a), we see that V_{BC} is positive, that is, the BC junction is forward biased. Since both the junctions are forward biased, the transistor is found to be in the saturation mode of operation.

Example 6.2

Calculate (a) V_{BC} and (b) V_{CE} when the collector terminal of transistor T_1 (see Example 6.1) is kept floating and $V_{BE} = 0.6$ V.

Solution: The expression for V_{BC} when $I_C = 0$ has been derived in HELP DESK 6.5. To calculate the open circuit base–collector voltage V_{BC}, the value of α_R should be known. Since the roles of emitter and collector are interchanged in the reverse mode of operation, Eq. (6.42) can be used to calculate α_R by replacing the parameters of the emitter with that of the collector. Therefore,

$$\alpha_R = \frac{1}{1 + \dfrac{D_{pC}N_AW_B}{D_nN_{DC}W_C} + \dfrac{W_B^2}{2L_n^2}} = \frac{1}{1 + \dfrac{10 \times 10^{11} \times 0.8}{2 \times 5 \times 10^{15} \times 3} + \dfrac{0.8^2}{2 \times 14^2}} = 0.2726$$

(a) Therefore, we have

$$V_{BC} = V_T \ln\left[\alpha_R \exp\left(\frac{V_{BE}}{V_T}\right)\right] = 0.0261 \ln\left[0.2726 \exp\left(\frac{0.6}{0.026}\right)\right] = 0.566\ \text{V}$$

Also from the HELP DESK 6.5,

(b) $$V_{CE} = V_T \ln\left(\frac{1}{\alpha_R}\right) = 0.026 \ln\left(\frac{1}{0.2726}\right) = 0.034\ \text{V}$$

Thus, we see that when the collector is floating, V_{BC} is only slightly less than V_{BE}.

6.7 STATIC OUTPUT *I–V* CHARACTERISTICS

The output characteristics of a BJT relate the output current to the output voltage for a fixed value of an input variable, usually the input current. Thus, for the common-emitter configuration, the output characteristics establish a relationship between I_C and V_{CE} for a constant I_B. In this section, we shall discuss the output characteristics of an npn transistor in the common base and common emitter configurations.

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6.7.1 Common Base Configuration

As shown in Fig. 6.12(a), the output characteristics of a BJT in the common-base configuration plots the variation of the output current I_C versus the output voltage V_{CB} for a constant input current I_E. We shall now explain the nature of the characteristics from our understanding of the operation of the BJT, as discussed in the previous sections.

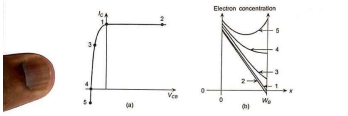

Figure 6.12 (a) Plot of I_C versus V_{CB} and the corresponding (b) minority carrier concentration profile in the base of an npn transistor for constant I_E.

We know from section 6.3 that I_B consists of two components, I_{eE} and I_{nE}. In a well-designed transistor with emitter efficiency nearly equal to unity,

$$I_B \approx -I_{nE} = \left(-AqD_n \frac{dn'_p}{dx}\right)_{x=0} \quad \text{(refer to Eq. (6.8))}$$

Thus, if I_E is fixed, $\left.\dfrac{dn'_p}{dx}\right|_{x=0}$ will also remain at a constant value. Figure 6.12(b) shows the minority carrier distribution in the base of a BJT for various values of V_{CB} while maintaining I_E as constant. We can see that $\left.\dfrac{dn'_p}{dx}\right|_{x=0}$ is constant in all the plots.

Now, from Eq. (6.1), $I_C = -(I_{nC} + I_{pC})$. We know that

$$-I_{nC} = \left(-AqD_n \frac{dn'_p}{dx}\right)_{x=W_B} = -I_{nE} - I_{BR} \approx I_E - \frac{Q_{nB}}{\tau_n}$$

Since I_E is held constant, this means that the rate of decrease in electron concentration $[-(dn'_p/dx)]$ at $x = W_B$ reduces with increase in stored charge in the base (Q_{nB}), which is actually the area under the minority concentration curve in Fig. 6.12(b). Also, it is clear that $-I_{nC}$, which is an important component of I_C, reduces with increase in Q_{nB}.

In fact, when $\dfrac{Q_{nB}}{\tau_n} = I_E$, $\left.\dfrac{dn}{dx}\right|_{x=W_B} = 0$ and $I_{nC} = 0$.

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With this background, the output characteristic curve and the plots of the minority carrier concentration in the base can now be explained in the way as shown in Fig. 6.12, where V_{CB} is varied in steps while keeping I_E constant.

Step 1: $V_{CB} = 0$.

Hence $n'_{pC} = 0$. Assuming recombination in the base to be small, the electron concentration in the base varies linearly as shown in plot 1 in Fig. 6.12(b). Since $I_{pC} = 0$, from Eq. (6.20), we can write $I_C = -I_{nC} = AqD_n n'_{pE}/W_B$. This value is shown as point 1 in the I_C versus V_{CB} plot shown in Fig. 6.12(a).

Step 2: $V_{CB} > 0$.

A large reverse bias is applied across the CB junction. Now, $n'_{pC} = -n_{pC}$ (plot 2 in Fig. 6.12(b)). The electron concentration in the base can again be approximated by a linear variation and at this point,

$$I_C = \frac{AqD_n(n'_{pE} + n_{pC})}{W_B}, \quad \text{as shown by point 2 in Fig. 6.12(a)}$$

The collector current I_C cannot increase any further with further increase in the magnitude of V_{CB}. As $n_{pC} \ll n'_{pE}$, the difference in I_C between points 1 and 2 is very small.

Step 3: $V_{CB} < 0$.

A forward bias is applied across CB junction. Also n'_{pC} is now positive and Q_{nB} increases. Therefore, as already discussed, the slope of the electron concentration distribution at $x = W_B$ reduces. Consequently $-I_{nC}$ and I_C decrease. In addition, due to an appreciable hole current (I_{pC}) from base to collector across the forward bias CB junction, I_C is further reduced. The situation is depicted by point 3 and plot 3 in Figs. 6.12(a) and (b) respectively.

Step 4: $V_{CB} = V_{CF}$.

where V_{CF} is the value of V_{CB} when the collector terminal is floating. So, when $V_{CB} = V_{CF}$, the collector current $I_C = 0$ [point 4 in Fig. 6.12(a)]. We have already seen in Example 6.2 that $|V_{CF}| = V_{BE}$. Since Q_{nB} is increased further, there is a further reduction in the slope of the electron concentration distribution at $x = W_B$ as shown in plot 4 in Fig. 6.12(b). At this point, $I_{nC} = -I_{nO}$ so that $I_C = 0$.

Step 5: $V_{CB} < 0$ and $|V_{CB}| > |V_{CF}|$.

The minority carrier distribution for this case is depicted as plot 5 in Fig. 6.12(b). As shown in the figure, the direction of the slope of the minority carrier concentration curve at $x = W_B$ changes resulting in injection of electrons from collector to the base. This, in addition to holes being injected from base to collector, results in a negative collector current as shown by point 5 in Fig. 6.12(a).

We have now obtained the output characteristic curve for a particular I_E. Figure 6.13 plots the output characteristic curves for various values of I_E. The figure also shows the different regions corresponding to the active, saturation, and cut-off modes of operation. As seen from the figure, in the active region, there is a linear relationship between I_C and I_E as predicted by Eq. (6.51). Since $\alpha \approx 1$, we see that $I_C \approx I_E$ in the active region. However in the saturation region, $I_C < \alpha I_E$.

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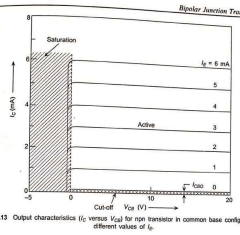

Figure 6.13 Output characteristics (I_C versus V_{CB}) for npn transistor in common base configuration for different values of I_E.

6.7.2 Common Emitter Configuration

Similar to the case of common base mode, we now proceed to explain the output characteristics (I_C versus V_{CE}) in the common emitter configuration by varying the output voltage (V_{CE}) in steps. The input current (I_B) is kept constant in this configuration. Figure 6.14(a) shows the output characteristics while Fig. 6.14(b) plots the minority carrier concentration curves in the base for various values of V_{CE}, while keeping I_B constant.

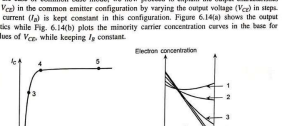

Figure 6.14 (a) Plot of I_C versus V_{CE} and (b) the corresponding minority carrier concentration profile in the base of an npn transistor keeping I_B constant.

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